INCH–POUND

MIL–STD–1651A
3 APRIL 1992
SUPERSEDING
MIL–STD–1651
27 JUNE 1975

MILITARY STANDARD

INSERT ARRANGEMENTS FOR MIL–C–5015, MIL–C–22992 (CLASSES C, J, AND R), AND MIL–C–83723 (SERIES II) ELECTRICAL CONNECTORS

AMSC NO DELIVERABLE DATA REQUIRED BY THIS STANDARD. FSC 5935

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FOREWORD

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, Code SR3, Lakehurst, NJ 08733-5100, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1. SCOPE

1.1 Scope. This standard gives the dimensional requirements for insert arrangements used in MIL-C-5015, MIL-C-22992 (Classes C, J and R), and MIL-C-83723 (Series II) electrical connectors.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.3).

SPECIFICATIONS

MILITARY

MIL-C-5015	Connectors, Electric, "AN" Type
MIL-C-39029	Contacts, Electrical Connector, General Specification for
MIL-C-22992	Connectors, Plugs and Receptacles, Electrical, Waterproof, Quick Disconnect, Heavy Duty Type, General Specification for
MIL-C-83723	Connectors, Electrical (Circular, Environment Resisting), Receptacles and Plugs, General Specification for

STANDARDS

MILITARY

MS3400	Connector, Receptacle Electric, Wall Mounting, Front Release, Crimp Contact, AN Type
MS3450	Connector, Receptacle Electric, Wall Mounting, Rear Release, Crimp Contact, AN Type
MS14056	Contact, Electric Connector, Shielded Cable Applications, Size 8
MS33695	Contact, Video Coaxial Connector

(Unless otherwise indicated, copies of military specifications and standards are available from the DODSSP, Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

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2.2 Non-Government publications. The following documents form a part of the document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation (see 6.3).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Y14.5M Dimensioning and Tolerancing (DOD Adopted)

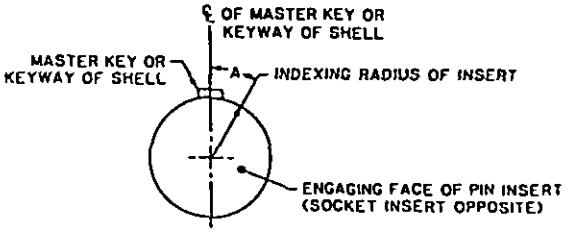
(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018-3308.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Definitions. The terms used in this standard are generally accepted by the electrical and electronics industries and commonly used in electrical connector engineering practice. An illustration of some of the terms used is shown below:



4. GENERAL REQUIREMENTS

Not applicable.

5. DETAIL REQUIREMENTS

5.1 Dimensions. Dimensions shall be in accordance with the applicable section of Appendix A of this standard and the following dimensional data:

a. Dimensioning and tolerancing in accordance with ANSI Y14.5-1982. Dimensions are true position and are in inches.

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b. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

c. Dimensions and markings shown are for engaging face of pin insert. Socket insert is opposite.

d. The following tolerances apply to insert installed in shell:

1. The center of each hole in insert for crimp contact connectors shall be located at true position within .014 dia.
⊕|Ø.014 dia.

2. The center of engaging end of each contact shall be located in true position within .024 dia. ⊕|Ø.024 dia.

e. Unless otherwise indicated, dimensions are symmetrical about centerline.
indicates C_L of insert arrangement.

f. Each insert arrangement is shown in the "normal position" in the shell.

5.2 Marking. Marking shall be in accordance with MIL-C-5015, MIL-C-22992, or MIL-C-83723, as applicable, and as shown in the applicable section of this military standard.

5.3 Contacts. Solder contacts shall be in accordance with MIL-C-5015, MIL-C-22992 or MIL-C-83723, as applicable. Crimp contacts shall be in accordance with table I. Contact sizes are designated by the symbols shown in table I.

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TABLE I. Contact designation.

Crimp or solder contacts		Crimp contact part numbers	
Size	Symbol	MIL-C-5015 MS3400's Connectors MIL-C-39029 Contacts	MIL-C-5015 and MIL-C-83723 Series II MS3450's Connectors MIL-C-39029 Contacts
0		MIL-C-39029/44	M39029/29-0-0
		MIL-C-39029/45	M39029/30-0-0
4		MIL-C-39029/44	M39029/29-4-4
		MIL-C-39029/45	M39029/30-4-4
8		MIL-C-39029/44	M39029/29-8-8
		MIL-C-39029/45	M39029/30-8-8
12		MIL-C-39029/44	M39029/29-12-12
		MIL-C-39029/45	M39029/30-12-12
16		MIL-C-39029/44	M39029/29-16-16
		MIL-C-39029/45	M39029/30-16-16
16S		NA	M39029/30-16S16
S (shielded)		MS14056 (MIL-C-5015 only)	
Video coax		MS33695 (MIL-C-5015 only)	

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This standard contains requirements for insert arrangements to be used in MIL-C-5015, MIL-C-22992 (Classes C, J, R) and MIL-C-83723 (Series II) Electrical Connectors.

6.2 Data requirements. No data is required by this standard.

6.3 Issue of DODISS. When this standard is used in acquisition, the applicable issue of the DODISS must be cited in the solicitation (see 2.1.1, 2.2).

* 6.4 Supersession data. Table II gives the insert arrangements which have been cancelled by Revision A to MIL-STD-1651. The insert patterns for these insert arrangements are contained in Appendix B. Manufacturers may continue to supply connectors using these insert patterns until 30 April 1994, or deplete their inventories, whichever comes first. Table III gives insert arrangements and their previous status which have been deleted from MIL-STD-1651 with the publication of Revision A.

* TABLE II. Insert arrangements cancelled by Revision A to MIL-STD-1651.

Insert Arrangement	Status in MIL-STD-1651	Replacement
12S, 13-1	Inactive for design after 8 March 1945	
12S, 13-2	Inactive for design after 17 April 1945	
14S, 15-10	Inactive for design after 8 December 1945	
14S, 15-11	Inactive for design after 17 April 1945	
14S, 15-12	Inactive for design after 17 April 1945	
14S, 15-13	Inactive for design after 17 April 1945	
16S, 17-14	Inactive for design after 27 April 1945	
16S, 17-15	Inactive for design after 27 April 1945	16S, 17-4

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*TABLE II. Insert arrangements cancelled by Revision A to MIL-STD-1651 - continued.

Insert Arrangement	Status in MIL-STD-1651	Replacement
16S, 17-16	Inactive for design after 27 April 1945	16S, 17-4
16S, 17-17	Inactive for design after 27 April 1945	16-14
18-17	Inactive for design after 19 May 1945	18-9
18-18	Inactive for design after 19 May 1945	18-9
18-19	Inactive for design after 19 May 1945	
18-23	Inactive for design after 19 May 1945	18-1
18-24	Inactive for design after 19 May 1945	18-1
18-25	Inactive for design after 19 May 1945	18-3
18-26	Inactive for design after 19 May 1945	18-3
18-27	Inactive for design after 19 May 1945	18-5
18-28	Inactive for design after 19 May 1945	18-5
18-30	Inactive for design after 19 May 1945	18-20
18-31	Inactive for design after 19 May 1945	18-20
20-10	Inactive for design after 1 June 1945	
20-25	Inactive for design after 1 June 1945	20-11
20-26	Inactive for design after 26 June 1945	

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*TABLE II. Insert arrangements cancelled by Revision A to MIL-STD-1651 - continued.

Insert Arrangement	Status in MIL-STD-1651	Replacement
20-30	Inactive for design after 1 June 1945	20-11
20-32	Inactive for design after 1 June 1945	20-7
22-30	Inactive for design after 26 June 1945	22-14
22-32	Inactive for design after 26 June 1945	22-5
22-35	Inactive for design after 26 June 1945	
24-15	Inactive for design after 26 June 1945	24-5
24-18	Inactive for design after 26 June 1945	
24-24	Inactive for design after 26 June 1945	24-5
24-25	Inactive for design after 26 June 1945	24-6
24-26	Inactive for design after 26 June 1945	24-6
28-7	Inactive for design after 13 May 1954	
28-13	Inactive for design after 26 June 1945	28-12
32-16	Inactive for design after 26 June 1945	32-6
32-18	Inactive for design after 26 June 1945	32-4
32-19	Inactive for design after 26 June 1945	32-1
32-20	Inactive for design after 26 June 1945	32-6

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*TABLE II. Insert arrangements cancelled by Revision A to MIL-STD-1651 - continued.

Insert Arrangement	Status in MIL-STD-1651	Replacement
36-2	Inactive for design after 28 December 1949	
36-11	Inactive for design after 15 June 1945	36-10
36-12	Inactive for design after 15 June 1945	36-10
36-16	Inactive for design after 15 June 1945	36-7
36-17	Inactive for design after 15 June 1945	36-7
36-18	Inactive for design after 15 June 1945	36-9
36-20	Inactive for design after 15 June 1945	
36-21	Inactive for design after 15 June 1945	36-9
40-12	Inactive for design after 30 May 1945	
40-13	Inactive for design after 30 May 1945	40-2
40-14	Inactive for design after 30 May 1945	
44-5	Inactive for design after 26 May 1945	44-1
44-6	Inactive for design after 26 May 1945	44-1
48-5	Inactive for design after 26 May 1945	

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*TABLE III. Insert arrangements removed from MIL-STD-1651 by Revision A.

Insert Arrangement	Status in MIL-STD-1651
10, 10S-5	Not to be used
10, 10S-6	Cancelled after 17 December 1958
12S, 13-6	Cancelled after 8 March 1945
14S, 15-8	Not to be used
18-2	Cancelled after 19 May 1945
18-21	Cancelled after 25 November 1949
20-1	Cancelled after 1 June 1945
20-28	Cancelled after 1 June 1945
22-26	Not to be used
22-31	Not to be used
22-37	Cancelled after 26 June 1945
24-8	Cancelled after 26 June 1945
24-13	Cancelled after 26 June 1945
24-19	Cancelled after 26 June 1945
32-11	Cancelled after 26 June 1945
40-8	Not to be used

6.5 International standardization agreements. Certain provisions of this standard are subject to international standardization agreements ASCC 12/44, NEPR 28 and NEPR 44 as listed below. When change notice, revision, or cancellation of this standard is proposed that will modify the international agreement concerned, the preparing activity will take appropriate

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action through international standardization channels, including Departmental Standardization offices, to change the agreement or make other appropriate accomodations.

INTL INTEREST

* **ASCC 12/44
#NEPR 28
/NEPR 44

6.6 Changes from previous issue. The margins of this standard are marked with asterisks or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and the relationship to the previous issue.

6.7 Subject term (keyword) listing.

Alternate insert positions
Insert position
Pin contact
Shell size
Socket contact

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR
Navy - EC
Air Force - 11, 15, 17, 99
DLA - ES

Preparing activity:
Navy - AS

(Project 5935-3560)

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APPENDIX A

INSERT ARRANGEMENTS FOR MIL-C-5015, MIL-C-22992 AND MIL-C-83723 SERIES II ELECTRICAL CONNECTORS

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APPENDIX A

INSERT ARRANGEMENTS FOR MIL-C-5015, MIL-C-22992 AND MIL-C-83723 SERIES II ELECTRICAL CONNECTORS

10. SCOPE

10.1 Scope. This appendix gives the data necessary for qualification of insert arrangements used in MIL-C-5015, MIL-C-22992 (Classes C, J, R) and MIL-C-83723 (Series II) Electrical Connectors. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. DEFINITIONS.

30.1 The terms used in this standard are generally accepted by the electrical and electronics industries and commonly used in electrical connector engineering practice.

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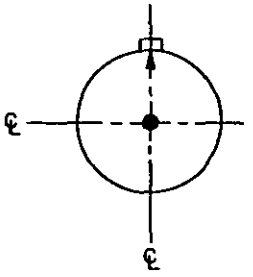
SECTION 100

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

SHELL SIZES 8, 8S

NOTES:

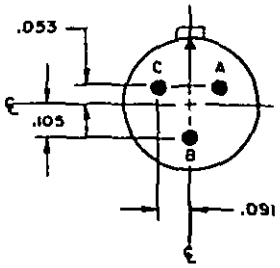
1. Shell size 8 is identical to shell size 8S and applies only to MIL-C-83723, series II.
2. Shell size 8S applies only to MIL-C-5015.
3. Section 10 is not applicable to MIL-C-22992.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
8, 8S	-1	A	1	16 16S	●	---	MS33680-1

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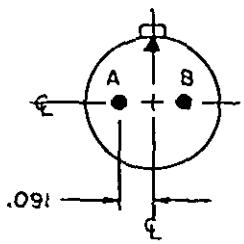
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

**#

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Connector class	Supersedes document
10SL, 11	-3	INST	3	16	●	All	K	MS33680-3
		A		16S			All others	

INACTIVE FOR NEW DESIGN

AFTER 17 DEC 1958



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
10SL, 11	-4	A	2	16	●	All	MS33680-4
				16S			

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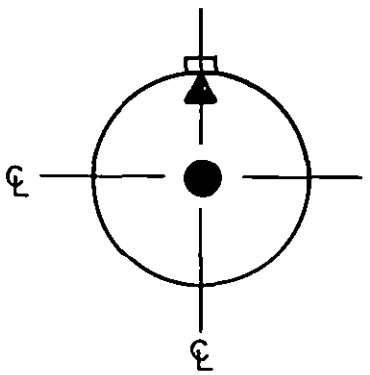
SECTION 200

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

SHELL SIZES 10, 10S, 10SL AND 11

NOTES:

1. Shell size 10 is identical to 10S and applies only to MIL-C-83723, series II.
2. Shell size 10S applies only to MIL-C-5015.
3. Shell size 10SL applies only to MIL-C-5015.
4. Shell size 11 is identical to 10SL and applies only to MIL-C-83723, series II.
5. Section 20 is not applicable to MIL-C-22992.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
10, 10S	-2	A	1	16 16S	●	--	MS33680-2

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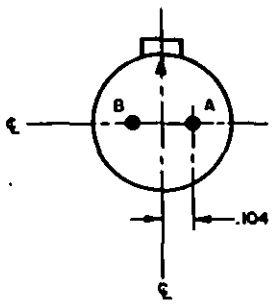
SECTION 300

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

SHELL SIZES 12, 12S AND 13

NOTES:

1. Shell size 12 applies to MIL-C-5015, MIL-C-22992 and MIL-C-83723, series II.
2. Shell size 12S applies only to MIL-C-5015.
3. Shell size 13 is identical to 12S and applies only to MIL-C-83723, series II.



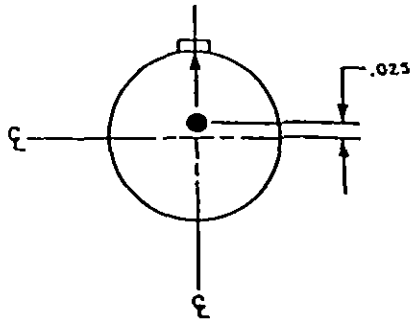
FOR OTHER INSERT POSITIONS SEE 12S-1 AND 12S-2 (APPENDIX A)

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

**#

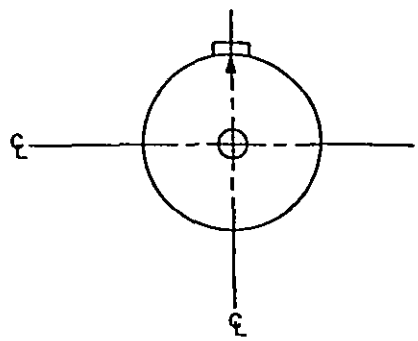
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
12S, 13	-3	A	2	16 16S	●	A11	MS33681-3

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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
12S, 13	-4	D	1	16 16S	●	--	MS33681-4



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
12	-5	D	1	12	⊕	--	MS33681-5

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SECTION 409

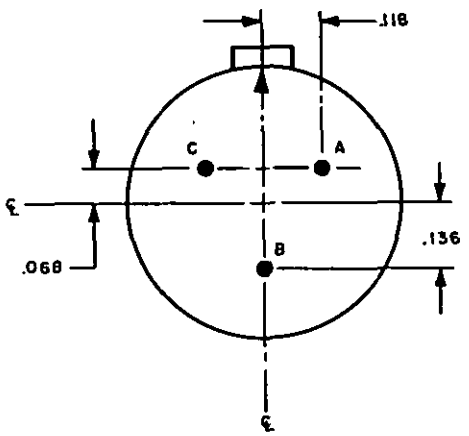
INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

SHELL SIZES 14, 14S AND 15

NOTES:

1. Shell size 14 applies to MIL-C-5015, MIL-C-22992 and MIL-C-83723, series II.
2. Shell size 14S applies only to MIL-C-5015.
3. Shell size 15 is identical to 14S and applies only to MIL-C-83723, series II.

INACTIVE FOR NEW DESIGN
AFTER 17 APR 1945

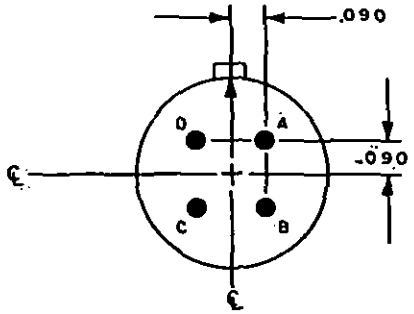


FOR OTHER INSERT POSITIONS SEE 14S-12 AND 14S-13.

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
14S, 15	-1	A	3	16 16S	●	All	MS33682-1

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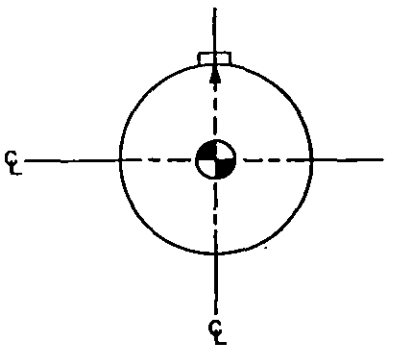
FOR OTHER INSERT POSITIONS SEE 14S-10 14S-11.

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	120	240	---

**#/

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
14S, 15	-2	INST	4	16 16S	●	All	MS33682-2

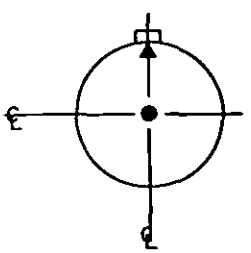
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
14	-3	A	1	8	◔	--	MS33682-3

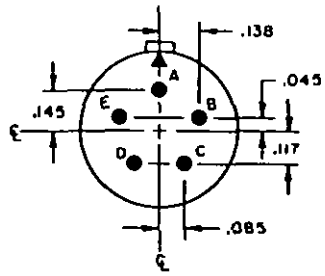
INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
14S, 15	-4	0	1	16 16S	●	--	MS33682-4

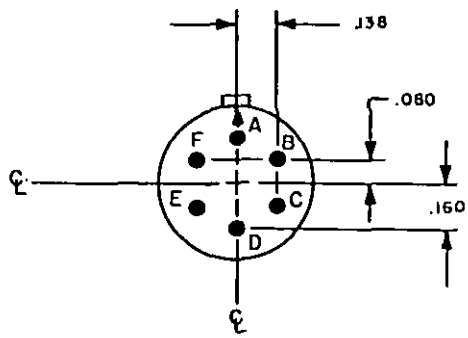
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	110	---	---

/

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
14S, 15	-5	INST	5	16 16S	●	All	MS33682-5

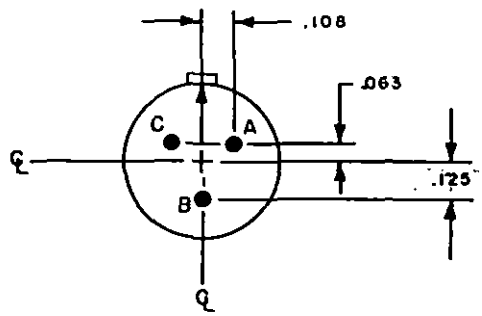


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

**#

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
14S, 15	-6	INST	6	16 16S	●	All	MS33682-6

Source: https://assist.dla.mil Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



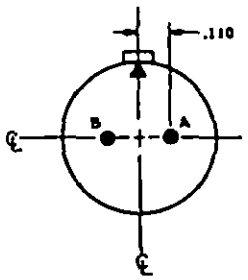
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
90	180	270	---

**#/

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
14S, 15	-7	A	3	16 16S	●	All	MS33682-7

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INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
14S, 15	-9	A	2	16 16S	●	All	MS33682-9

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

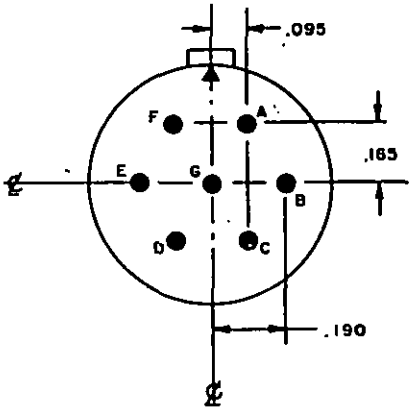
SECTION 500

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

SHELL SIZES 16, 16S AND 17

NOTES:

1. Shell size 16 applies to MIL-C-5015, MIL-C-22992 and MIL-C-83723, series II.
2. Shell size 16S applies only to MIL-C-5015.
3. Shell size 17 is identical to 16S and applies only to MIL-C-83723, series II.

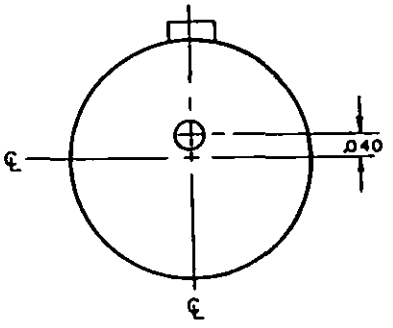


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	---	---	280

**#

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16S, 17	-1	A	7	16 16S	●	All	MS33683-1

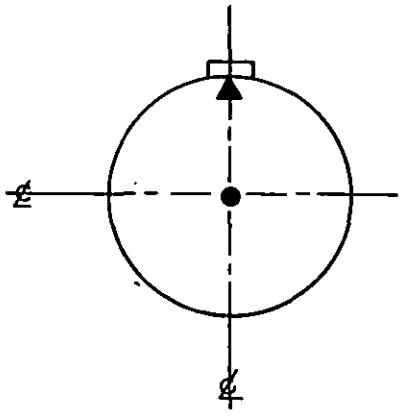
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16	-2	E	1	12	⊕	--	MS33683-2

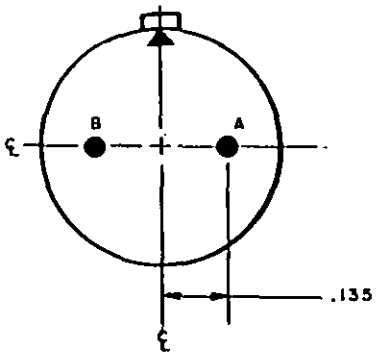
Source: https://assist.dla.mil - Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16S, 17	-3	B	1	16 16S	●	--	MS33683-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

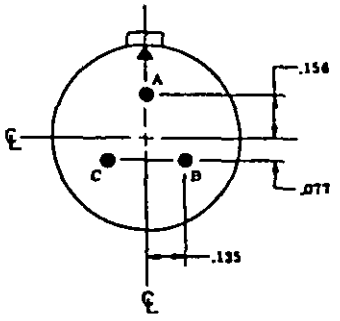


FOR OTHER INSERT POSITIONS SEE 16S-15 AND 16S-16. (SEE APPENDIX B)

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16S, 17	-4	D	2	16 16S	●	All	MS33683-4

INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954

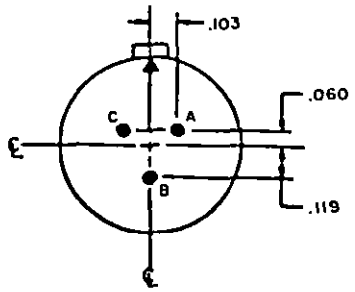


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16S, 17	-5	A	3	16 16S	●	All	MS33683-5

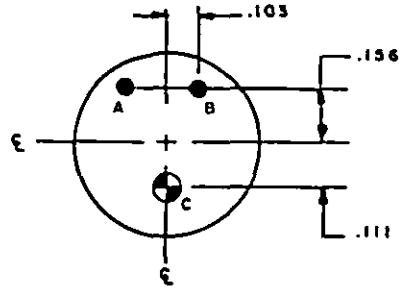
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR NEW DESIGN
AFTER 30 JUL 1951



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
90	180	270	---

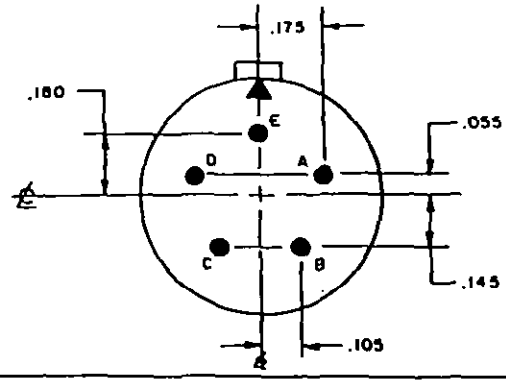
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16S, 17	-6	A	3	16 16S	●	All	MS33683-6



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

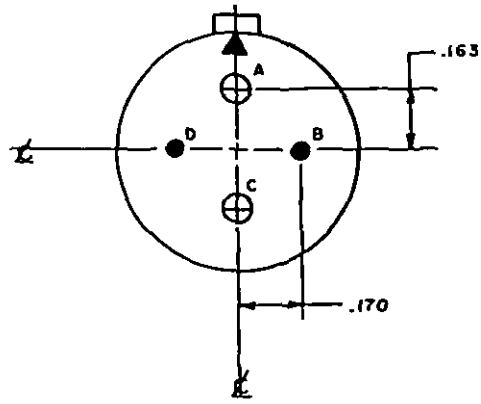
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16	-7	A	2	16	●	All	MS33683-7
			1	8	◕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	170	265	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16S, 17	-8	A	5	16 16S	●	All	MS33683-8

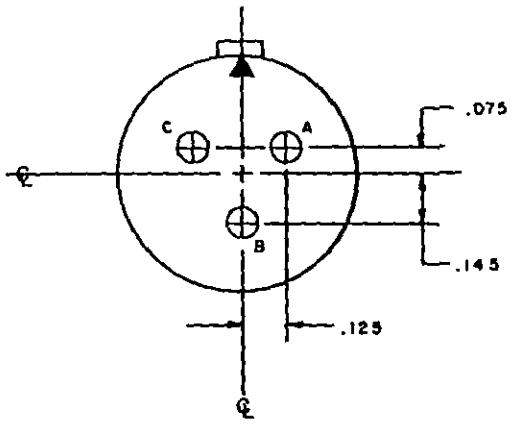


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

**#f

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16	-9	A	2	16	●	All	MS33683-9
			2	12	⊕		

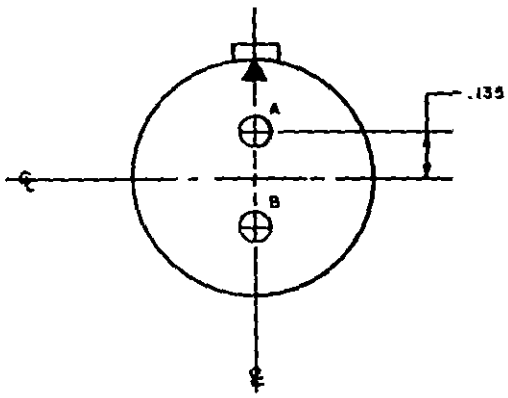
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
90	180	270	---

**#/

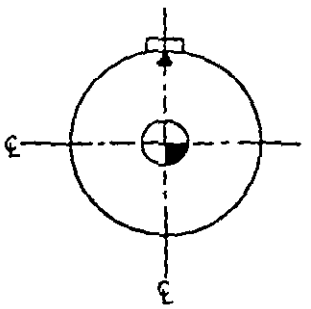
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16	-10	A	3	12	⊕	A11	MS33683-10



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

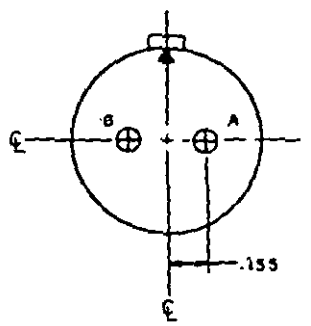
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16	-11	A	2	12	⊕	A11	MS33683-11

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16	-12	A	1	4	⊕	--	MS33683-12



CONTACT A - IRON
CONTACT B - CONSTANTAN

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16	-13	A	2	12	⊕	All	MS33683-13

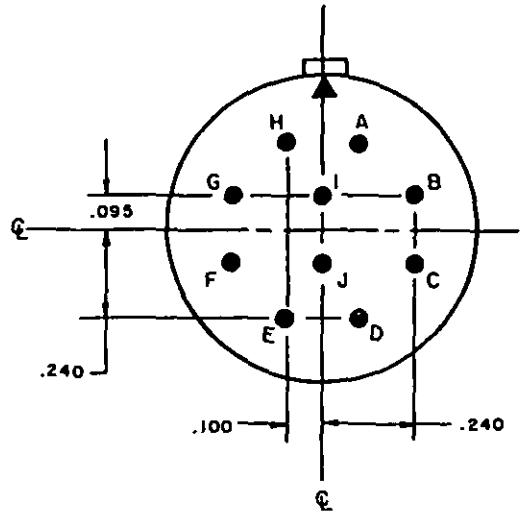
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SECTION 600

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

SHELL SIZE 18

INACTIVE FOR NEW DESIGN OF AIR FORCE
AIRBORNE EQUIPMENT AFTER MAY 1971.



FOR OTHER INSERT POSITIONS SEE 18-23 AND 18-24 (APPENDIX A)

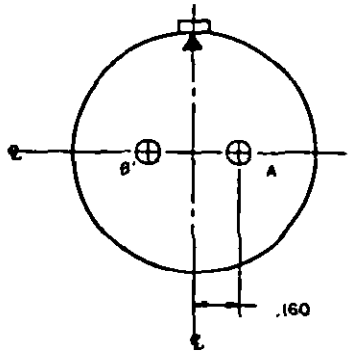
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

**#/

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-1	A	4	16	●	B,C,F,G	MS33684-1
		INST	6			All others	

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INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954



FOR OTHER INSERT POSITIONS SEE 18-25 AND 18-26 (APPENDIX A)

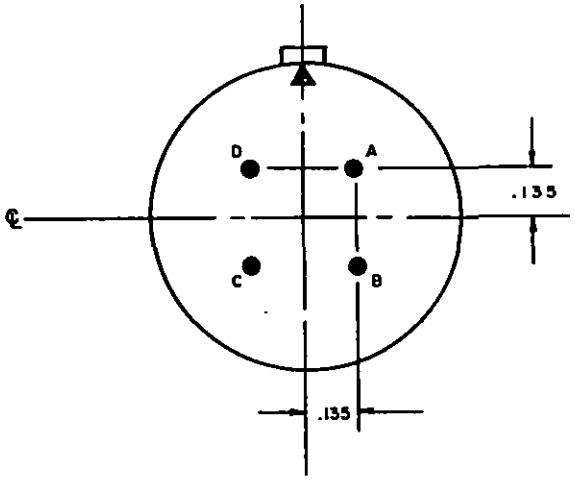
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-3	D	2	12	⊕	A11	MS33684-3

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Check the source to verify that this is the current version before use.

INACTIVE FOR NEW DESIGN OF AIR FORCE AIRBORNE EQUIPMENT AFTER 28 MAY 1971

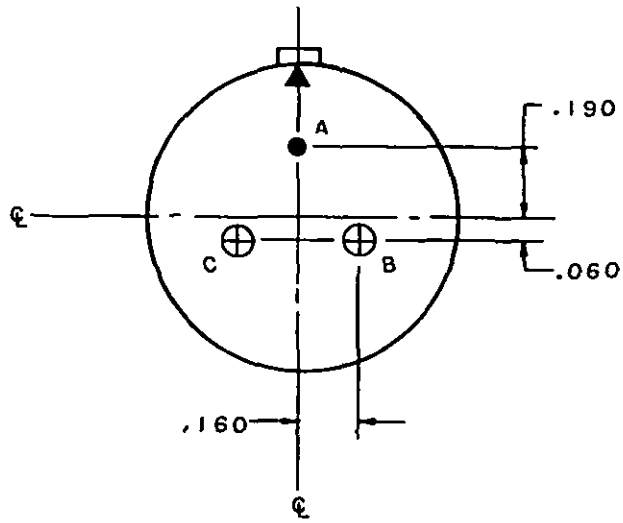


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-4	D	4	16	●	A11	MS33684-4

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Check the source to verify that this is the current version before use.

INACTIVE FOR NEW DESIGN OF AIR FORCE AIRBORNE EQUIPMENT AFTER 28 MAY 1971

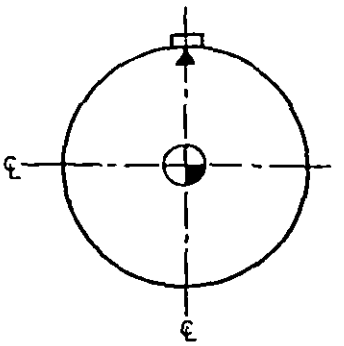


FOR OTHER INSERT POSITIONS SEE 18-27 AND 18-28. (APPENDIX B)

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

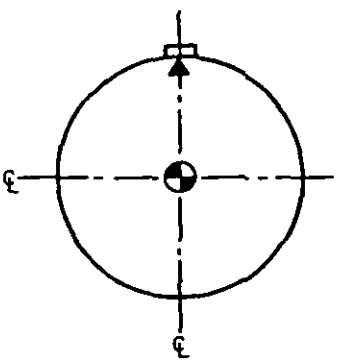
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-5	D	1	16	●	A11	MS33684-5
			2	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-6	D	1	4	⊕	--	MS33684-6

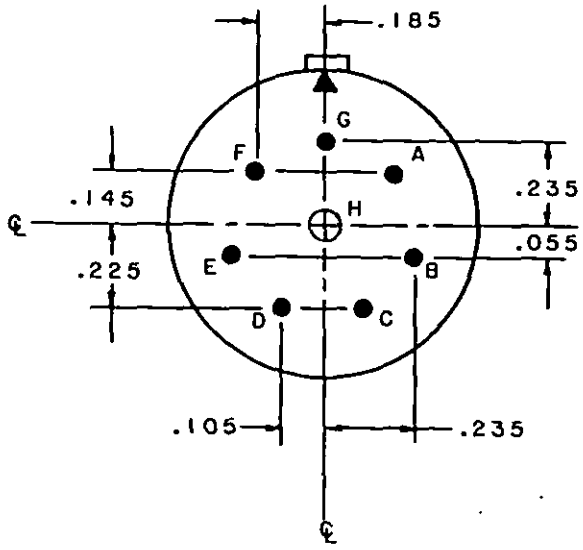


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-7	B	1	8	◒	--	MS33684-7

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:50Z
Check the source to verify that this is the currentversion before use.

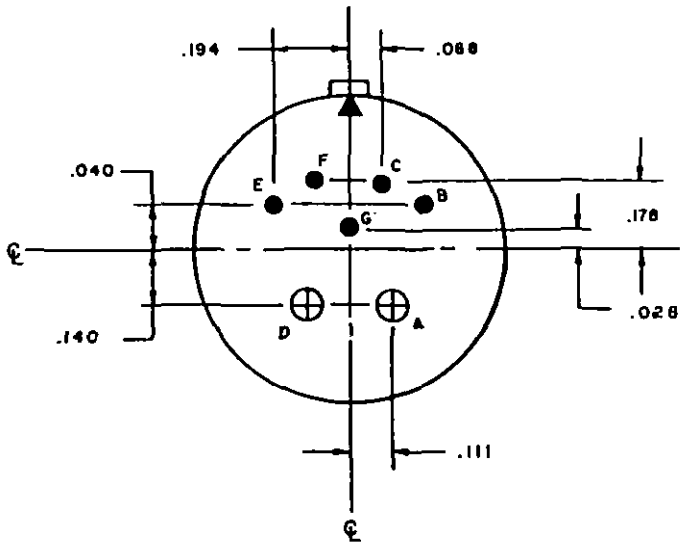
INACTIVE FOR NEW DESIGN OF AIR FORCE AIRBORNE EQUIPMENT AFTER 28 MAY 1971



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	---	---	290

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-8	A	7	16	●	All	MS33684-8
			1	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



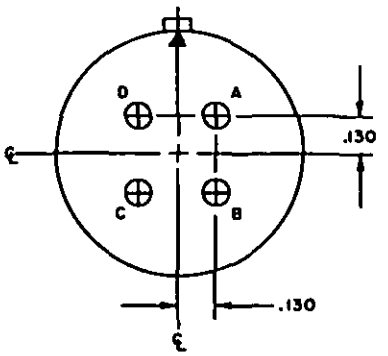
FOR OTHER INSERT POSITIONS SEE 18-17 AND 18-18.

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-9	INST	5	16	●	All	MS33684-9
			2	12	⊕		

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INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954



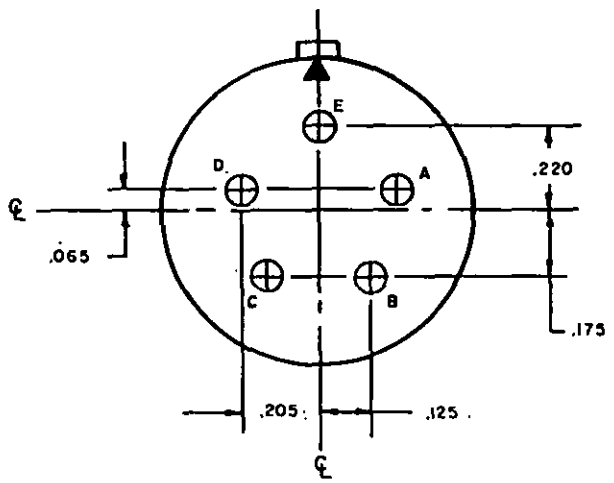
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	120	240	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-10	A	4	12	⊕	A11	MS33684-10

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:50Z
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INACTIVE FOR NEW DESIGN OF AIR FORCE AIRBORNE EQUIPMENT AFTER 28 MAY 1971.



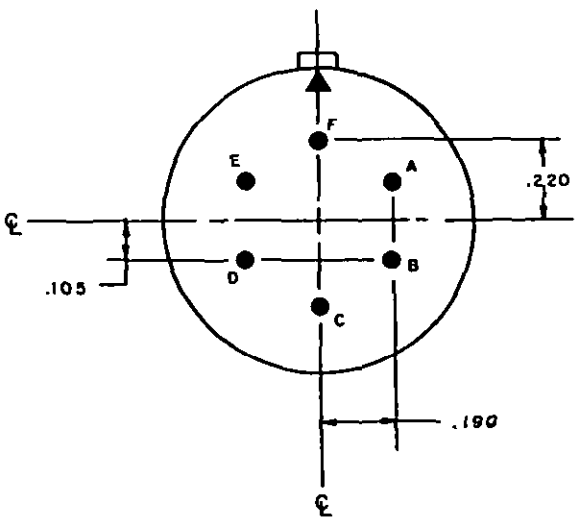
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	170	265	---

**#/

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-11	A	5	12	⊕	A11	MS33684-11

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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INACTIVE FOR NEW DESIGN OF AIR FORCE AIRBORNE EQUIPMENT AFTER 13 MAY 1954.



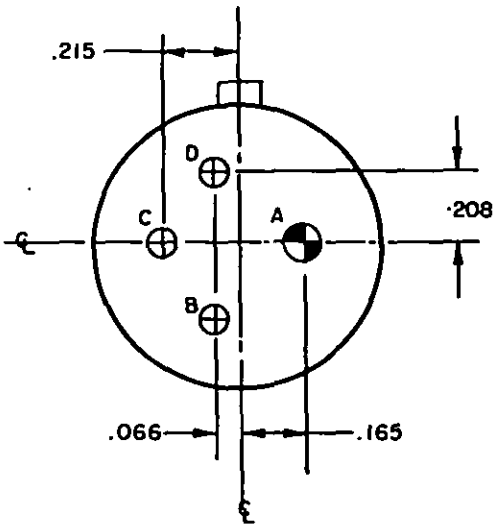
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	---	---	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-12	A	6	16	●	All	MS33684-12

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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INACTIVE FOR NEW DESIGN OF AIR FORCE
EQUIPMENT AFTER 28 MAY 1971.

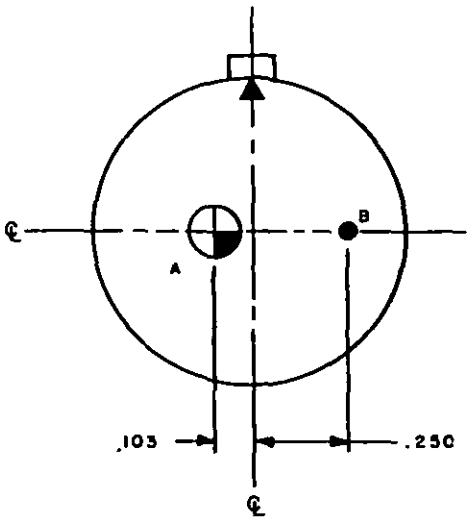


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-13	A	3	12	⊕	All	MS33684-13
			1	8	◒		

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INACTIVE FOR NEW DESIGN OF AIR FORCE AIRBORNE EQUIPMENT AFTER 28 MAY 1971.

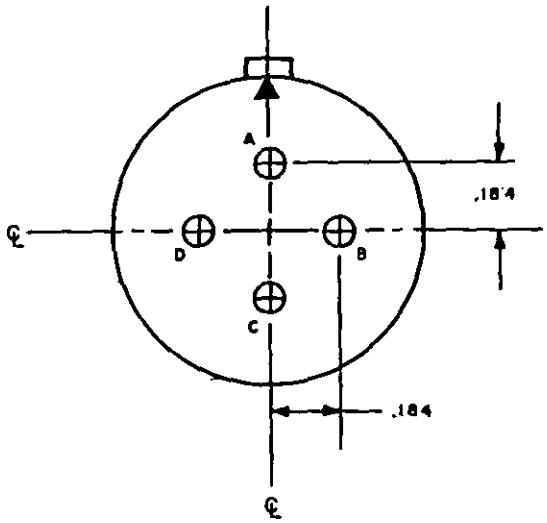


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-14	A	1	16	●	A11	MS33684-14
			1	4	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR NEW DESIGN OF AIR FORCE
AIRBORNE EQUIPMENT AFTER 28 MAY 1971.



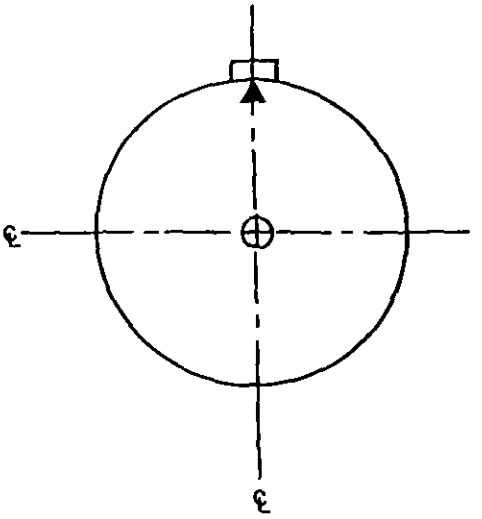
CONTACTS A AND C-IRON
CONTACTS B AND D-CONSTANTAN

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	120	240	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-15	A	4	12	⊕	All	MS33684-15

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Check the source to verify that this is the current version before use.

INACTIVE FOR NEW DESIGN OF AIR FORCE
AIRBORNE EQUIPMENT AFTER 28 MAY 1971.



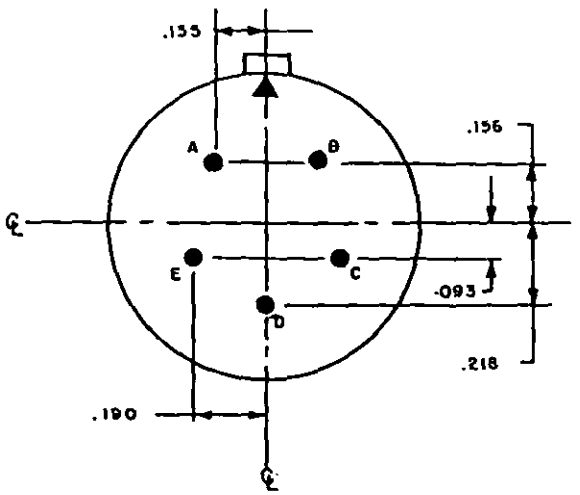
FOR INSERT DESIGN SEE MS33694.

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-16	C	1	12	⊕	---	MS33684-16

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INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954



FOR OTHER INSERT POSITIONS SEE 18-30 AND 18-31. (APPENDIX B)

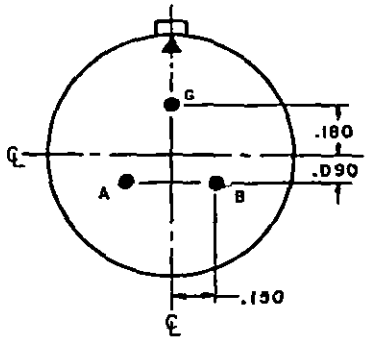
(Not applicable to MIL-C-22992)

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
90	180	270	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-20	A	5	16	●	A11	MS33684-20

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954



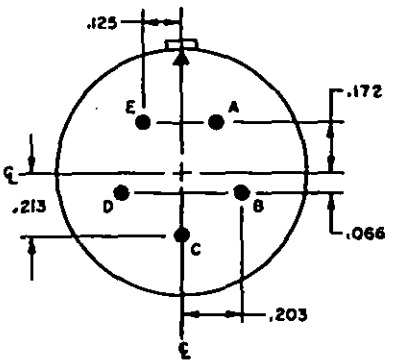
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-22	D	3	16	●	All	MS33684-22

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Check the source to verify that this is the current version before use.

INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
90	180	270	---

(Not applicable to MIL-C-22992)

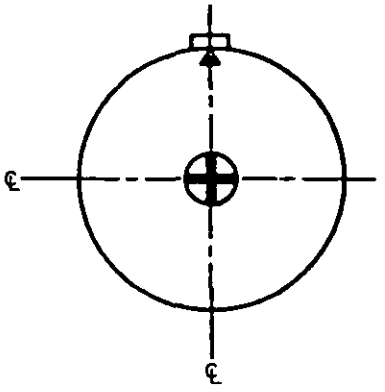
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-29	A	5	16	●	All	MS33684-29

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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SECTION 700

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

SHELL SIZE 20

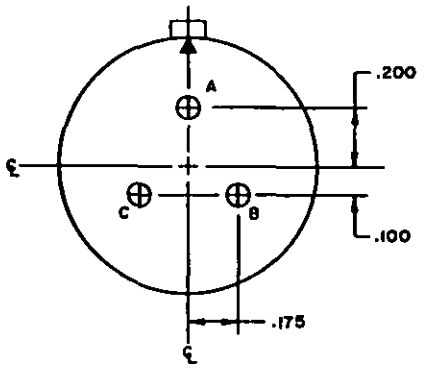


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-2	D	1	0	⊕	---	MS33685-2

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INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954

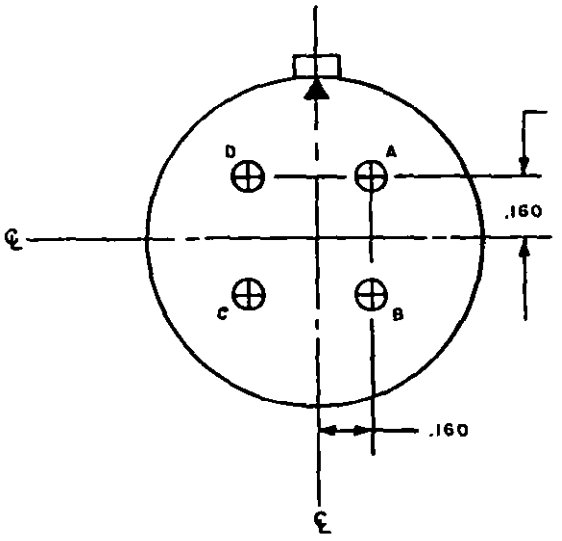


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-3	D	3	12	⊕	All	MS33685-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

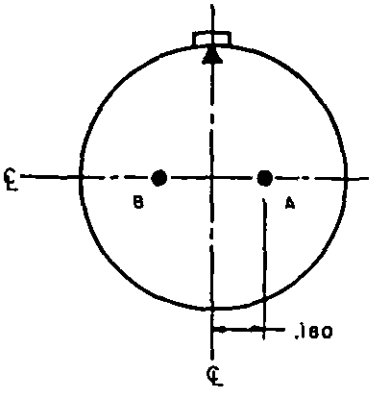


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
45	110	250	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-4	D	4	12	⊕	-A11	MS33685-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR NEW DESIGN
AFTER 17 DEC 1958

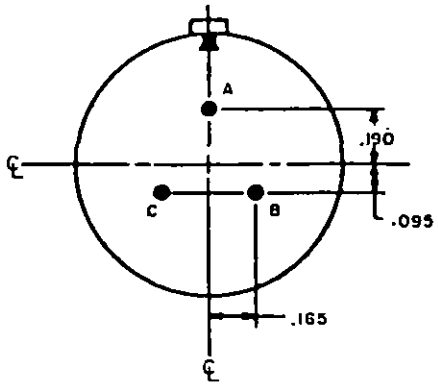


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-5	E	2	16	●	All	MS33685-5

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Check the source to verify that this is the current version before use.

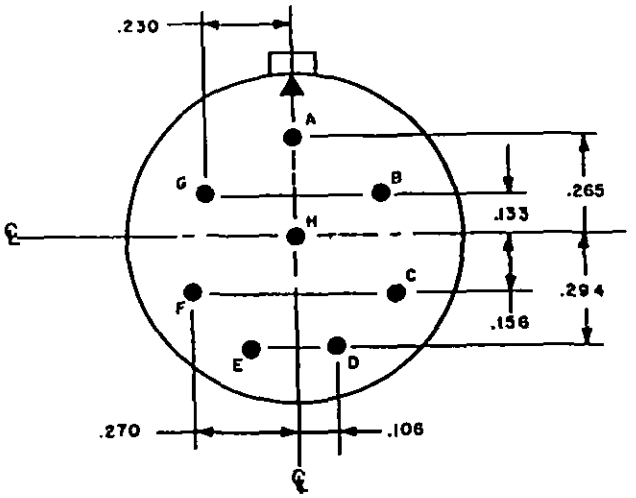


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-6	D	3	16	●	A11	MS33685-6

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

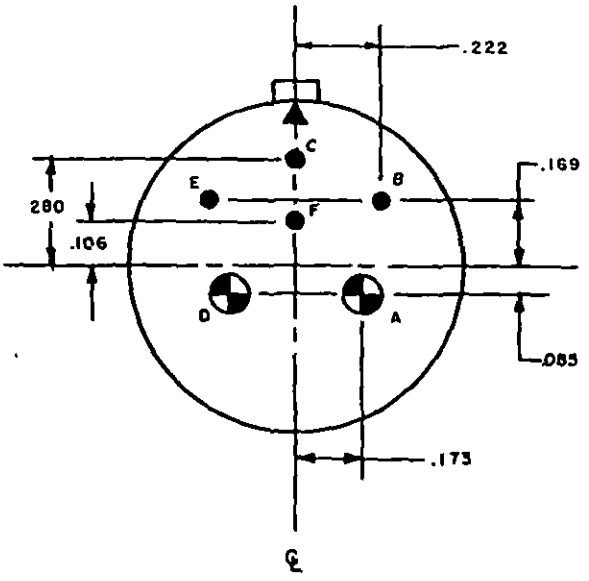


FOR OTHER INSERT POSITION SEE 20-22.

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-7	D	4	16	●	A,B,G,H	MS33685-7
		A	4			All others	

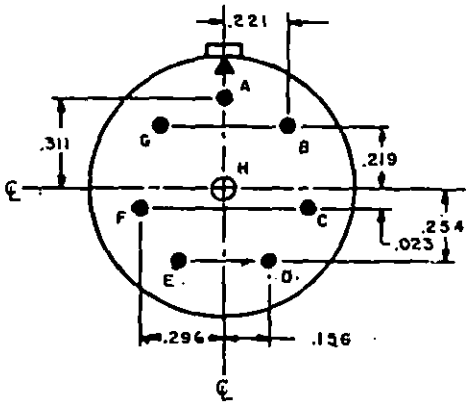
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-8	INST	4	16	●	All	MS33685-8
			2	8	⊗		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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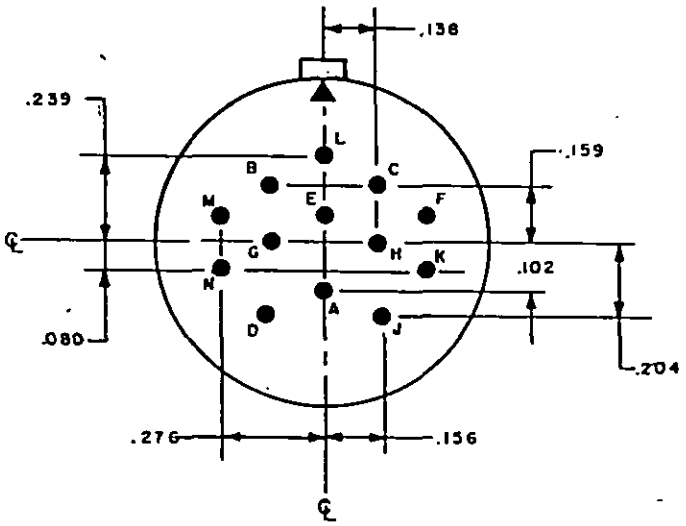


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-9	D	1	12	⊕	H	MS33685-9
		A	7	16	●	All others	

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INACTIVE FOR NEW DESIGN
AFTER 1 JUNE 1945



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

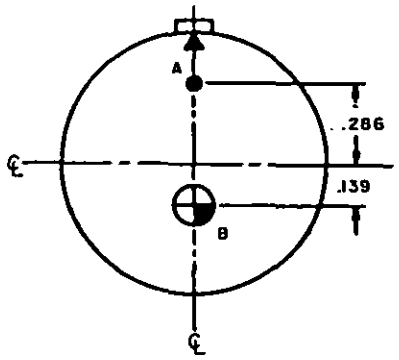
FOR OTHER INSERT POSITIONS SEE 20-25 AND 20-30.

(Not applicable to MIL-C-22922)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-11	INST	13	16	●	All	MS33685-11

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INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954



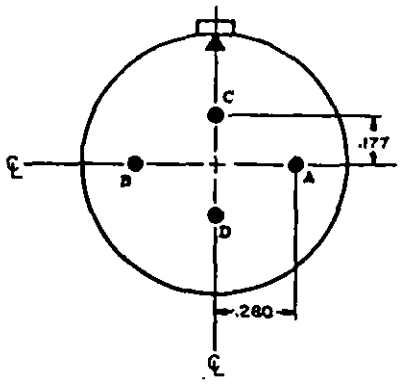
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-12	A	1	16	●	All	MS33685-12
			1	4	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR NEW DESIGN
AFTER 1 JUNE 1945



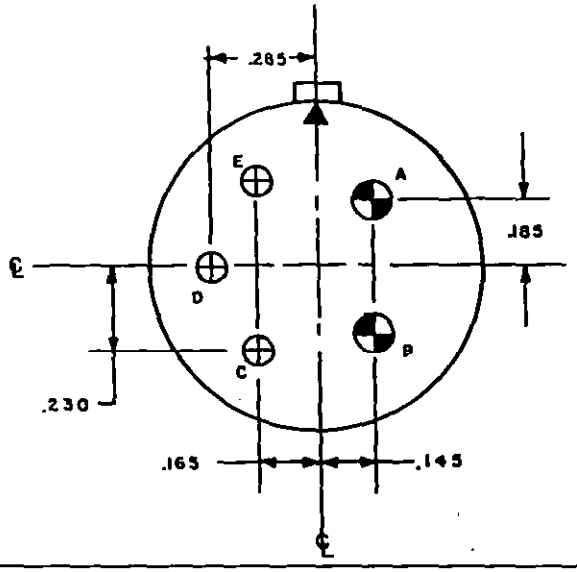
SOCKETS C AND D TO MAKE METALLIC CONTACT
TO EACH OTHER WHEN PINS ARE REMOVED.

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-13	A	4	16	●	All	MS33685-13

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

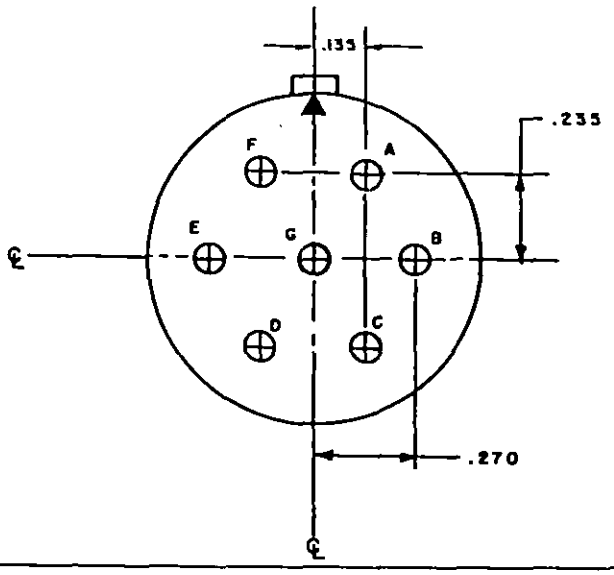


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

**#/

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-14	A	3	12	⊕	All	MS33685-14
			2	8	◔		

Source: https://assist.dla.mil - Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

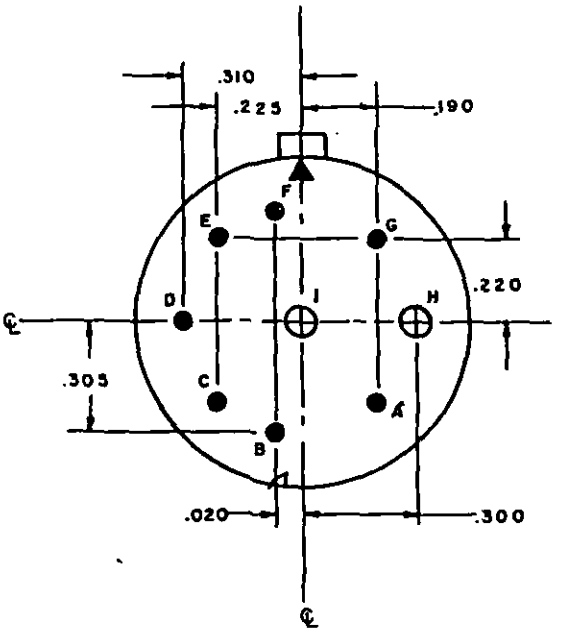


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	---	---	280

**#/

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-15	A	7	12	⊕	All	MS33685-15

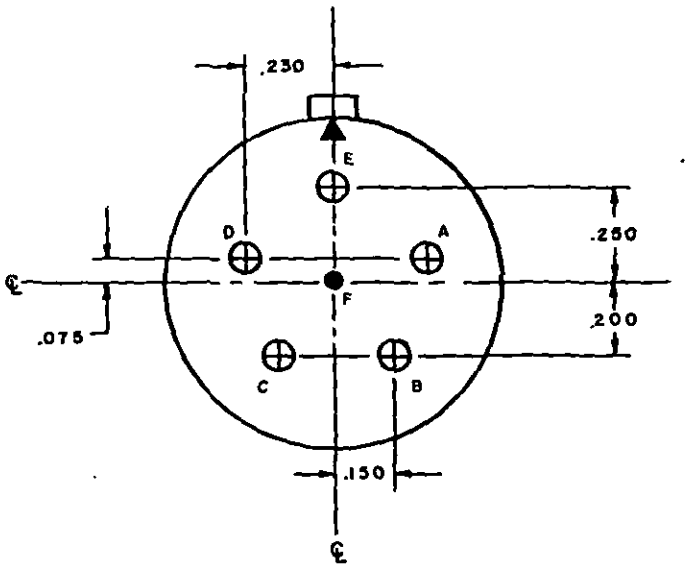
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-16	A	7	16	●	All	MS33685-16
			2	12	⊕		

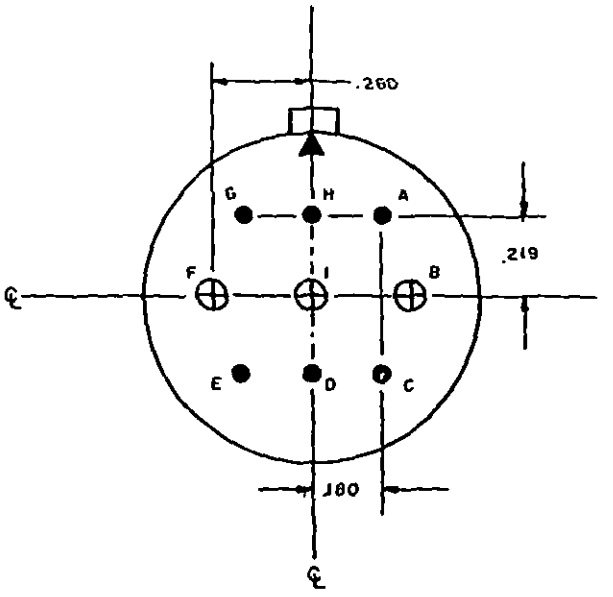
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
90	180	270	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-17	A	1	16	●	All	MS33685-17
			5	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



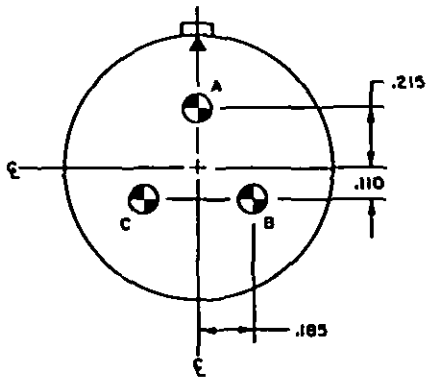
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

**#/

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-18	A	6	16	●	All	MS33685-18
			3	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:50Z
Check the source to verify that this is the currentversion before use.

INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954



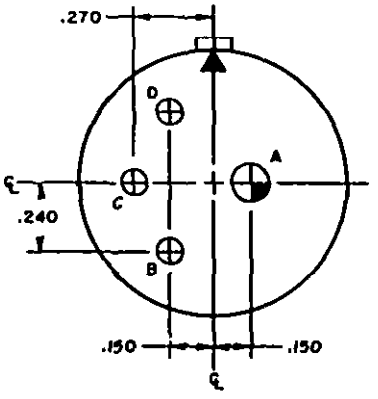
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
90	180	270	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-19	A	3	8		All	MS33685-19

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:50Z
Check the source to verify that this is the currentversion before use.

INACTIVE FOR NEW DESIGN
AFTER 17 DEC 1958

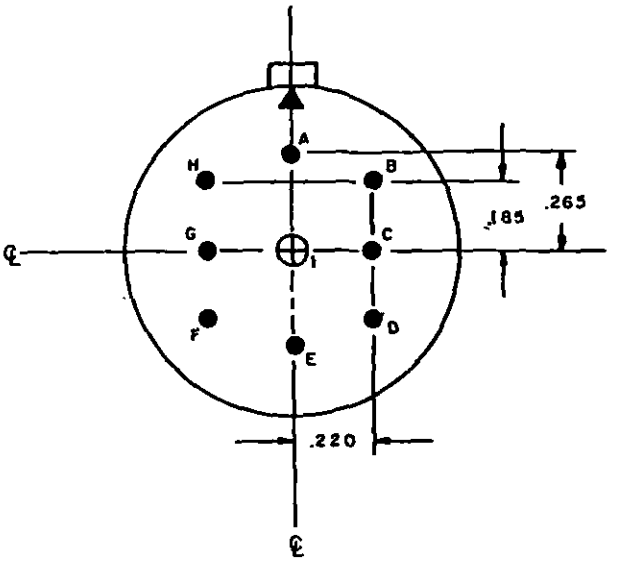


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-20	A	3	12	⊕	All	MS33685-20
			1	4	◒		

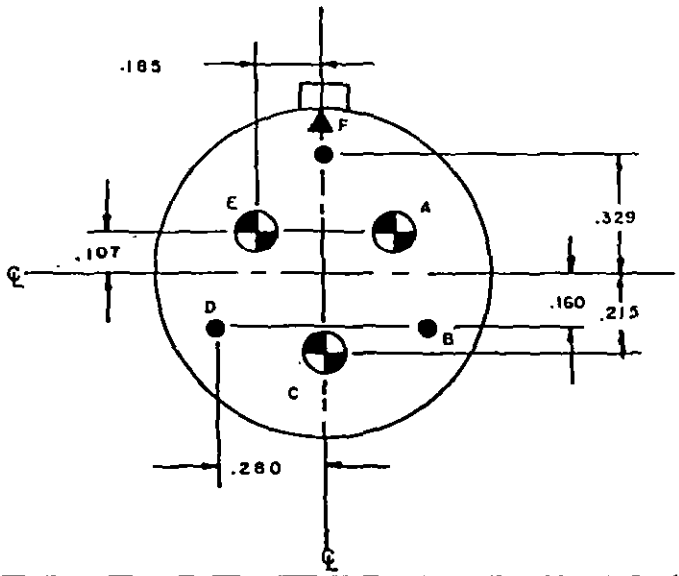
Source: https://assist.dla.mil -- Downloaded: 202610-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-21	A	8	16	●	All	MS33685-21
			1	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



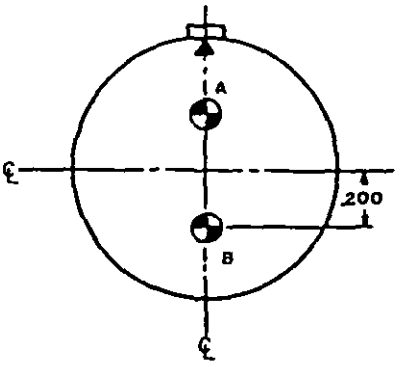
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

**#/

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-22	A	3	16	●	All	MS33685-22
			3	8	◕		

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INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954



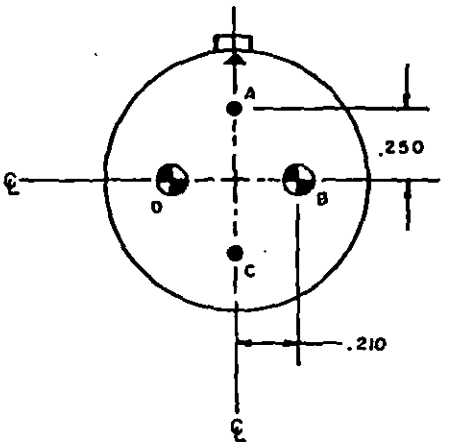
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-23	A	2	8		All	MS33685-23

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:50Z
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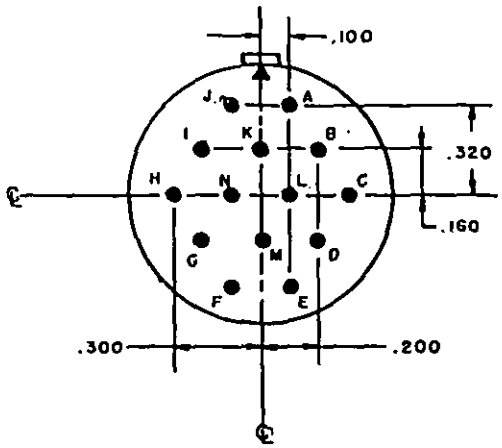
INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-24	A	2	16	●	All	MS33685-24
			2	8	◒		

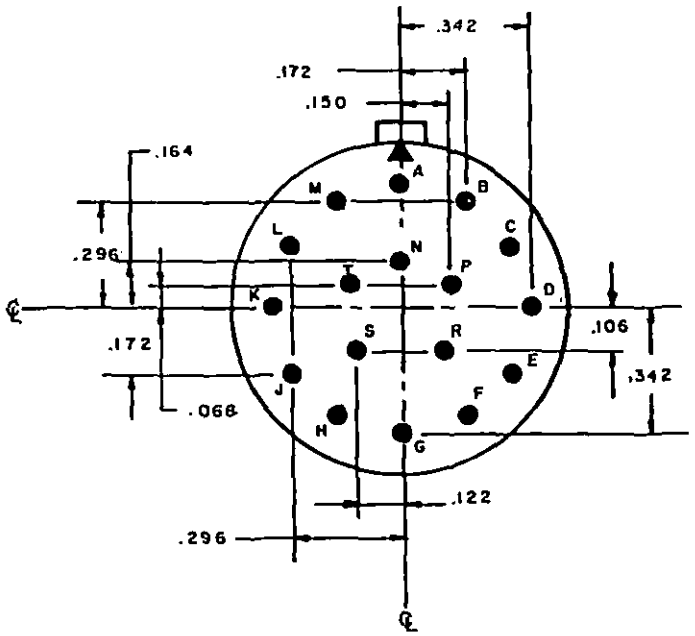
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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-27	A	14	16	●	A11	MS33685-27

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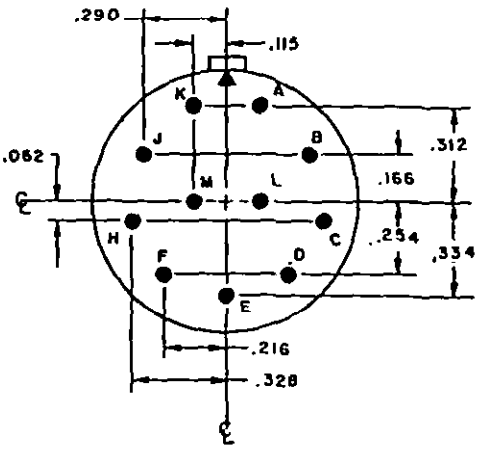
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	---	---	280

**#/

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-29	A	17	16	●	All	MS33685-29

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INACTIVE FOR NEW DESIGN
AFTER 18 MAY 1951

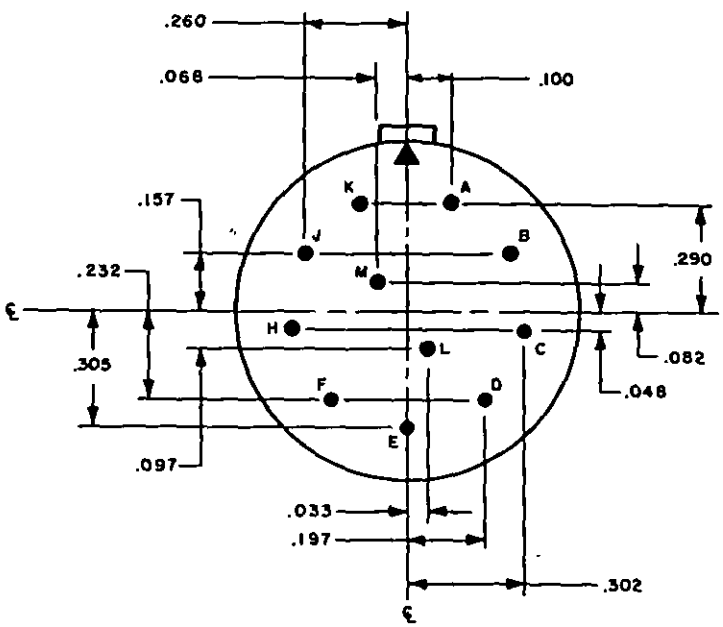


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-31	A	11	16	●	All	MS3368b-31

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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-33	A	11	16	●	A11	MS33685-33

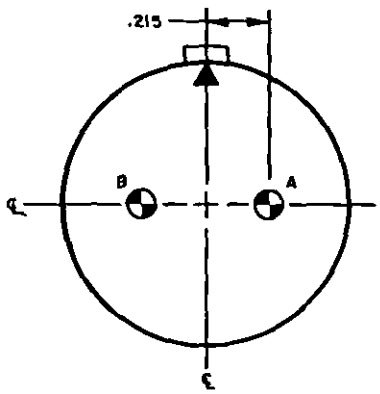
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SECTION 800

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

SHELL SIZE 22

**INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954**

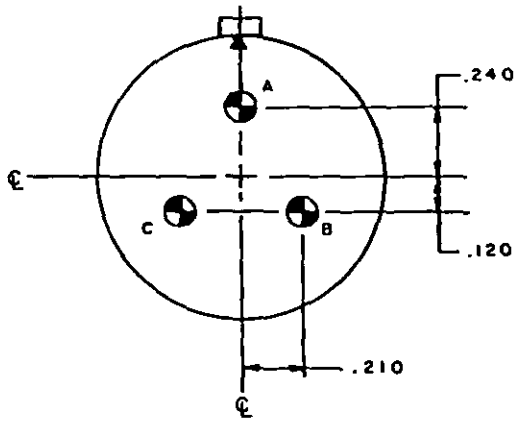


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-1	D	2	8		A11	MS33686-1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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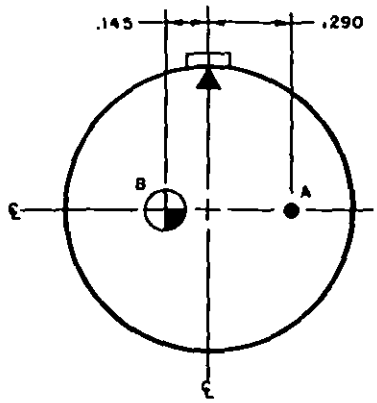


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-2	D	3	8		All	MS33686-2

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INACTIVE FOR NEW DESIGN
AFTER 17 DEC 1958



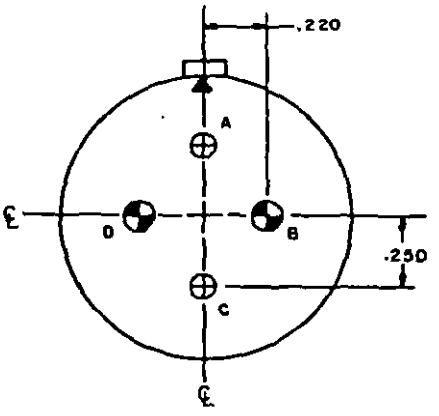
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-3	D	1	16	●	All	MS33686-3
			1	4	⊕		

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INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954

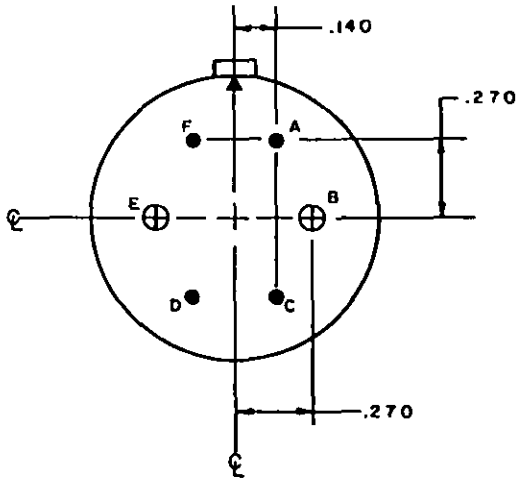


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-4	A	2	12	⊕	All	MS33686-4
			2	8	◒		

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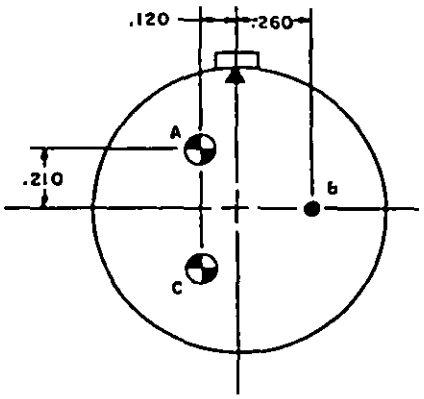


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-5	D	4	16	●	All	MS33686-5
			2	12	⊕		

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INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954

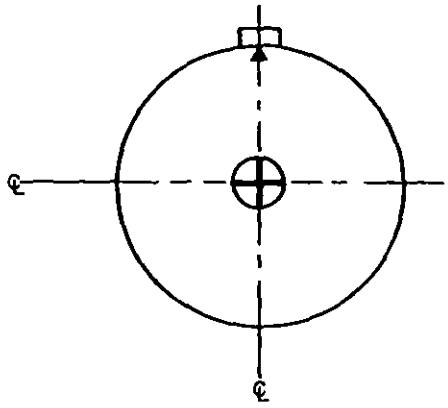


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-6	D	1	16	●	A11	MS33686-6
			2	8	⊕		

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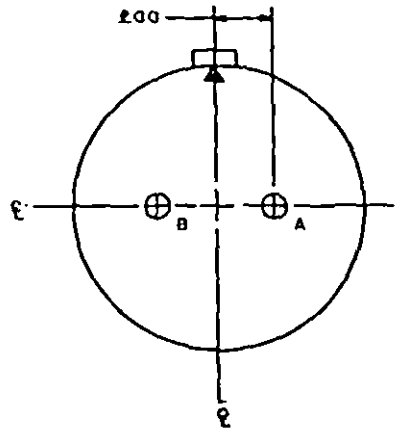


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-7	E	1	0	⊕	---	MS33686-7

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INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954

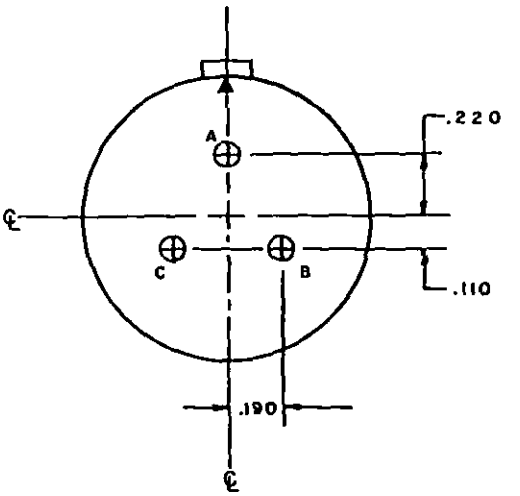


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-8	E	2	12	⊕	All	MS33686-8

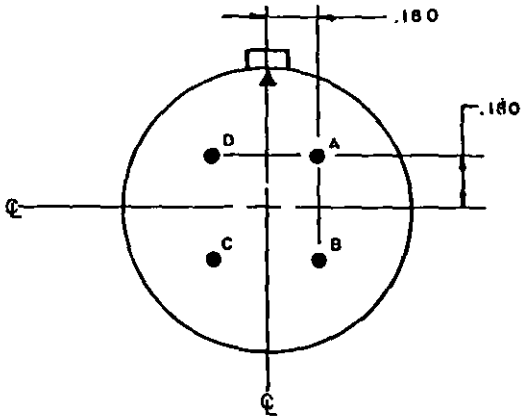
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

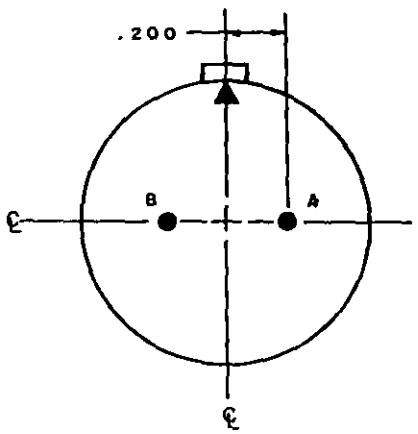
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-9	E	3	12	⊕	A11	MS33686-9

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Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

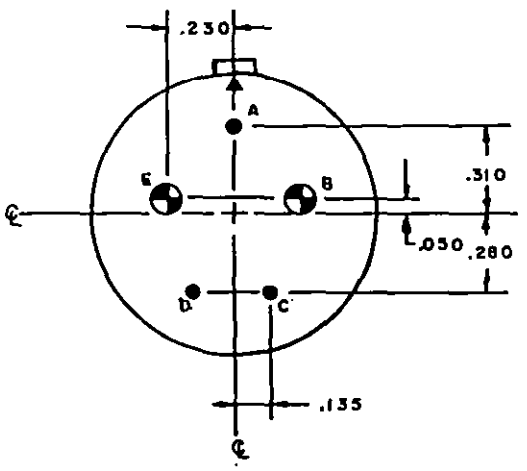
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-10	E	4	16	●	All	MS33686-10



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-11	B	2	16	●	All	MS33686-11

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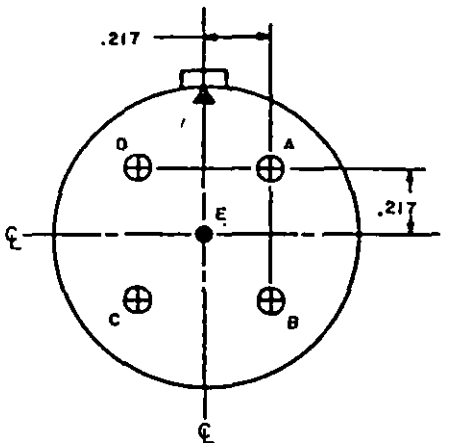


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-12	D	3	16	●	All	MS33686-12
			2	8	⊕		

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INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954



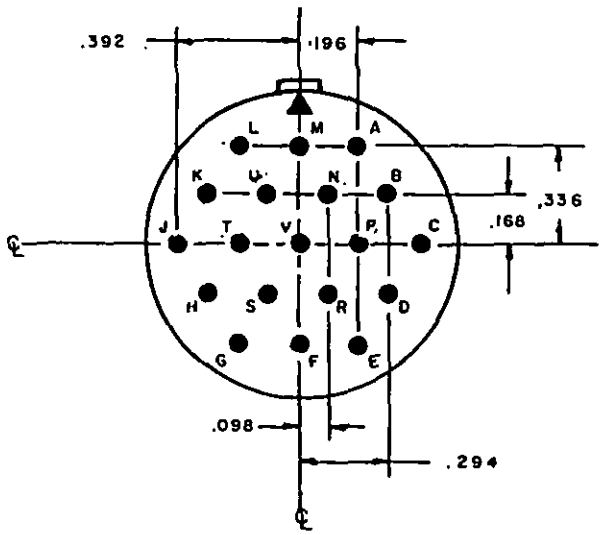
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-13	D	1	16	●	E	MS33686-13
		A	4	12	⊕	All others	

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ALTERNATE POSITIONS X AND Y
CANCELLED AFTER 26 JUN 1968

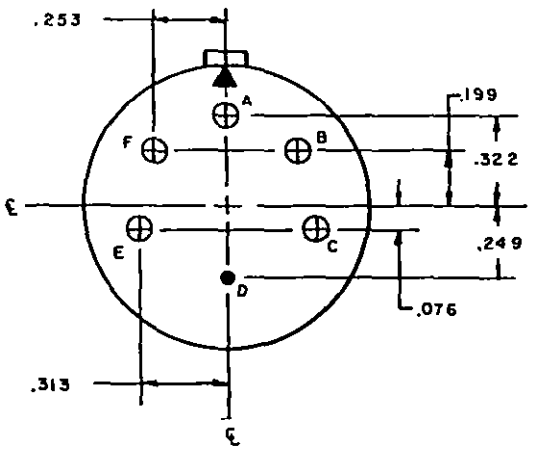


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

**#/

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-14	A	19	16	●	All	MS33686-14

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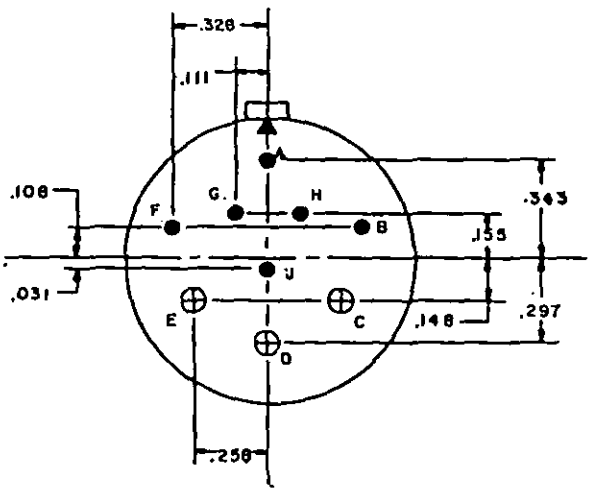


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-15	E	1	16	●	D	MS33686-15
		A	5	12	⊕	All others	

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INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954

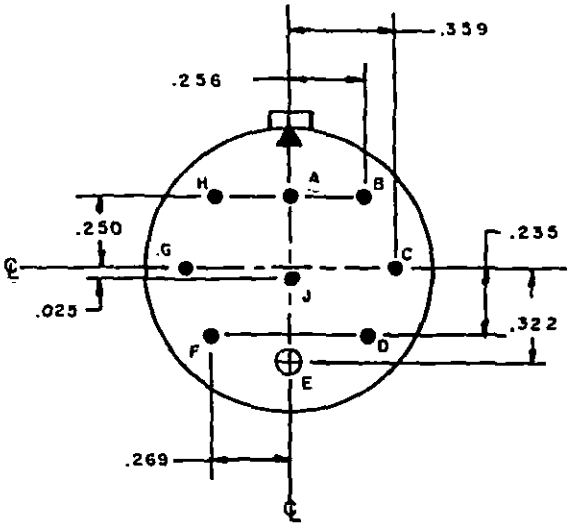


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-16	A	6	16	●	All	MS33686-16
			3	12	⊕		

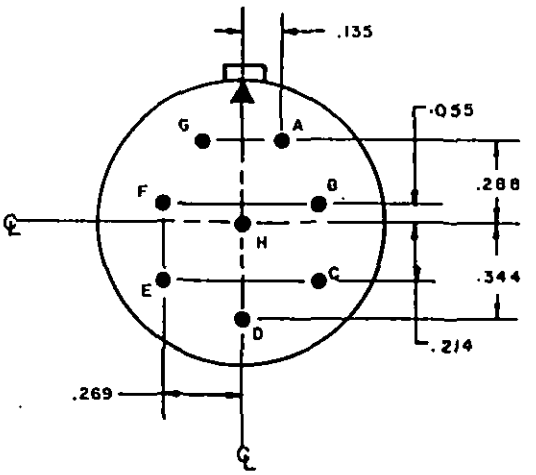
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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-17	D	1	16	●	A	MS33686-17
			7			All others	
		A	1	12	⊕		

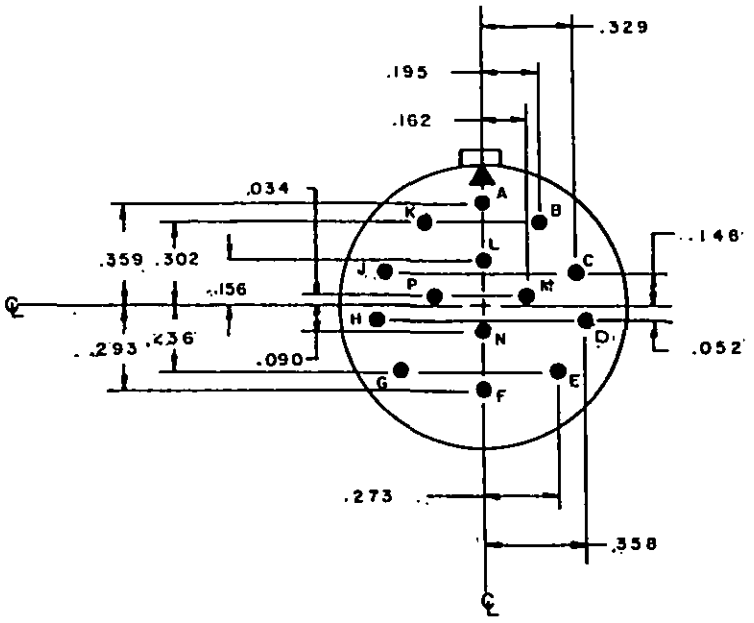
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Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-18	A	3	16	●	C,D,E	MS33686-18
		D	5			All others	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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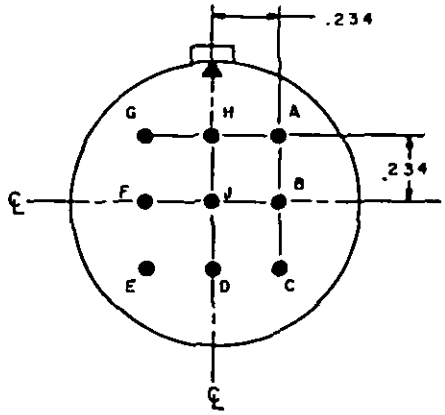


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-19	A	14	16	●	A11	MS33686-19

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INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954

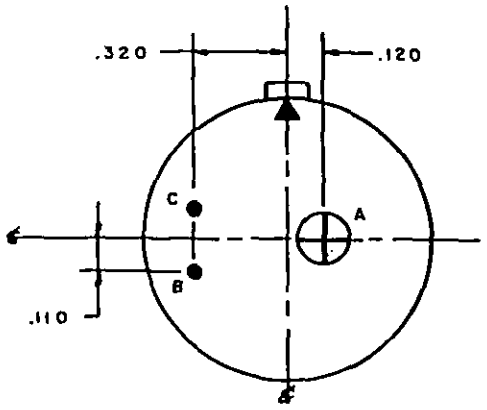


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-20	A	9	16	●	All	MS33686-20

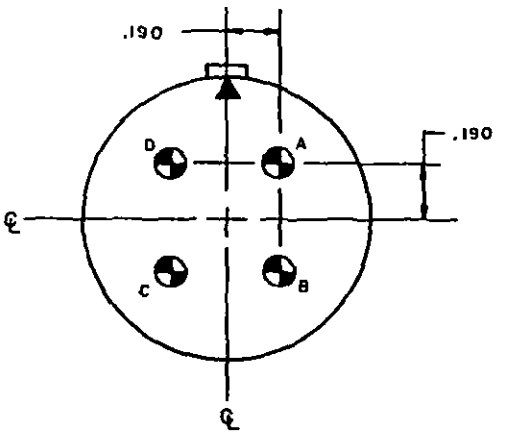
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-21	A	2	16	●	A11	MS33686-21
			1	0	⊕		

Source: https://assist.dla.mil - Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

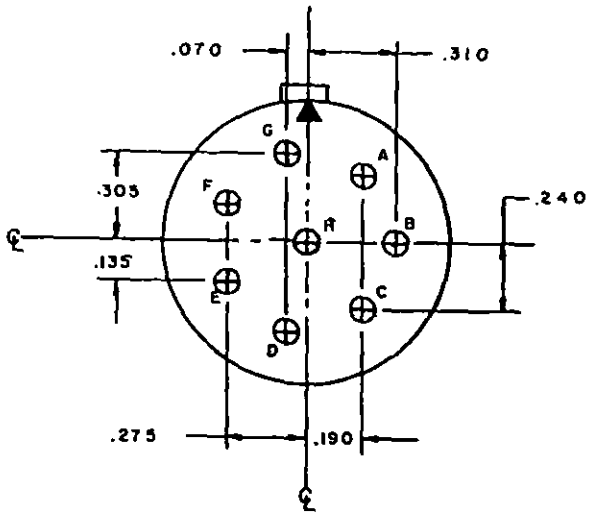


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	110	250	---

**#/

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-22	A	4	8	◔	A11	MS33686-22

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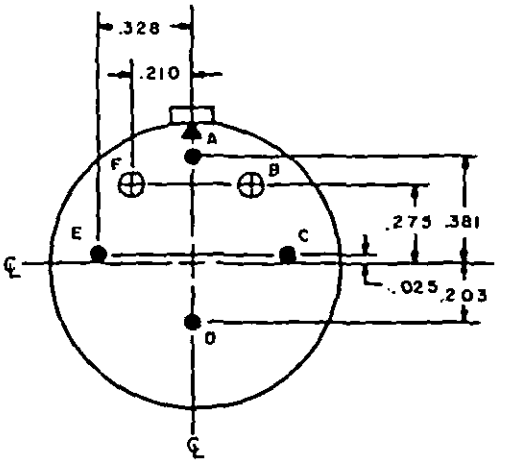
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	---	250	---

**#/

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-23	D	1	12	⊕	H	MS33686-23
		A	7			All others	

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INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954



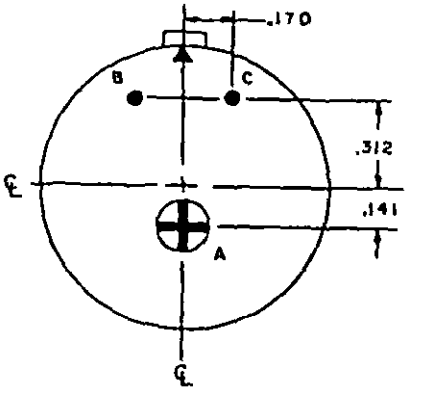
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-24	D	3	16	●	C,D,E	MS33686-24
		A	1			All others	
			2	12	⊕		

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INACTIVE FOR NEW DESIGN
AFTER 31 JUL 1951

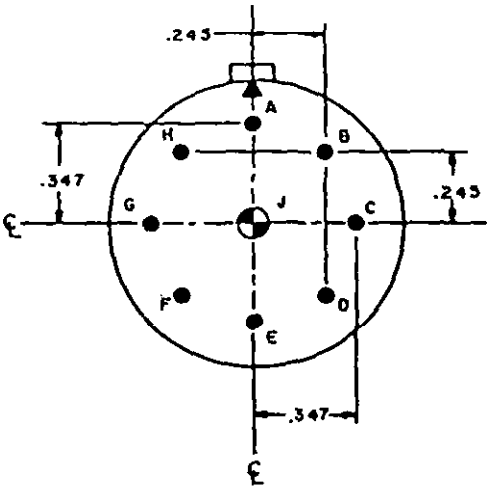


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-25	A	2	16	●	All	MS33686-25
			1	0	⊕		

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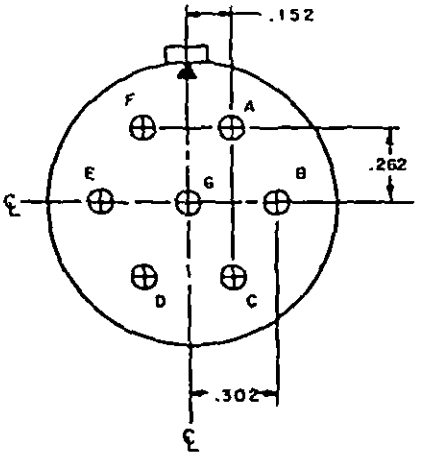


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80		250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-27	D	1	8	◔	J	MS33686-27
		A	8	16	●	All others	

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INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954



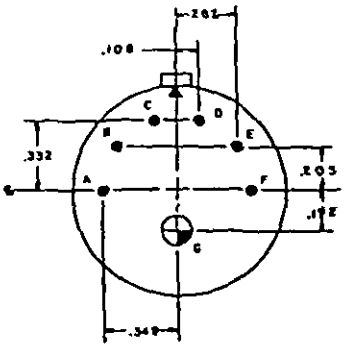
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80			280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-28	A	7	12	⊕	All	MS33686-28

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Check the source to verify that this is the currentversion before use.

INACTIVE FOR NEW DESIGN
AFTER 17 DEC 1958



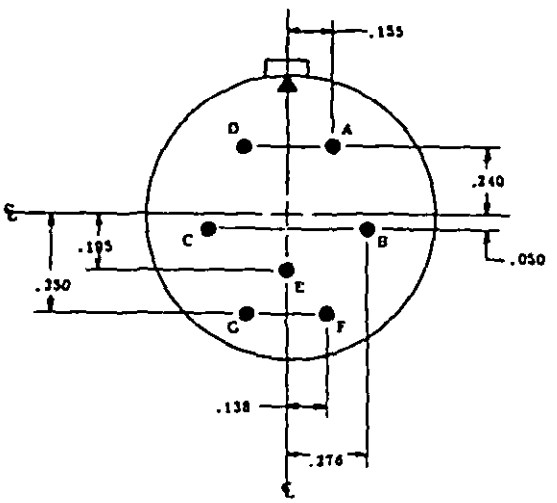
(Not applicable to MIL-C-22992)

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-29	A	6	16	●	All	MS33686-29
			1	4	⊕		

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INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954



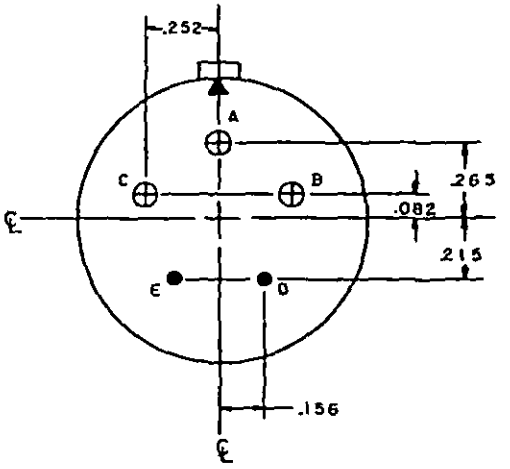
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-33	A	3	16	●	E,F,G	MS33686-33
		D	4			All others	

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Check the source to verify that this is the current version before use.

INACTIVE FOR NEW DESIGN
AFTER 17 DEC 1958

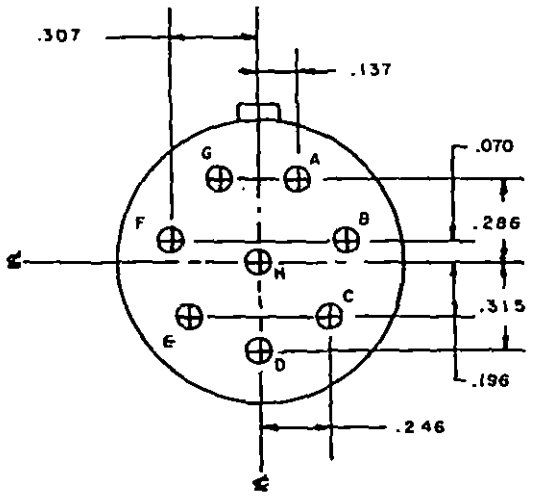


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-34	D	2	16	●	All	MS33686-34
			3	12	⊕		

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CONTACTS A, C, E AND G - IRON
CONTACTS B, D, F AND H - CONSTANTAN

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
90	---	270	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-36	D	1	12	⊕	H	MS33686-36
		A	7			All others	

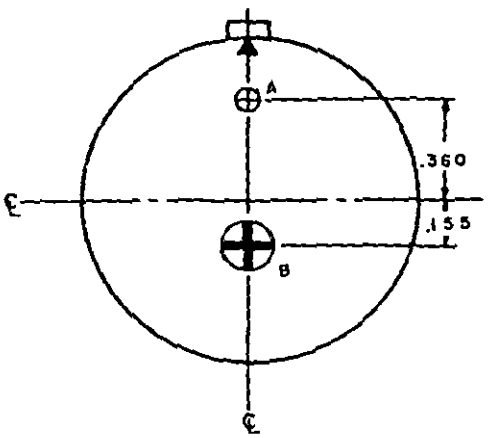
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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SECTION 900

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

SHELL SIZE 24

INACTIVE FOR NEW DESIGN
AFTER 17 DEC 1958

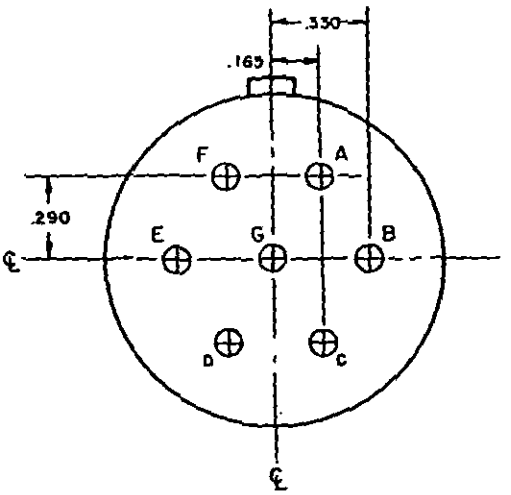


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-1	D	1	12	⊕	All	MS33687-1
			1	0	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

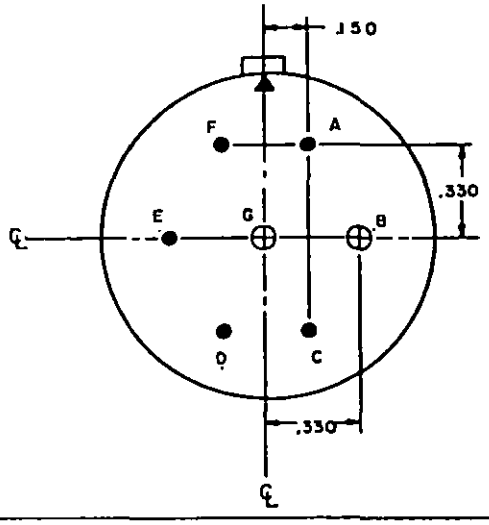


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	---	---	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-2	D	7	12	⊕	A11	MS33687-2

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INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954

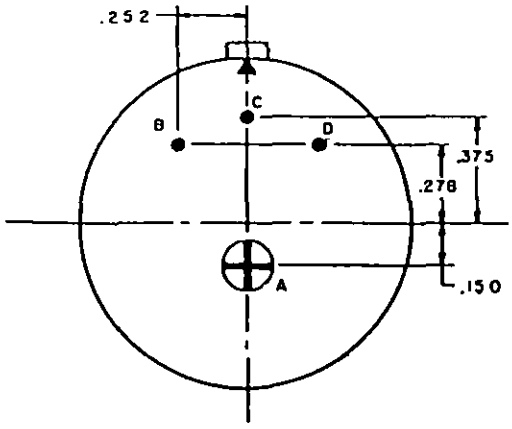


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-3	D	5	16	●	A11	MS33687-3
			2	12	⊕		

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Check the source to verify that this is the current version before use.

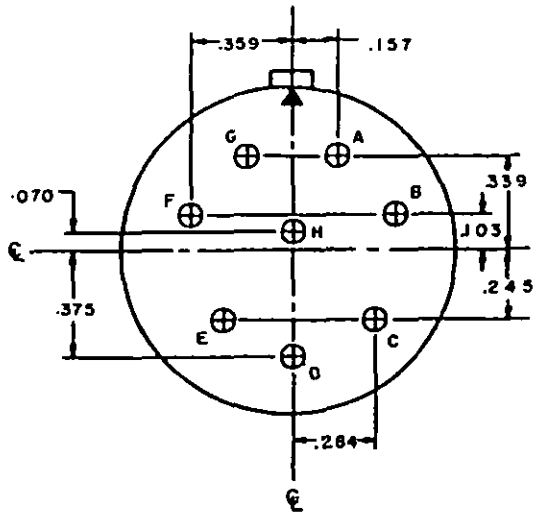


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-4	D	3	16	●	All	MS33687-4
			1	0	⊕		

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ALTERNATE INSERT POSITIONS

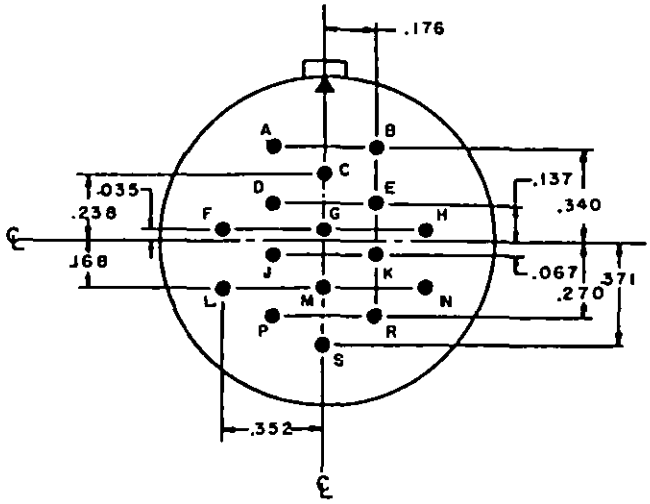
A DEGREES			
W	X	Y	Z
80	110	250	280

FOR OTHER INSERT POSITIONS SEE 24-25 AND 24-26.

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-6	D	3	12	⊕	A,G,H	MS33687-6
		A	5			All others	

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INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954



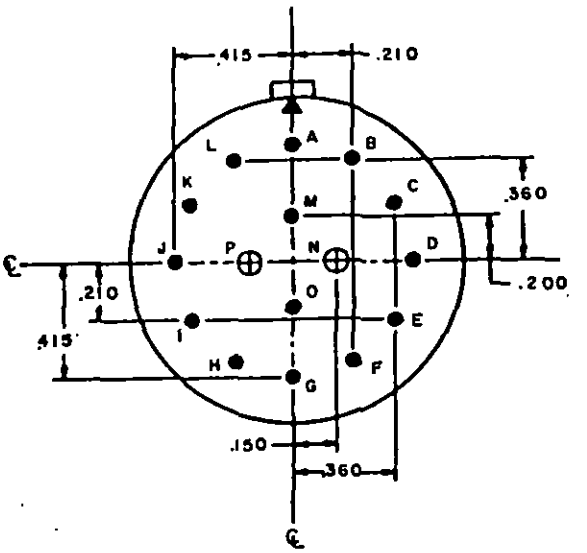
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

FOR OTHER INSERT POSITIONS SEE 24-15 and 24-24.

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-5	A	16	16	●	All	MS33687-5

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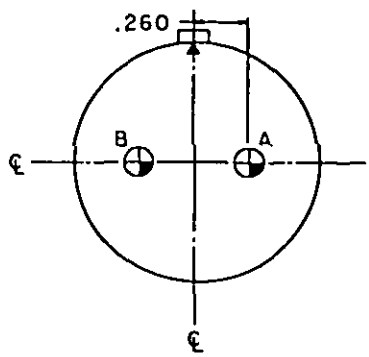


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-7	A	14	16	●	All	MS33687-7
			2	12	⊕		

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INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954

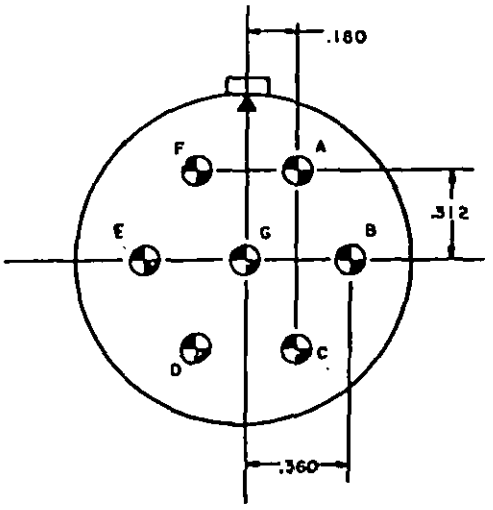


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-9	A	2	4	⊕	All	MS33687-9

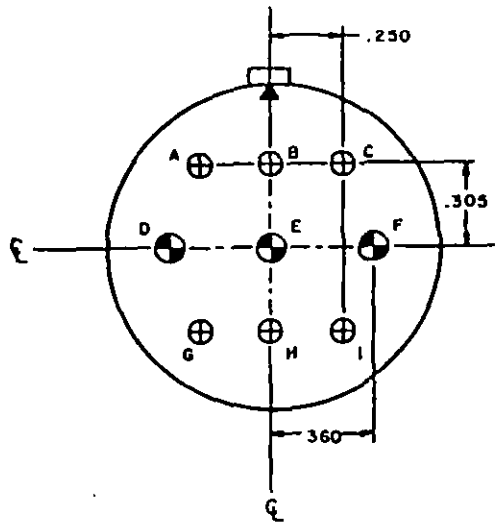
Source: https://assist.dla.mil - Downloaded: 2026-10-26T07:50Z
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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	---	---	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-10	A	7	8		All	MS33687-10

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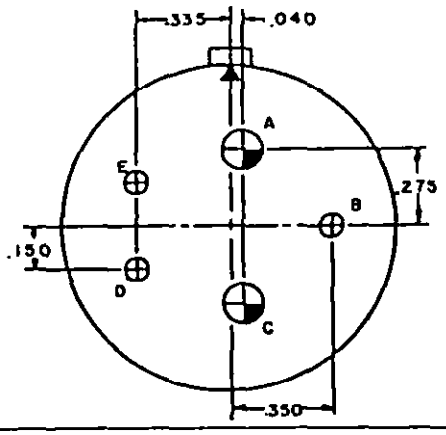


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

**#/

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-11	A	6	12	⊕	All	MS33687-11
			3	8	◒		

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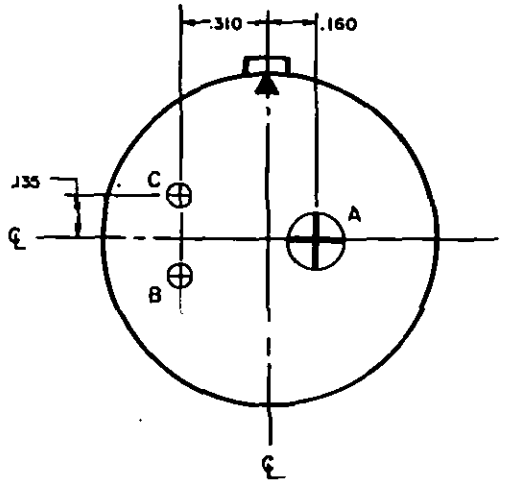


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-12	A	3	12	⊕	All	MS33687-12
			2	4	◒		

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INACTIVE FOR NEW DESIGN
AFTER 17 DEC 1958

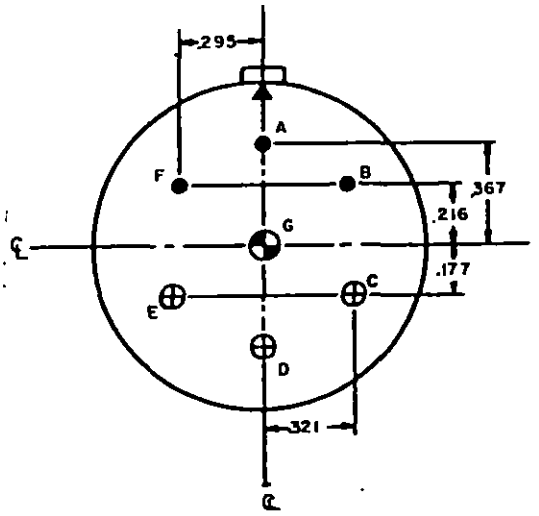


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-14	A	2	12	⊕	All	MS33687-14
			1	0	⊕		

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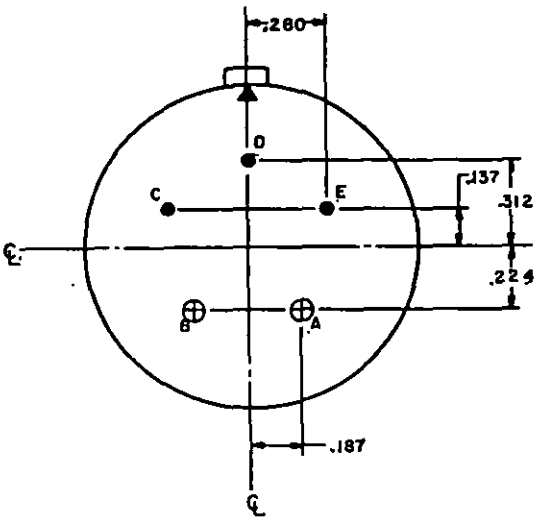


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-16	D	3	16	●	A,B,F,G	MS33687-16
			1	8	◕		
		A	3	12	⊕	All others	

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INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954

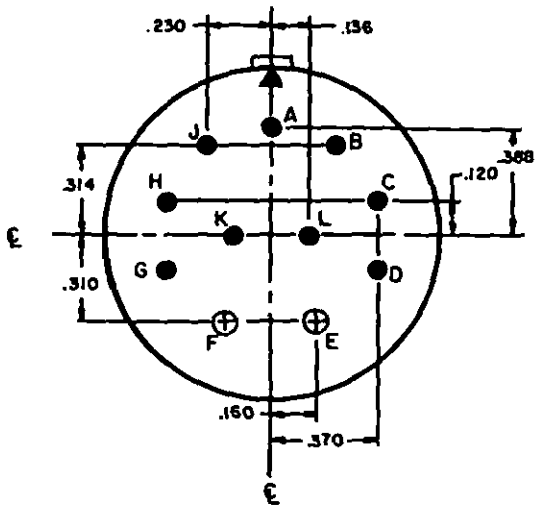


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-17	D	3	16	●	A11	MS33687-17
			2	12	⊕		

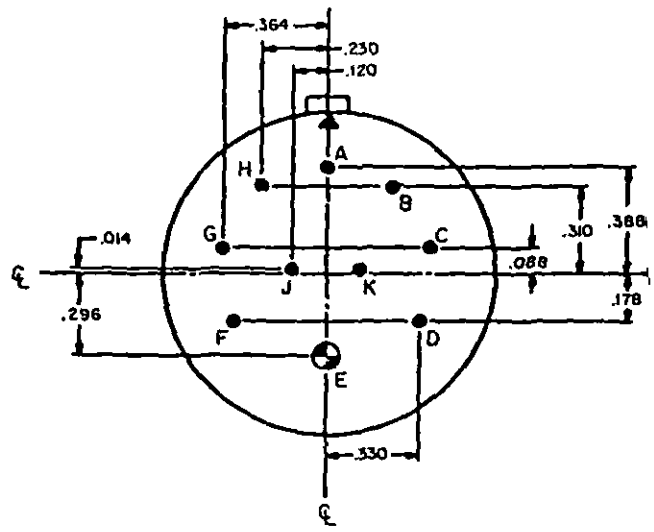
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-20	D	9	16	●	All	MS33687-20
			2	12	⊕		

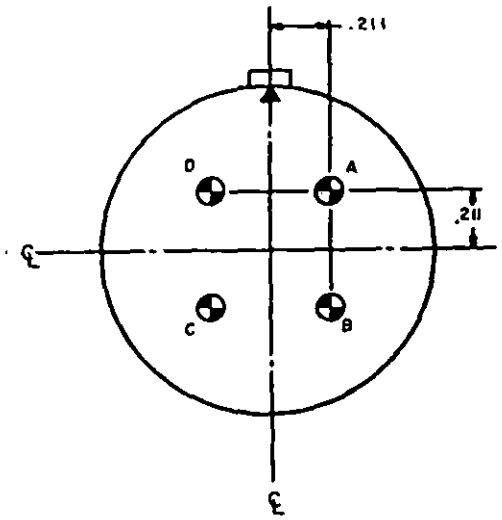
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-21	D	9	16	●	A11	MS33687-21
			1	8	◔		

Source: https://assist.dla.mil Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

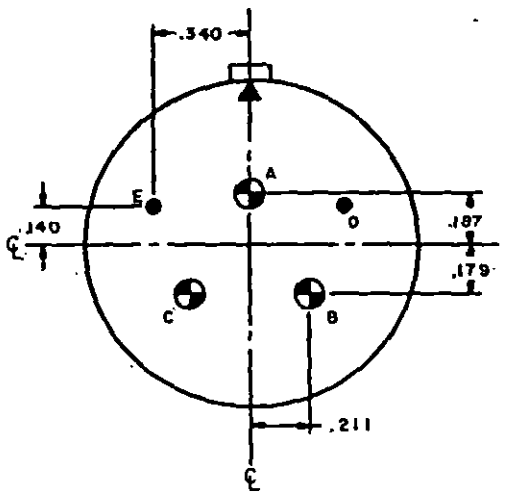


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
45	110	250	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-22	D	4	8		All	MS33687-22

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INACTIVE FOR NEW DESIGN
AFTER 17 DEC 1958

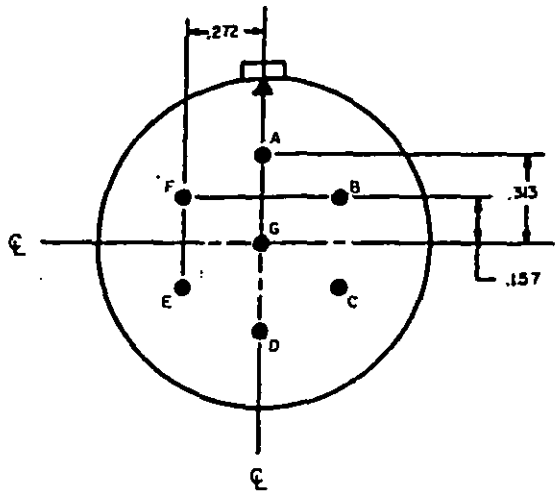


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-23	D	2	16	●	All	MS33687-23
			3	8	◕		

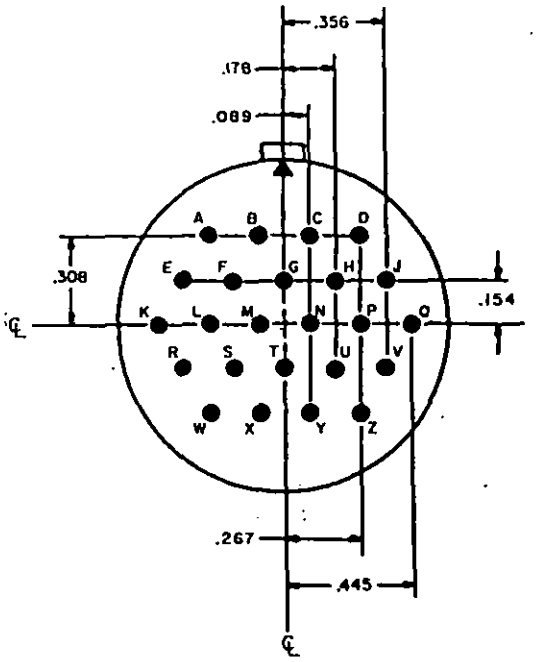
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	---	---	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-27	E	7	16	●	A11	MS33687-27

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



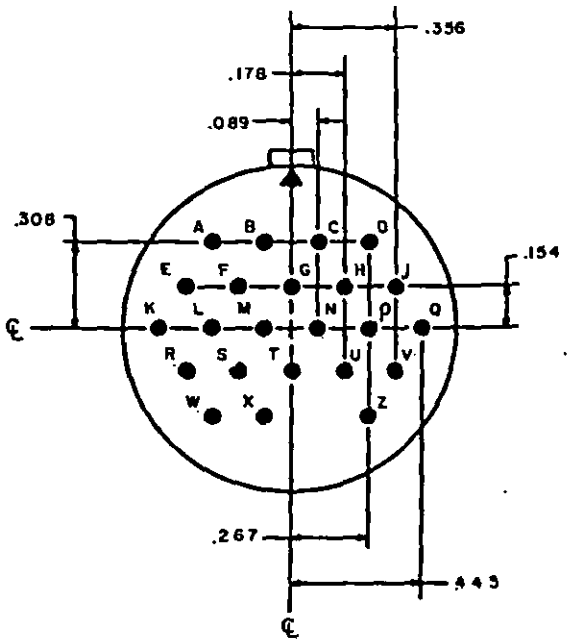
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

**#/

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-28	INST	24	16	●	-A11	MS33687-28

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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NOT APPLICABLE FOR DESIGN OF
AIR FORCE AIRBORNE EQUIPMENT.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	145	240	300

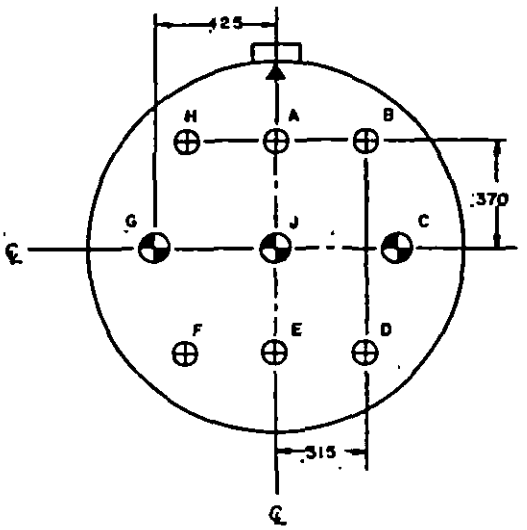
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-80	INST	23	16	●	All	MS33687-80

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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SECTION 1000

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

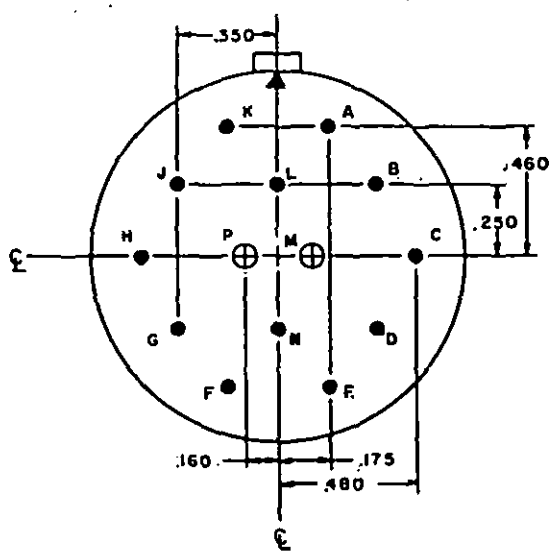
SHELL SIZE 28



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-1	D	2	12	⊕	A,E	MS33688-1
		A	4			B,D,F,H	
		D	1	8	◕	J	
		A	2			G,C	

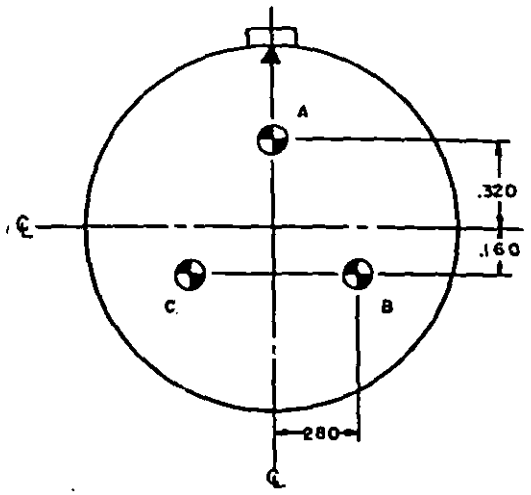
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-2	D	12	16	●	All	MS33688-2
			2	12	⊕		

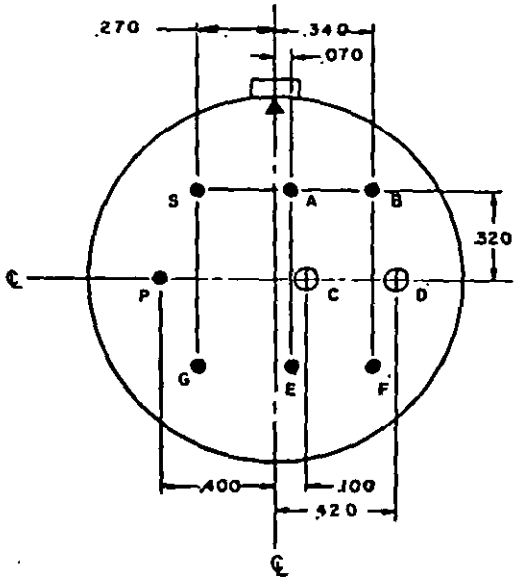
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-3	E	3	8		All	MS33688-3

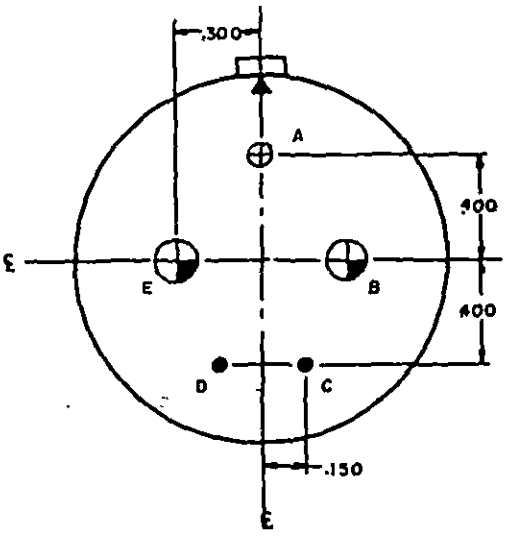
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-4	E	3	16	●	G,P,S	MS33688-4
		D	4			A,B,E,F	
			2	12	⊕	C,D	

Source: https://assist.dla.mil - Downloaded: 2026-10-26T07:50Z
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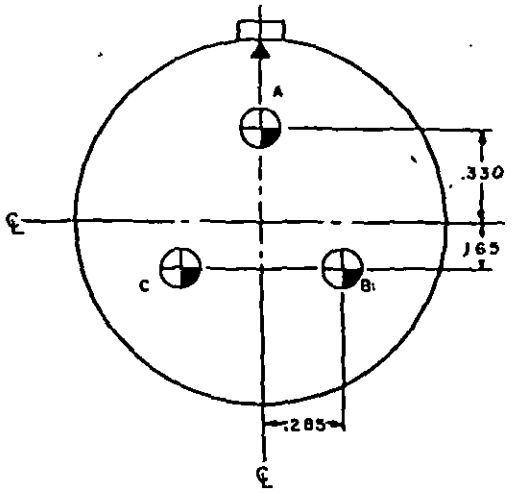


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-5	D	2	16	●	A11	MS33688-5
			1	12	⊕		
			2	4	◒		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954

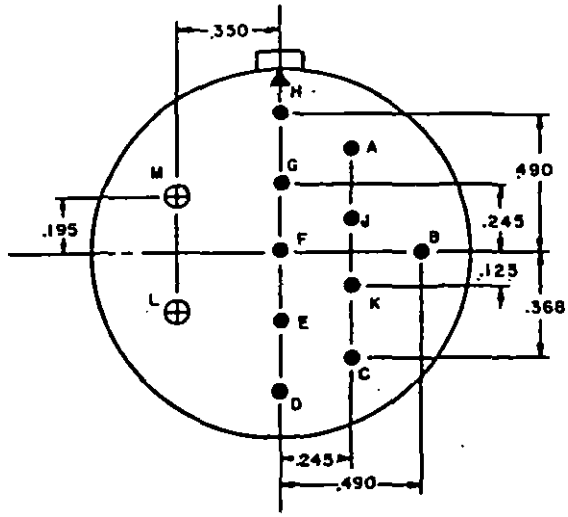


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-6	D	3	4	⊕	All	MS33688-6

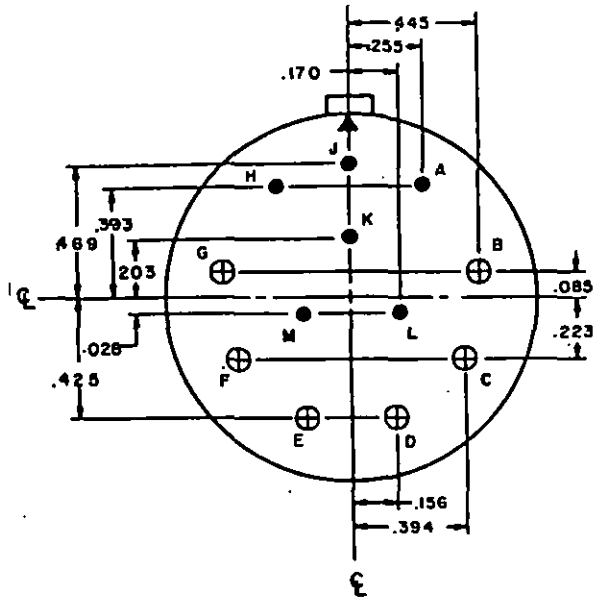
Source: https://assist.dla.mil/ -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-8	E	2	12	⊕	L,M	MS33688-8
		D	1	16	●	B	
		A	9			All others	

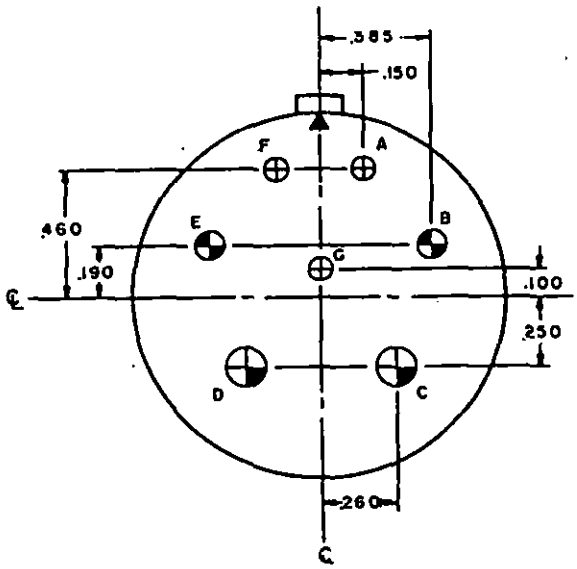
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-9	D	6	16	●	All	MS33688-9
			6	12	⊕		

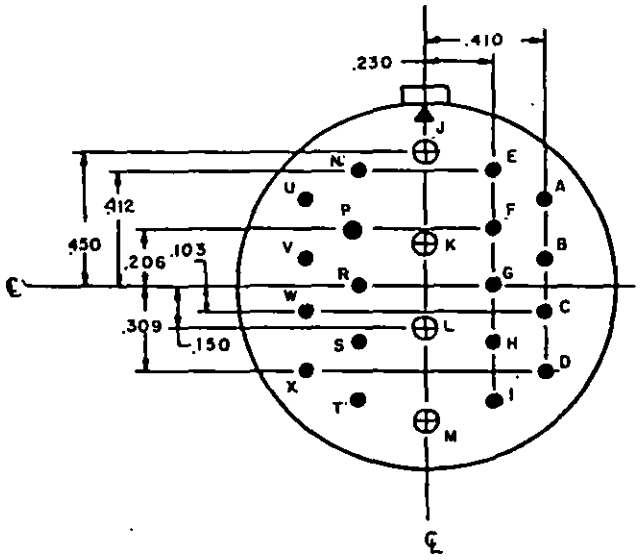
Source: https://assist.dla.mil -- Downloaded: 202610-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-10	D	1	12	⊕	G	MS33688-10
		A	2			A,F	
			2	8	◕	B,E	
			2	4	◕	C,D	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

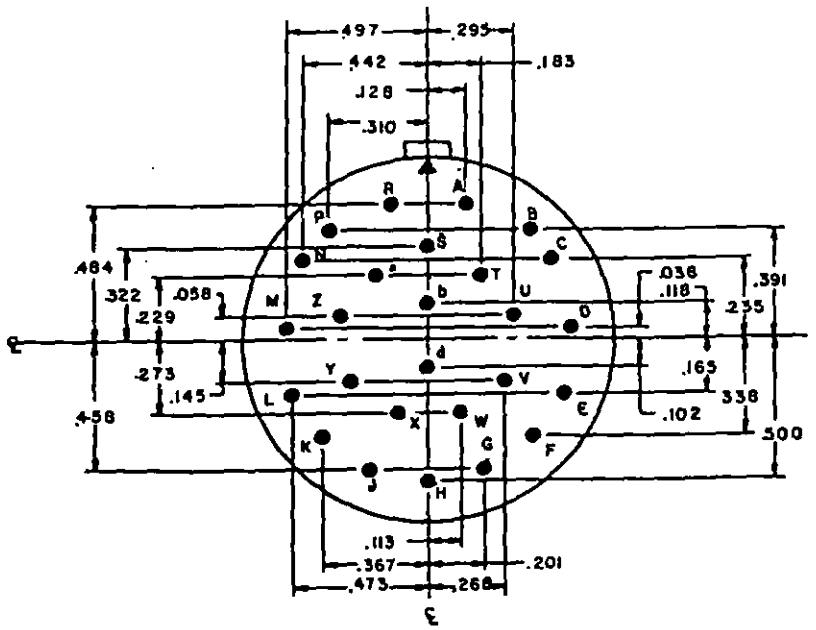


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

**#/

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-11	A	18	16	●	All	MS33688-11
			4	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:50Z
Check the source to verify that this is the current version before use.



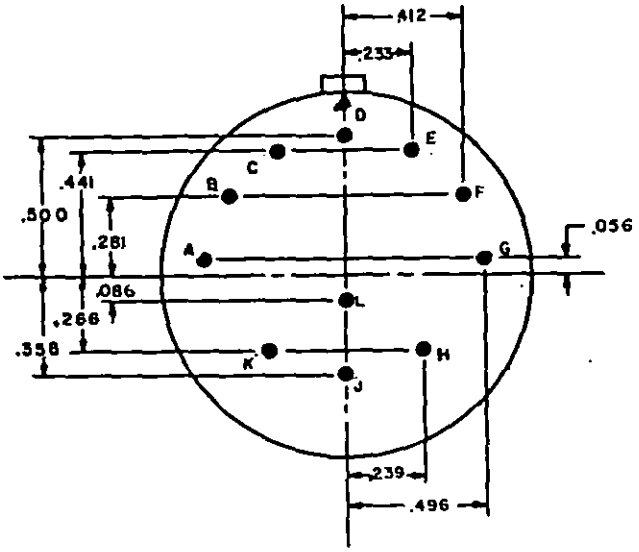
FOR OTHER INSERT POSITION SEE 28-13.

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
90	180	270	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-12	A	26	16	●	A11	MS33688-12

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:50Z
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INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954



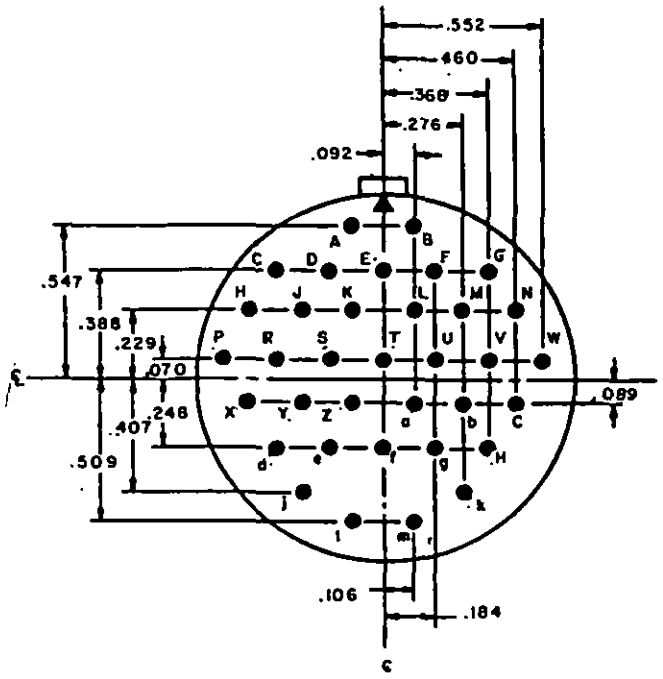
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-14	D	11	16	●	All	MS33688-14

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Check the source to verify that this is the current version before use.

INACTIVE FOR NEW DESIGN OF AIR FORCE AIRBORNE EQUIPMENT AFTER 13 MAY 1954.

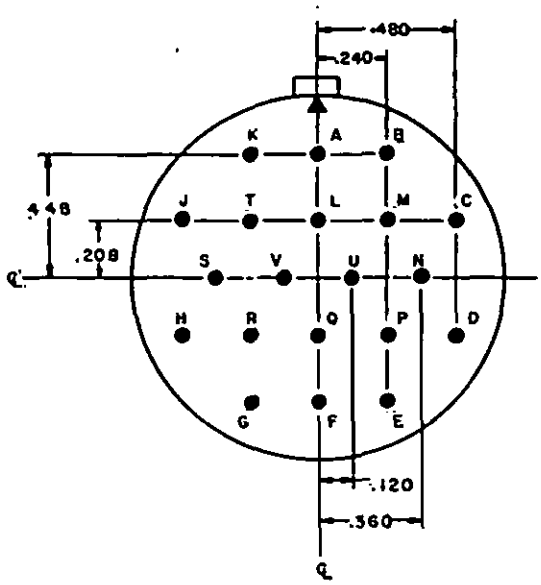


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-15	A	35	16	●	All	MS33688-15

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954

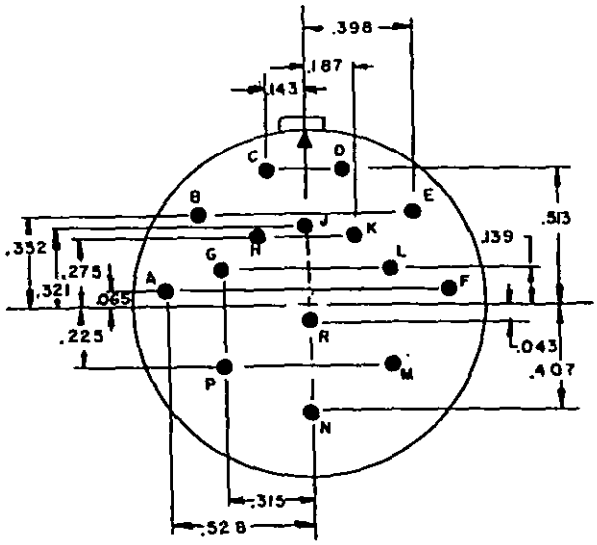


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-16	A	20	16	●	All	MS33688-16

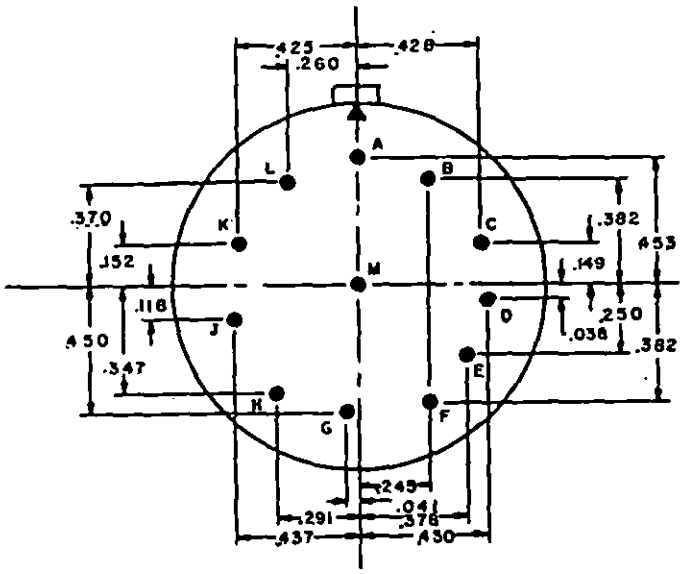
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-17	B	1	16	●	R	MS33688-17
		D	3			M,N,P	
		A	11			All others	

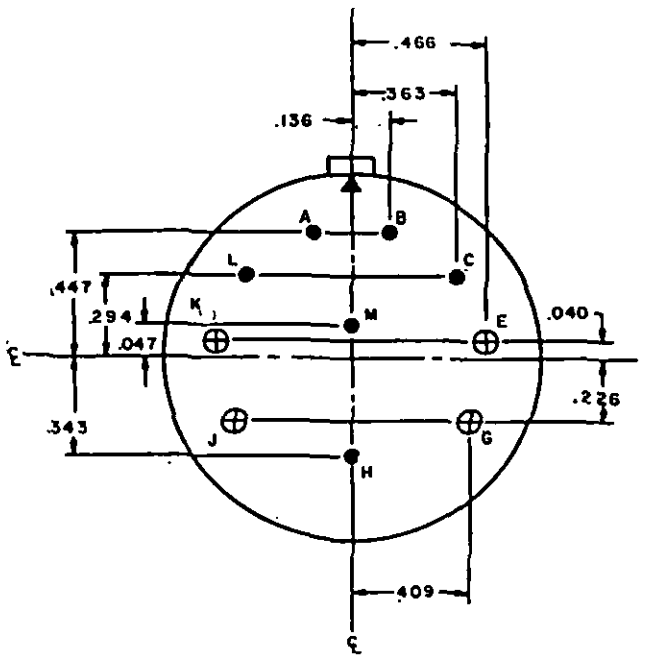
Source: https://assist.dla.mil -- Downloaded: 202610-26T07:50Z
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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-18	C	1	16	●	M	MS33688-18
		A	2			A,B	
		INST	4			C,D,E,F	
		D	5			All others	

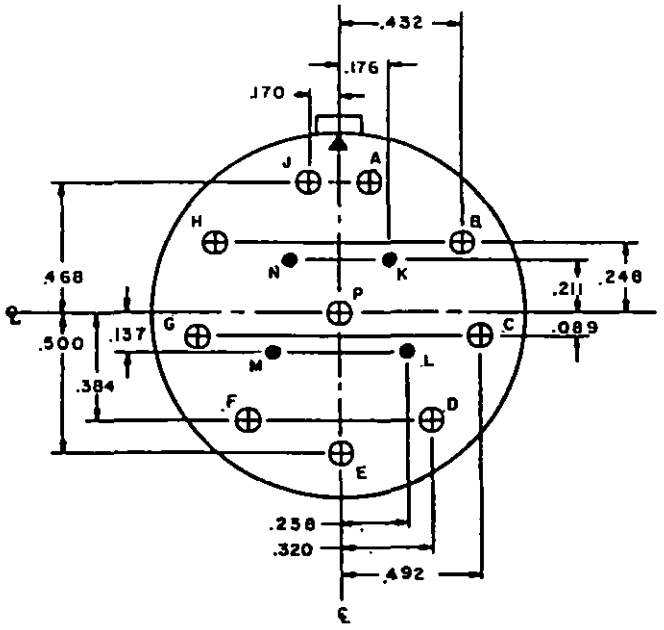
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-19	B	2	16	●	H,M	MS33688-19
		D	2			A,B	
		A	2			All others	
			4	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:50Z
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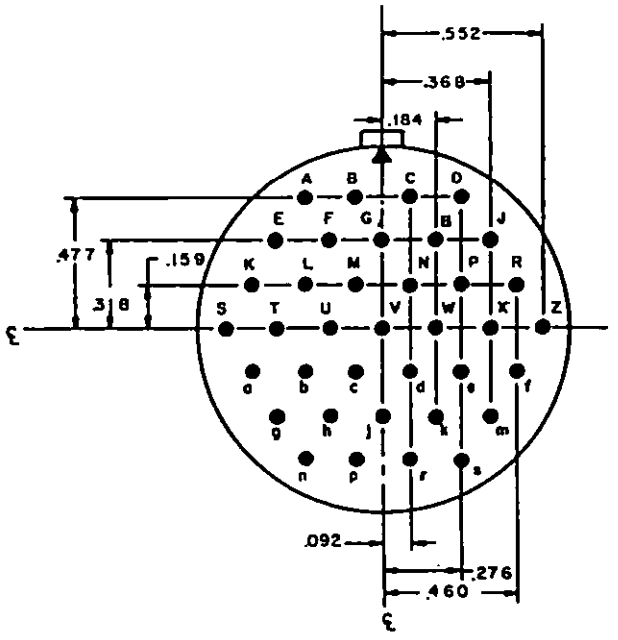


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

**#/

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-20	A	4	16	●	All	MS33688-20
			10	12	⊕		

Source: https://assist.dla.mil/ - Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

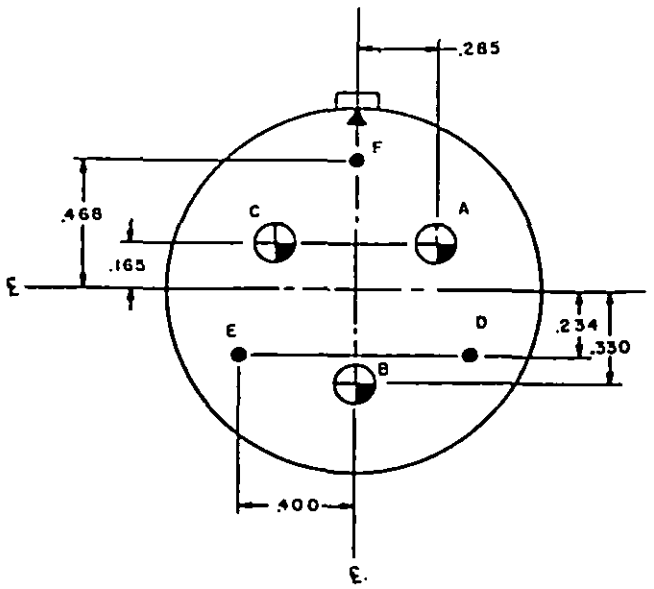


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

**#/

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-21	A	37	16	●	All	MS33688-21

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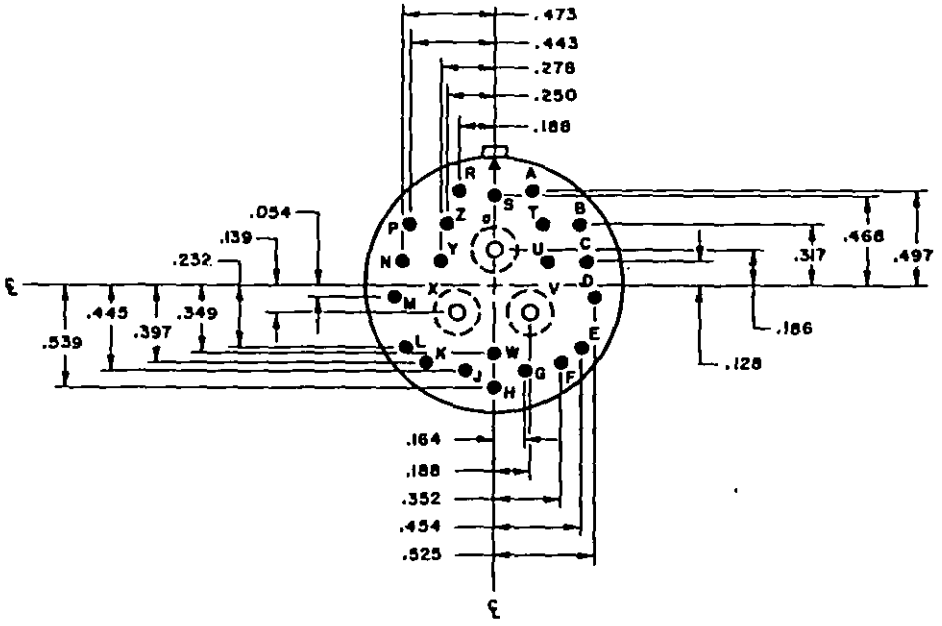


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

**#/

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-22	D	3	16	●	All	MS33688-22
			3	4	⊕		

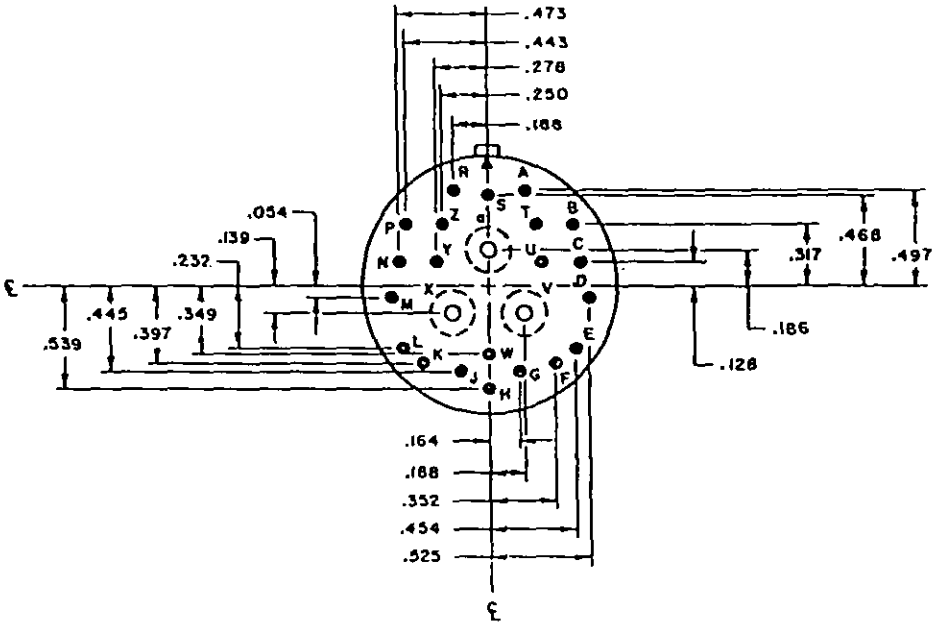
Source: https://assist.dla.mil -- Downloaded: 202610-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-23		3	8 (shielded) MS14056-2	⊙	V,X,a	MS33688-23
		INST	21	16	●	All others	

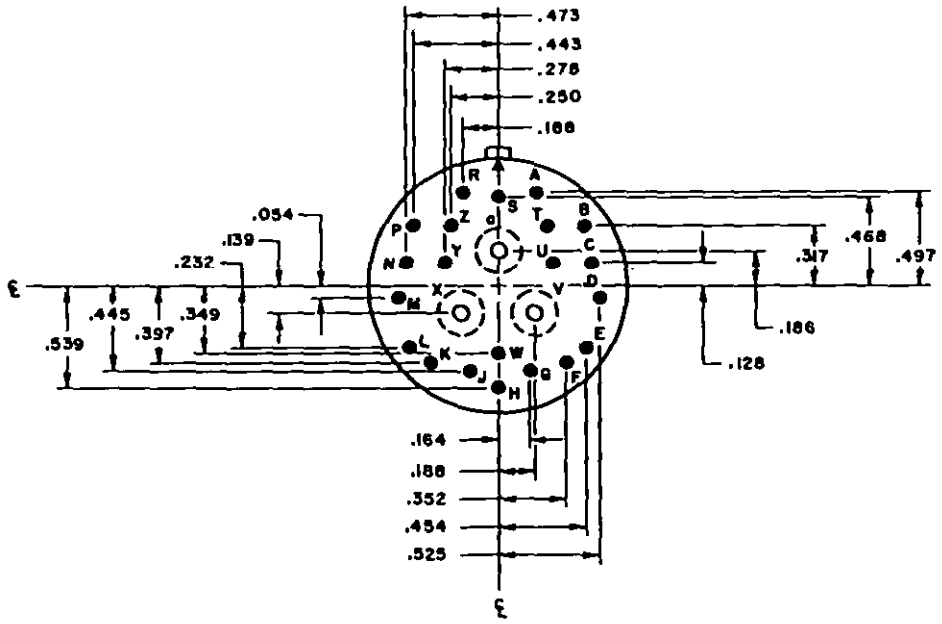
Source: https://assist.dla.mil -- Downloaded: 202610-26T07:50Z
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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-24	INST	3	8 (shielded) MS14056-1	◎	V,X,a	MS33688-24
			21	16	●	All others	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

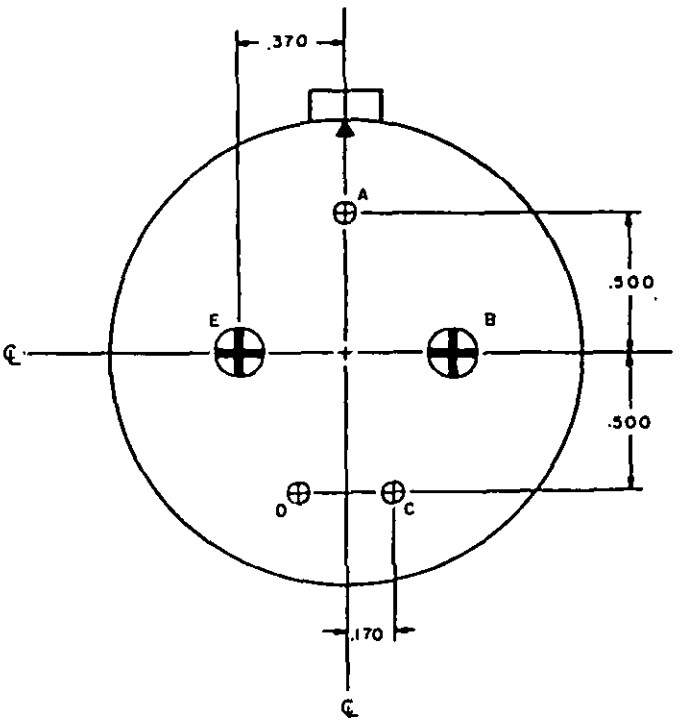
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-24	INST	3	8 (shielded) MS14056-1	⦶	V,X,a	MS33688-24
			21	16	●	All others	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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SECTION 1100

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

SHELL SIZE 32

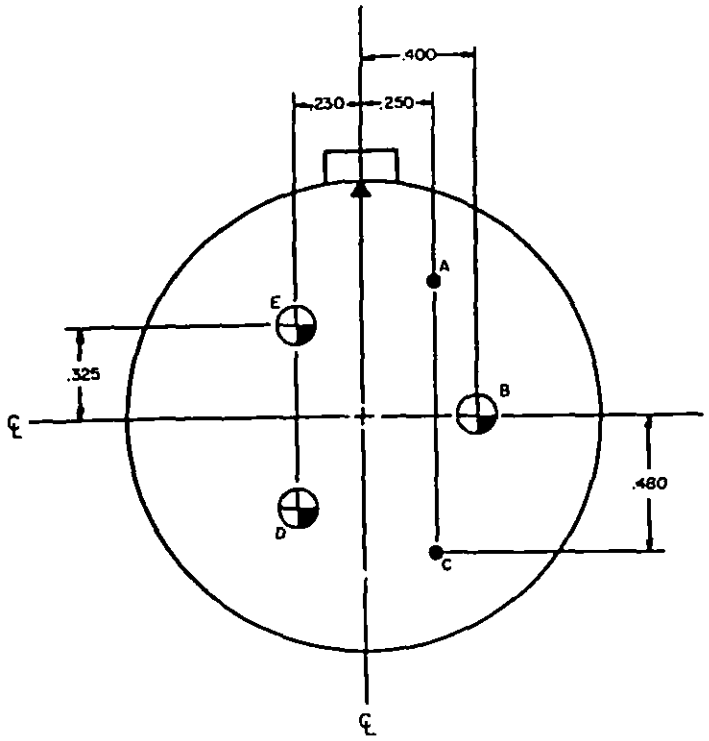


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

FOR OTHER INSERT POSITION SEE 32-19.

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-1	E	1	12	⊕	A	MS33689-1
			2			D,C	
		D	2	0	⊕	B,E	

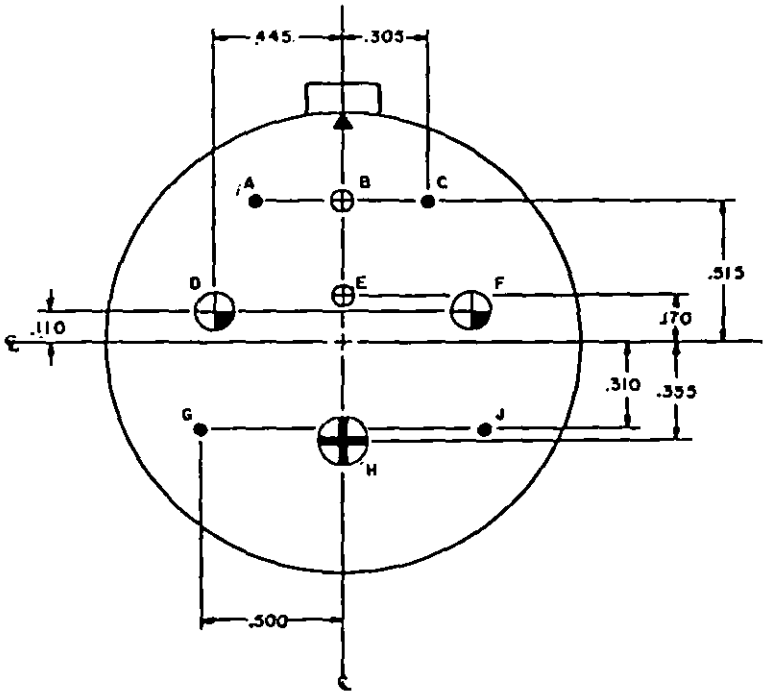
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-2	E	2	16	●	All	MS33689-2
			3	4	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

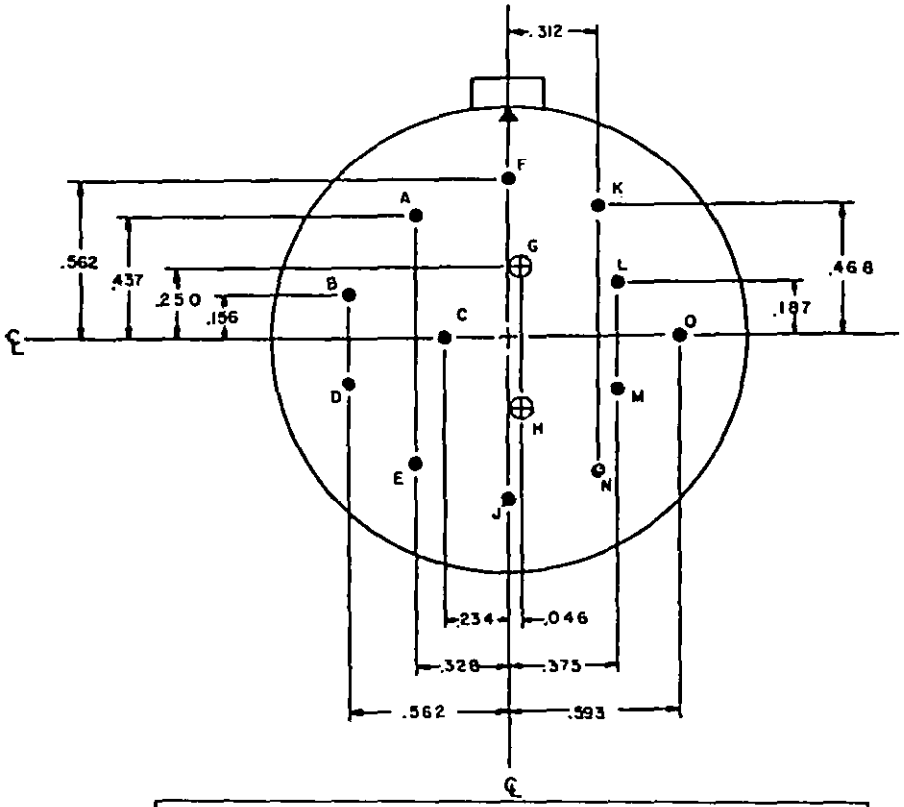


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-3	D	4	16	●	All	MS33689-3
			2	12	⊕		
			2	4	◒		
			1	0	⊕		

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Check the source to verify that this is the current version before use.

INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954



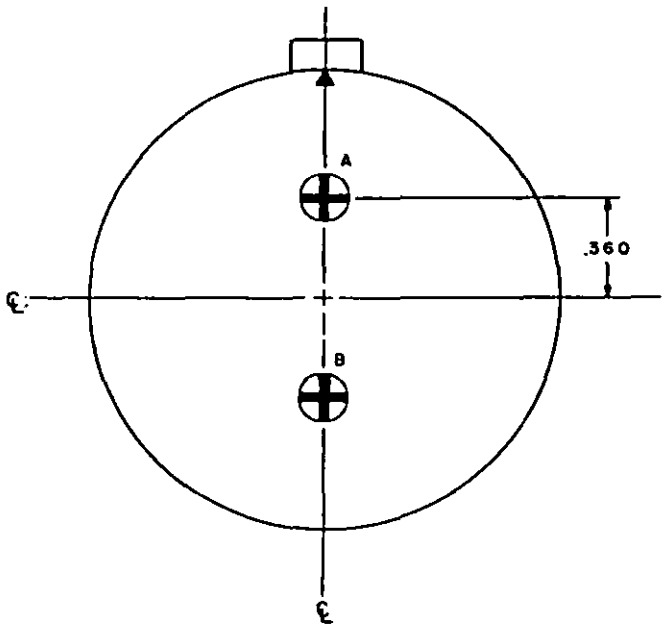
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992) (For other insert position see 32-18)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-4	D	2	12	⊕	G,H	MS33689-4
		A	4	16	●	F,J,K,N	
		D	8			All others	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954

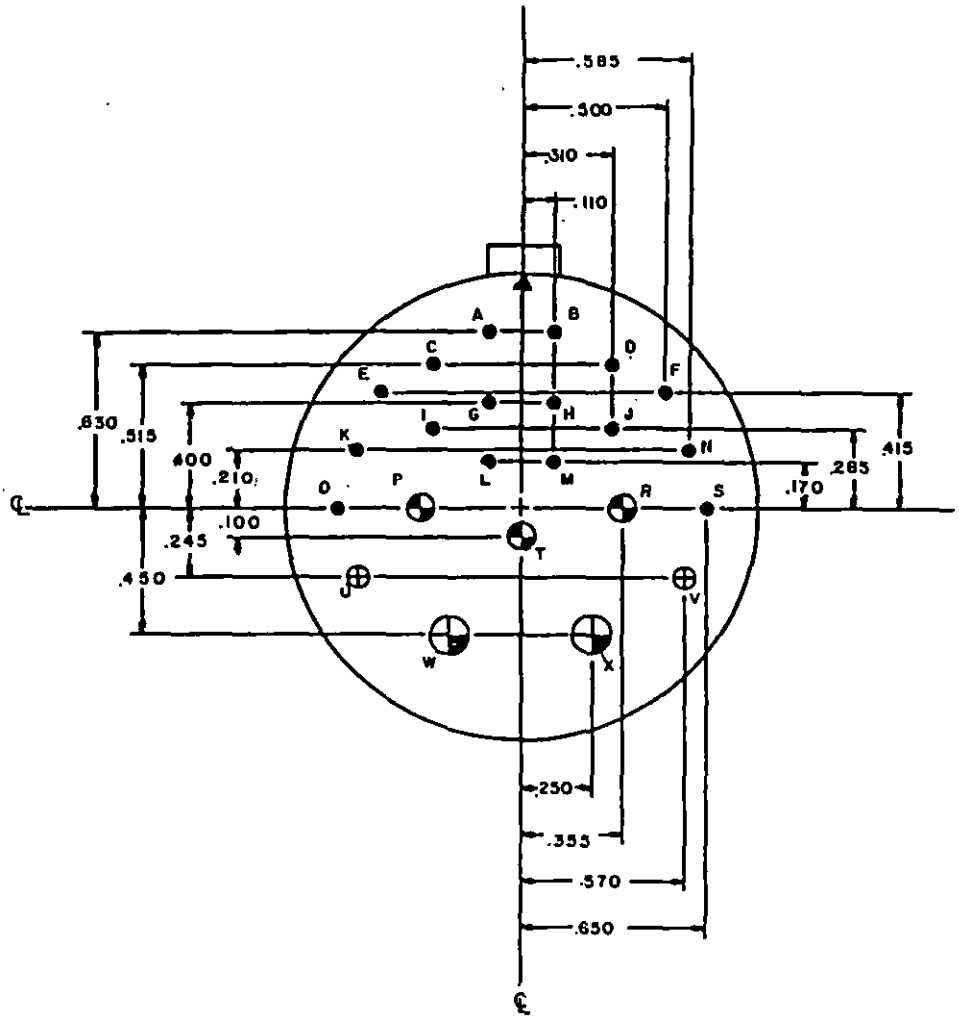


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	325

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-5	D	2	0	⊕	All	MS33689-5

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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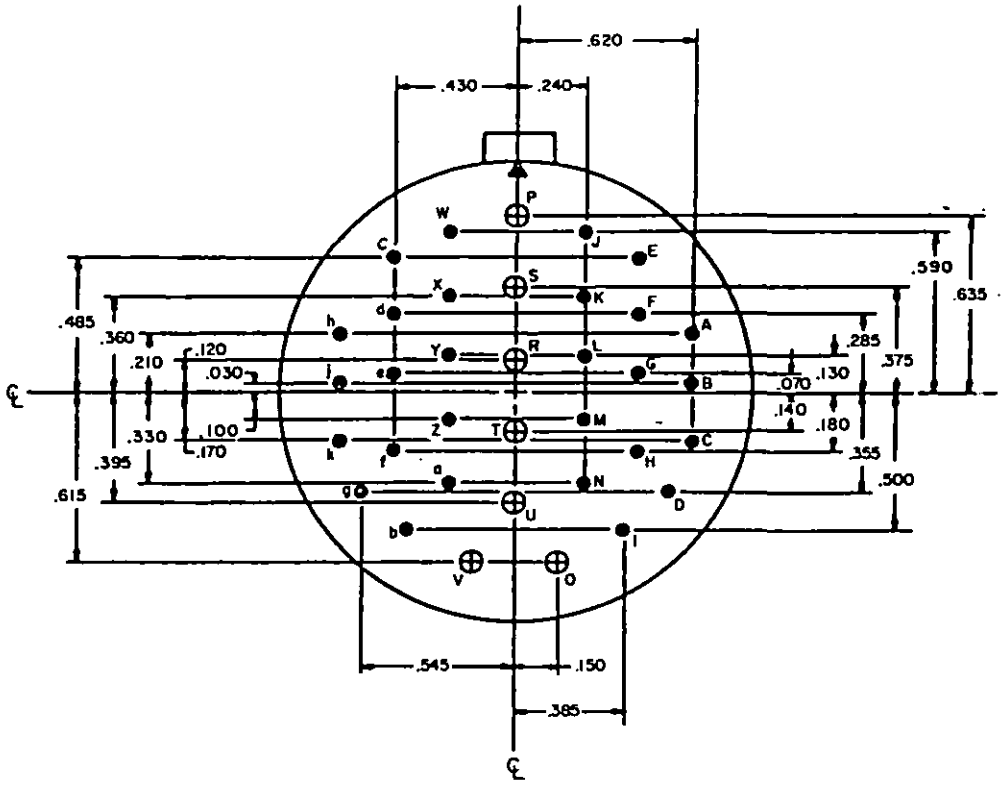


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

**#/

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-6	A	16	16	●	All	MS33689-6
			2	12	⊕		
			3	8	◕		
			2	4	◕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



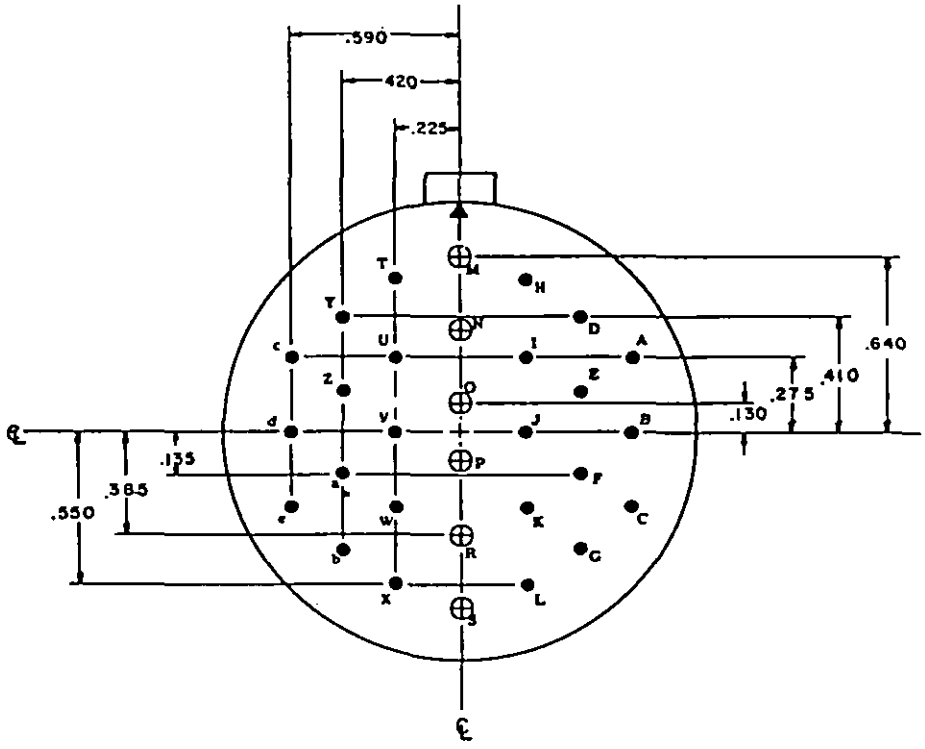
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	125	235	280

**#/

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-7	INST	4	16	●	A,B,h,j	MS33689-7
		A	24			All others	
			7	12	⊕		

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Check the source to verify that this is the current version before use.

INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954

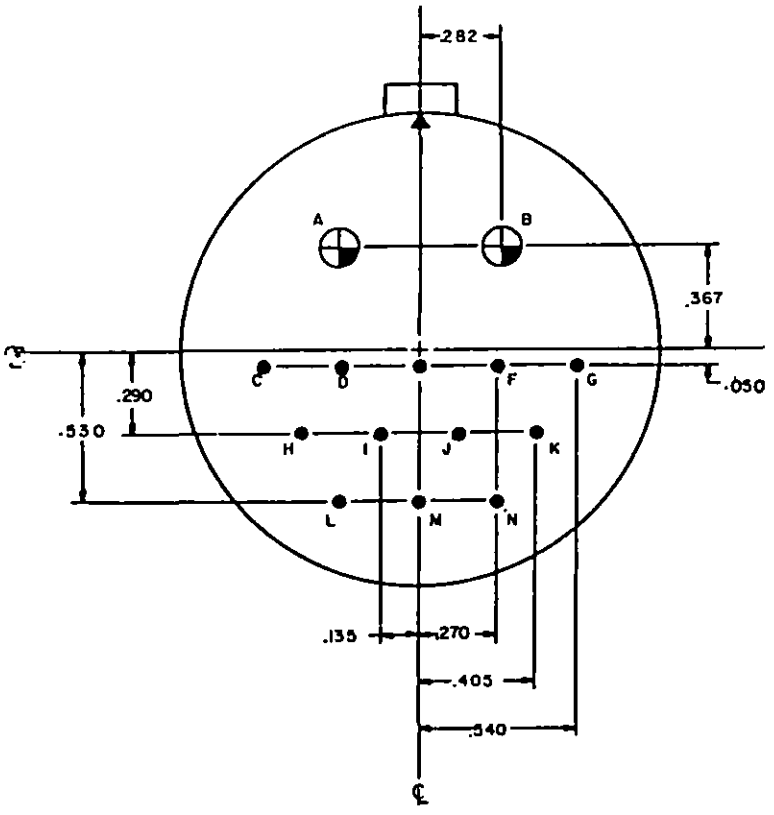


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	125	235	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-8	A	24	16	●	All	MS33689-8
			6	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

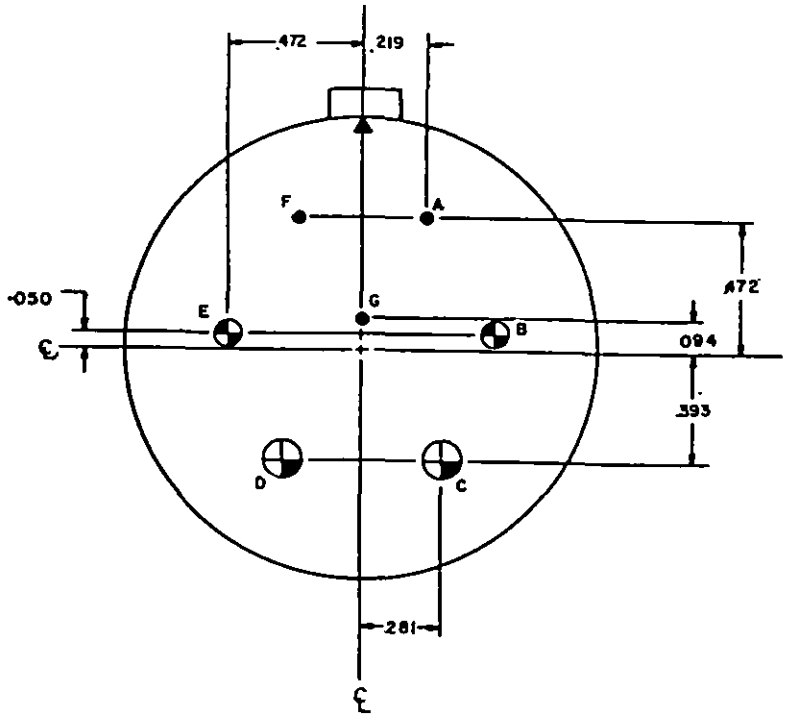


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-9	D	12	16	●	All	MS33689-9
			2	4	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR NEW DESIGN
AFTER 17 DEC 1958



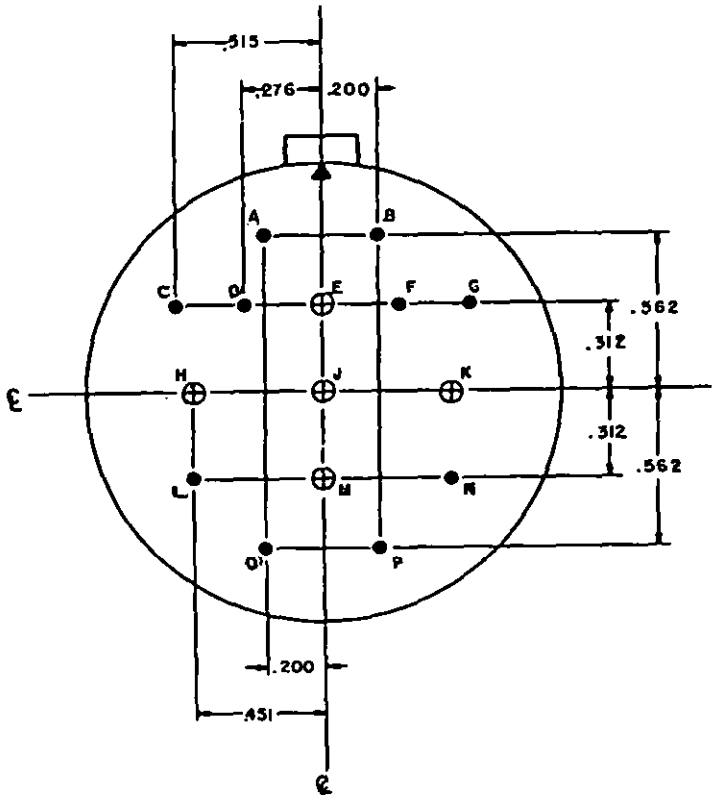
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-10	E	2	16	●	A,F	MS33689-10
		B	1			G	
		D	2	8	⊕	B,E	
		A	2	4	⊕	C,D	

Source: https://assist.dla.mil/ Downloaded: 2026-04-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954



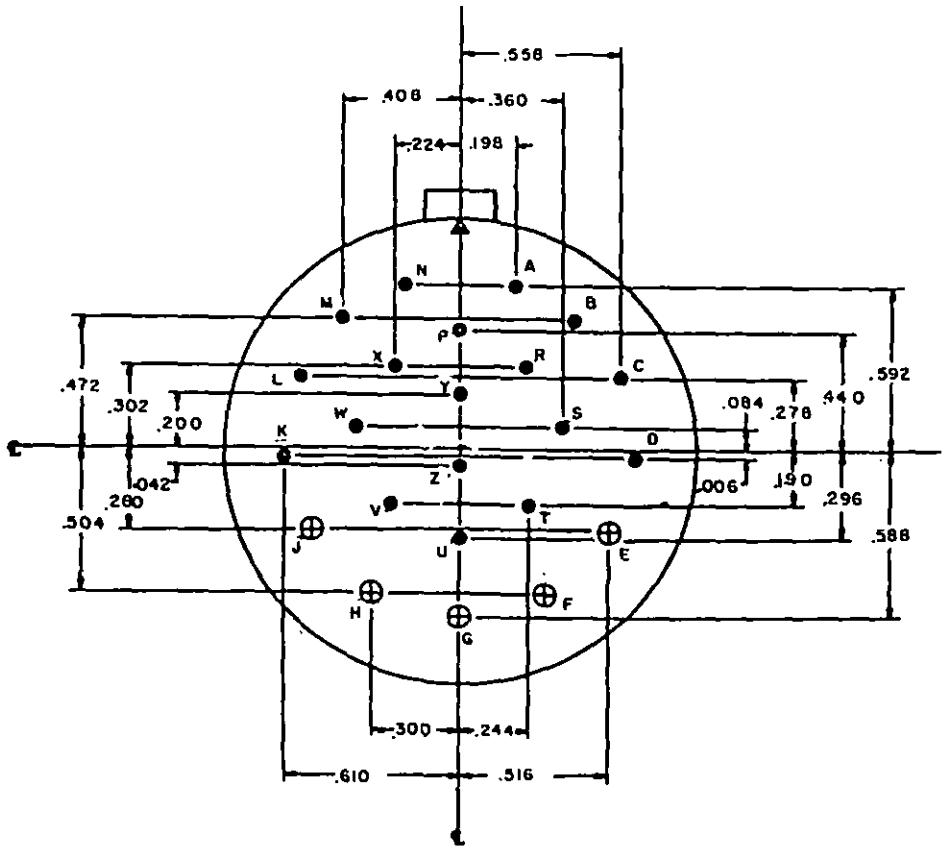
(Not applicable to MIL-C-22992)

ALTERNATE INSERT POSITIONS

A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-12	A	1	12	⊕	E	MS33689-12
		D	4			H,J,K,M	
		A	4	16	●	C,D,F,G	
		D	6			All others	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

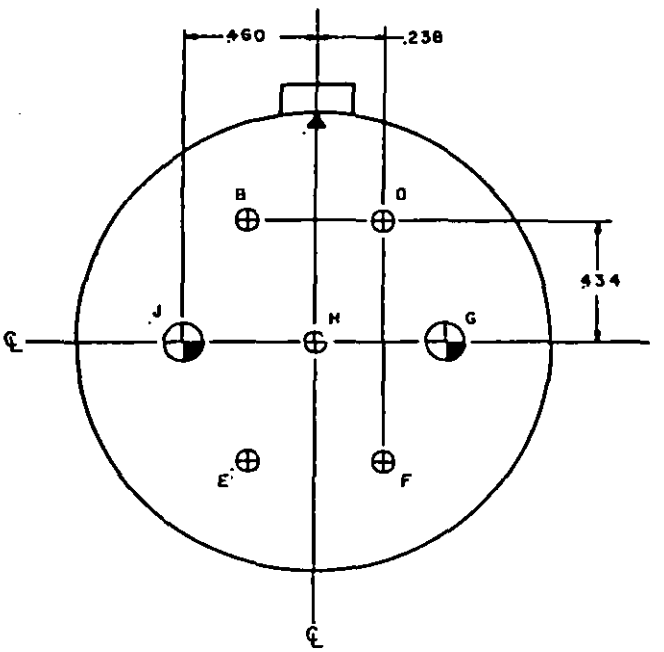


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-13	D	18	16	●	All	MS33689-13
			5	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR NEW DESIGN
AFTER 17 DEC 1958

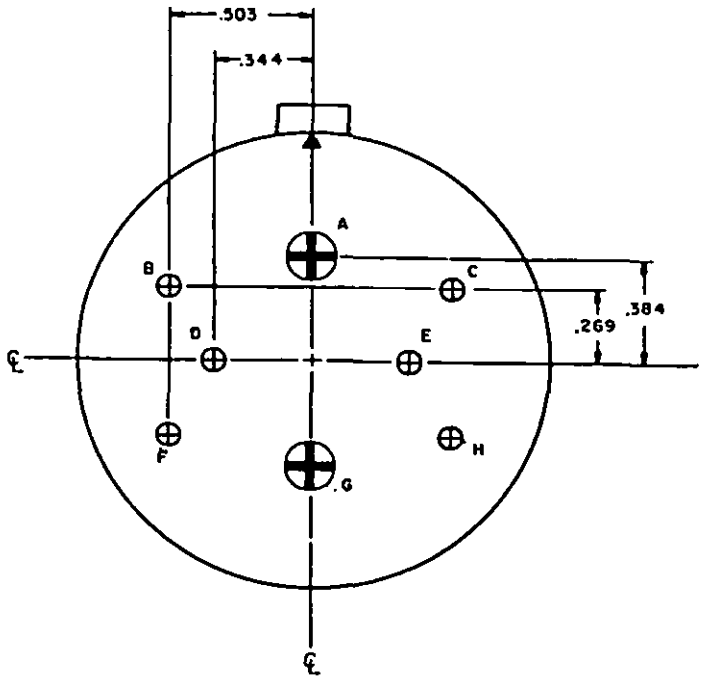


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-14	D	5	12	⊕	All	MS33689-14
			2	4	⊕		

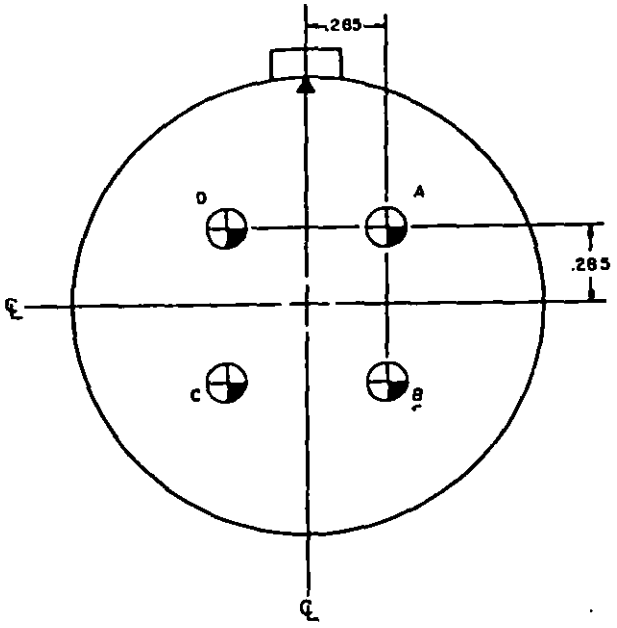
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-15	D	6	12	⊕	All	MS33689-15
			2	0	⊕		

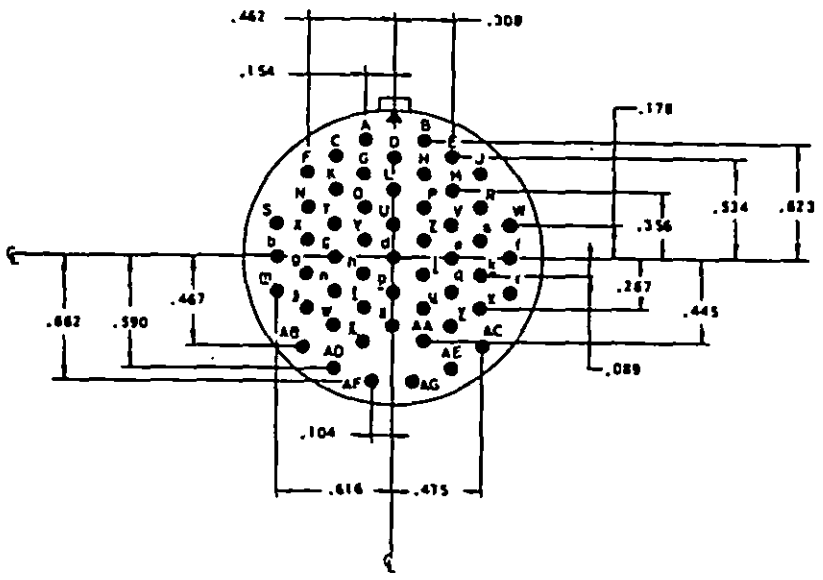
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
45	110	250	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-17	D	4	4	⊕	All	MS33689-17

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

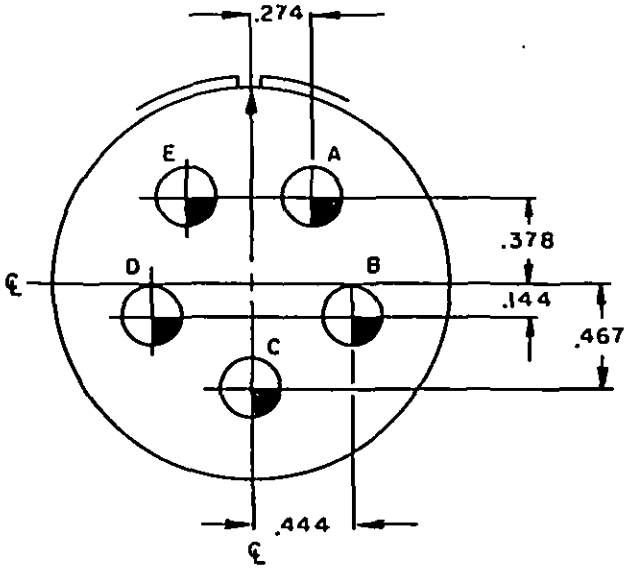


ALTERNATE INSERT POSITIONS					
A DEGREES					
U	V	W	X	Y	Z
100	260	80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-22	A	54	16	●	A11	MS33689-22

Source: https://assist.dla.mil. Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



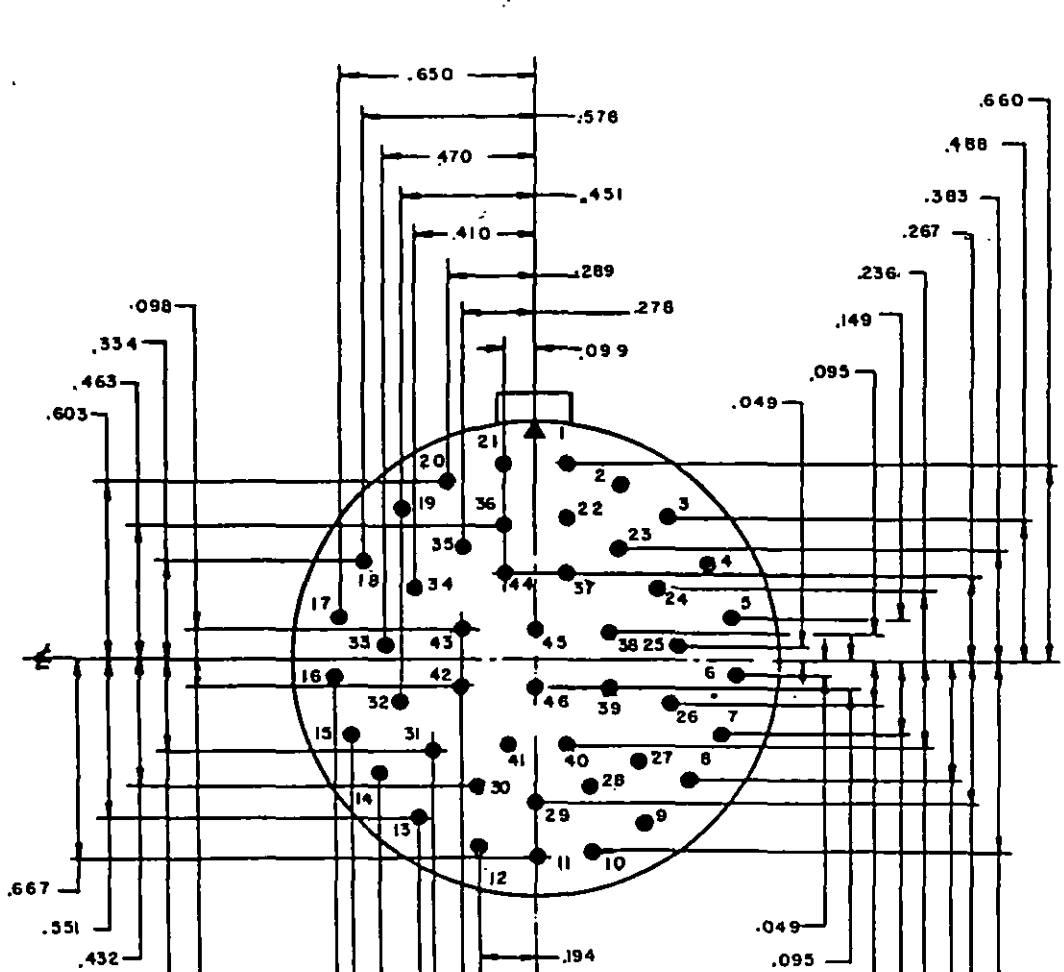
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
-	-	-	-

(Not applicable to MIL-C-22992 or MIL-C-83723)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-63	D	5	4	⊕	All	---

Source: https://assist.dla.mil - Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

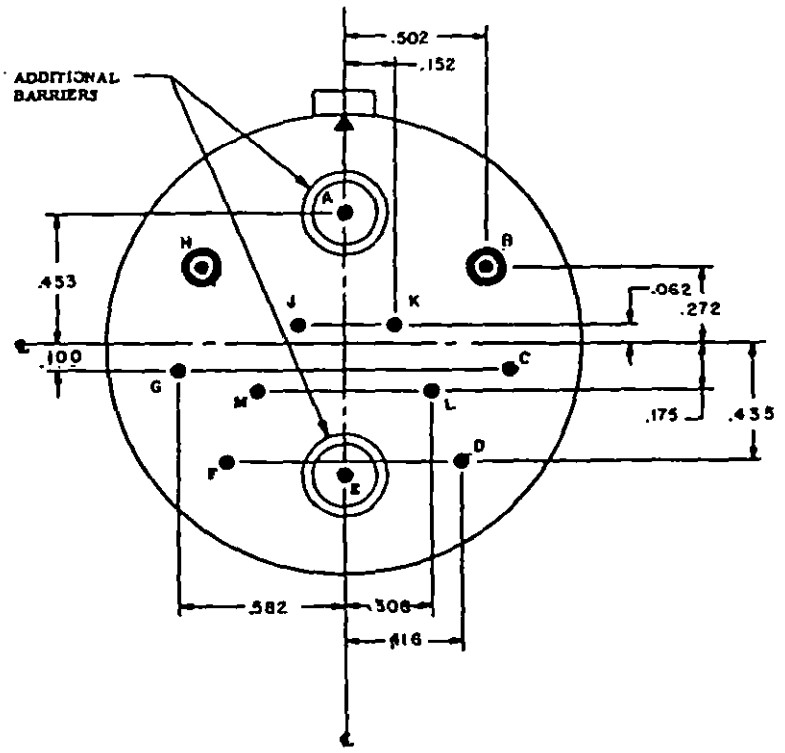
NOT APPLICABLE FOR DESIGN OF
AIR FORCE AIRBORNE EQUIPMENT.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
36	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-73	A	46	16	●	A11	MS33689-73

Source: https://assist.dla.mil/ Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

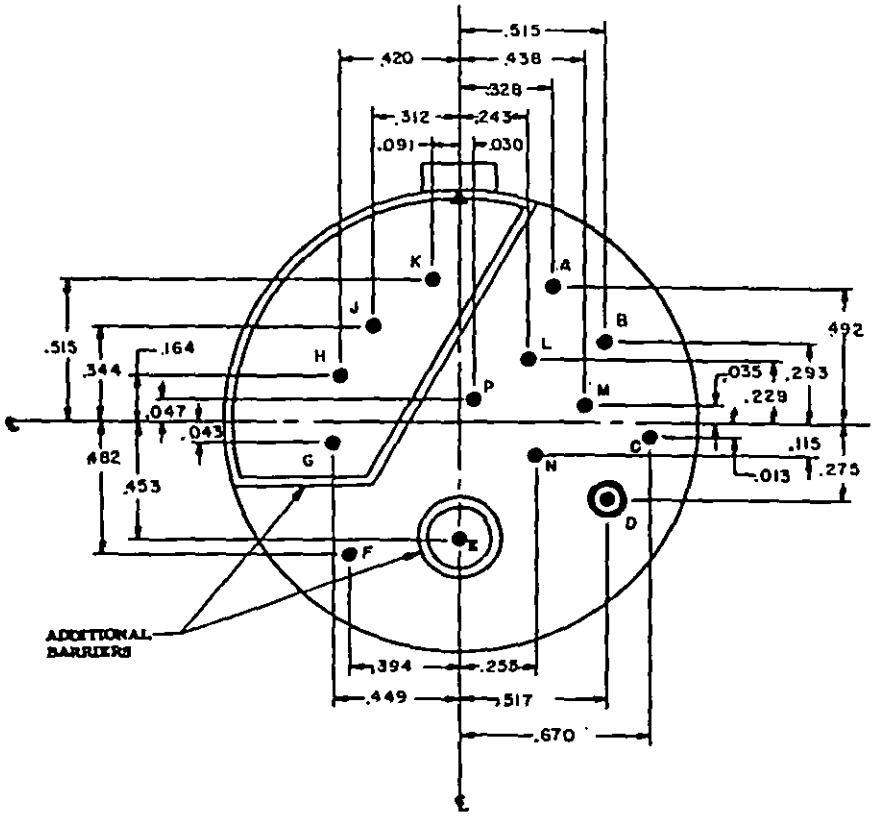


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
65	125	225	310

(Not applicable to MIL-C-22992 or MIL-C-83723)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-101		2	Coax (MS33695)	◉	B,H	MS33689-101
		C	2	16	●	A,E	
		D	8			All others	

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Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
65	125	225	310

(Not applicable to MIL-C-22992 or MIL-C-83723)

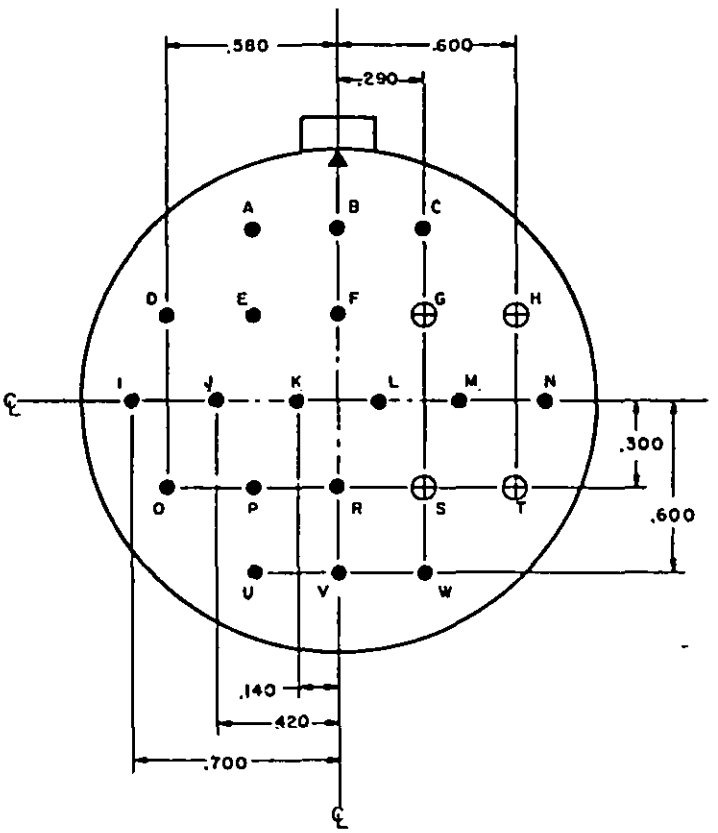
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-102		1	Coax (MS33695)	◎	D	MS33689-102
		A	6	16	●	C,G,H, J,M,N	
		B	1			E	
		D	6			All others	

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SECTION 1200

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS
SHELL SIZE 36

INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954

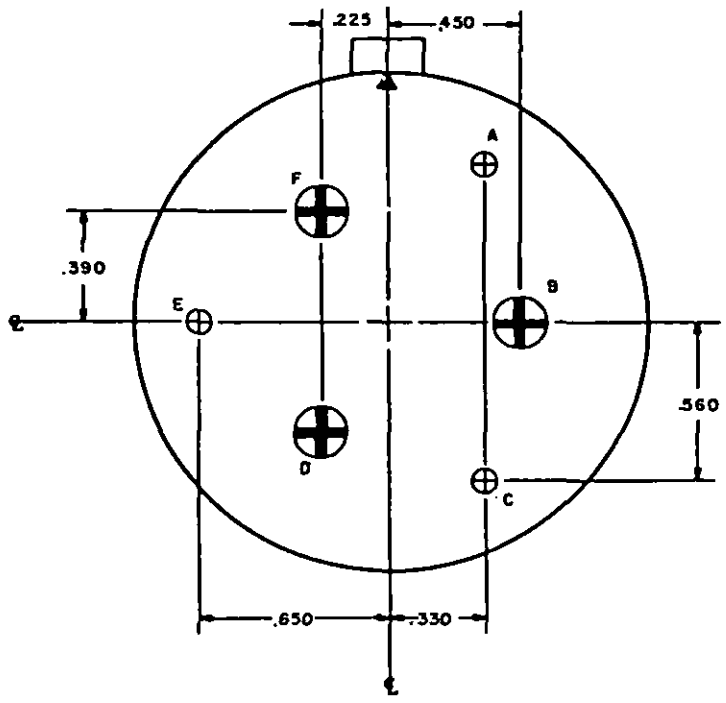


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-1	D	18	16	●	A11	MS33690-1
			4	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

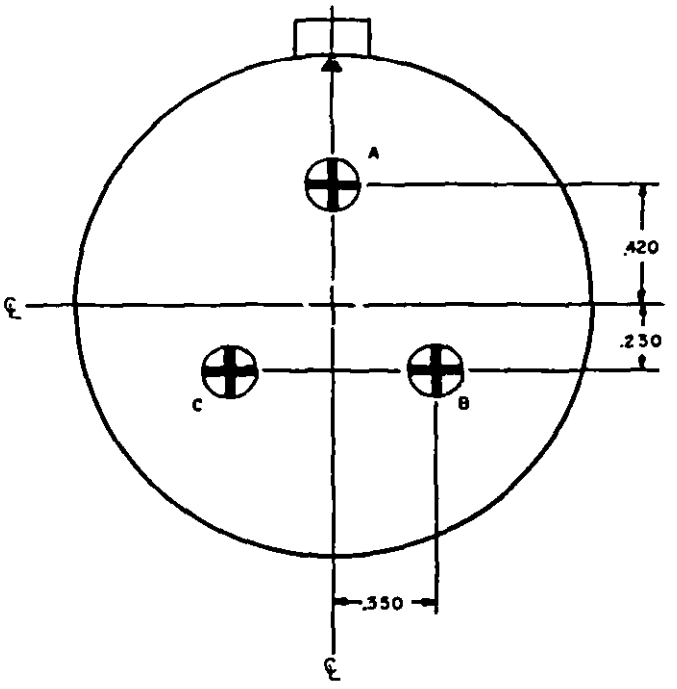


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-3	D	3	12	⊕	All	MS33690-3
			3	0	⊕		

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Check the source to verify that this is the current version before use.

INACTIVE FOR NEW DESIGN
AFTER 13 MAY 1954

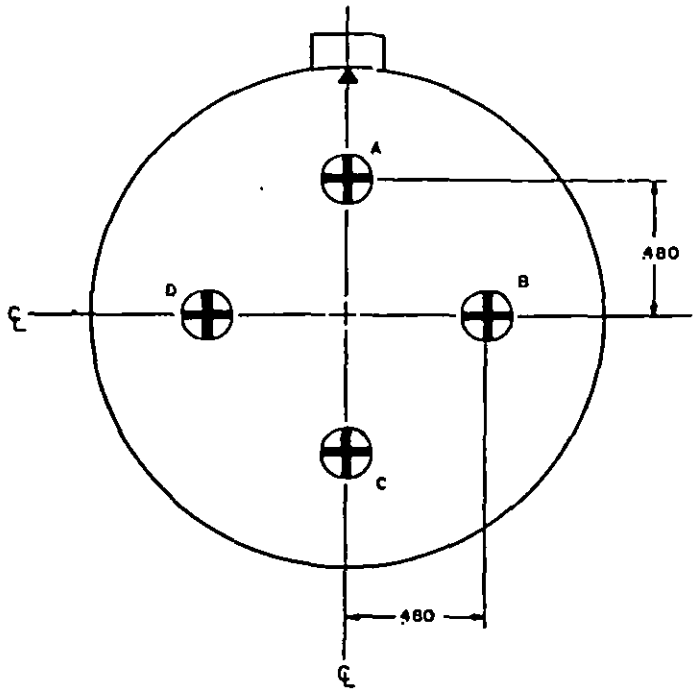


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
70	145	215	290

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-4	D	1	0	⊕	A	MS33690-4
		A	2			B,C	

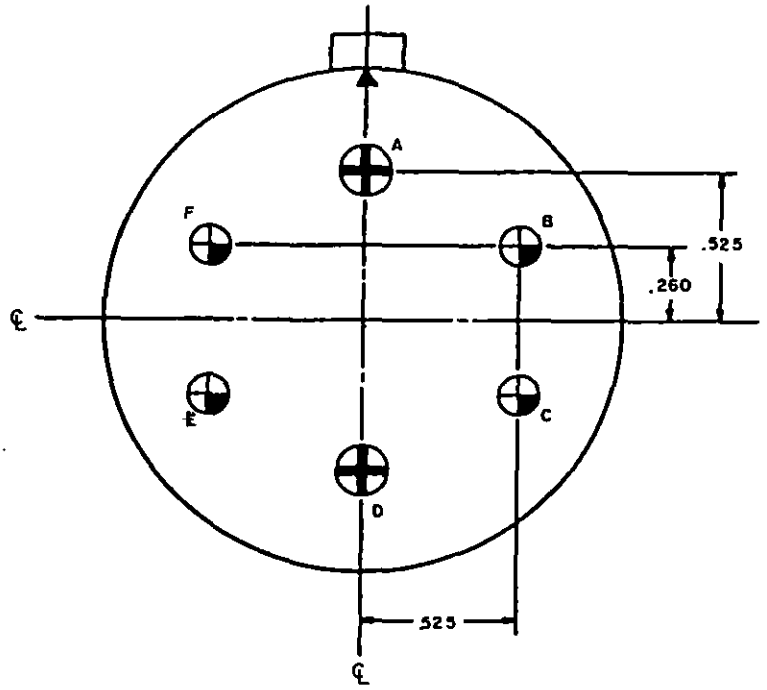
Source: https://assist.dla.mil -- Downloaded: 202610-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	120	240	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-5	A	4	0	⊕	All	MS33690-5

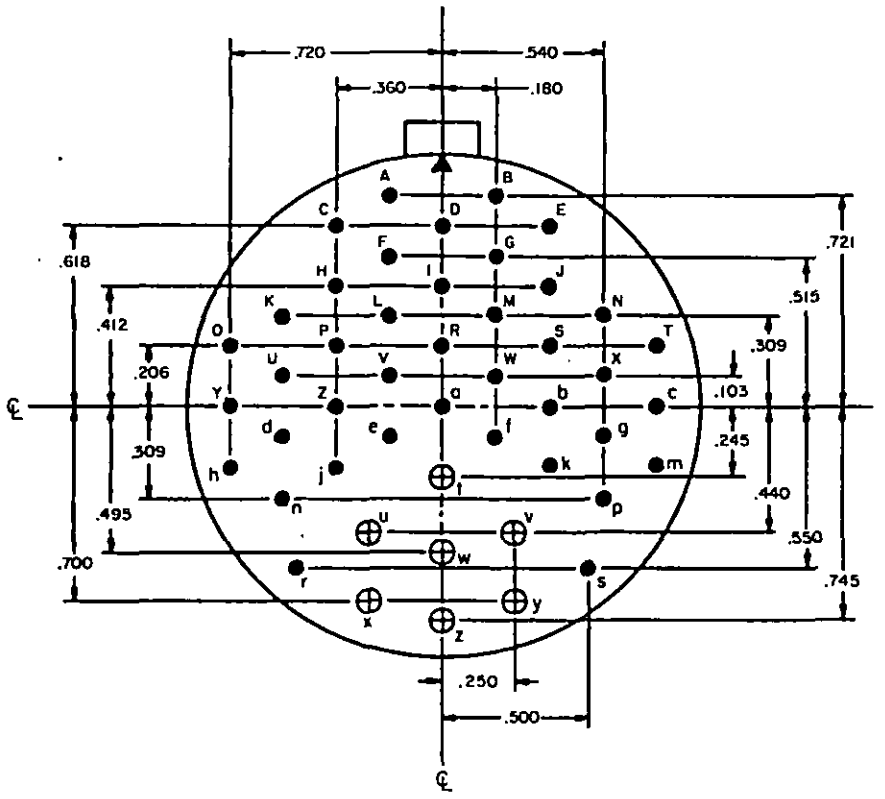
Source: https://assist.dla.mil/ - Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-6	A	4	4	⊕	All	MS33690-6
			2	0	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



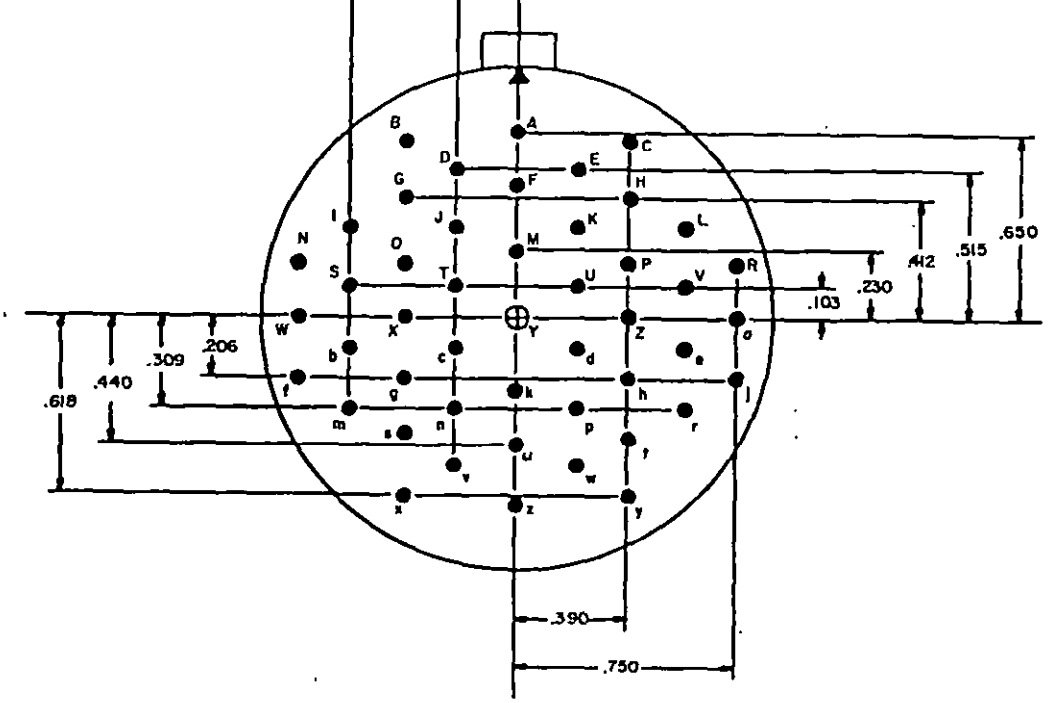
FOR OTHER INSERT POSITIONS SEE 36-16 AND 36-17.

ALTERNATE INSERT POSITIONS

A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-7	A	40	16	●	All	MS33690-7
			7	12	⊕		

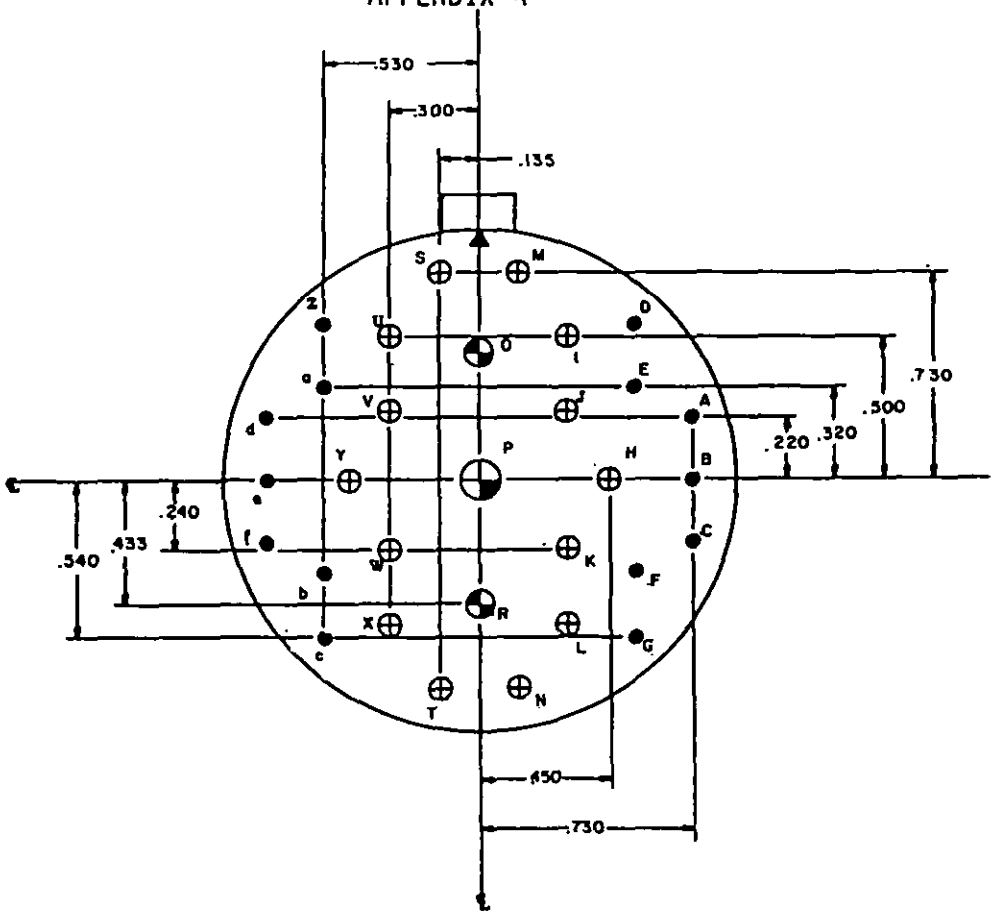
Source: https://assist.dla.mil -- Downloaded: 202640-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-8	A	46	16	●	All	MS33690-8
			1	12	⊕		

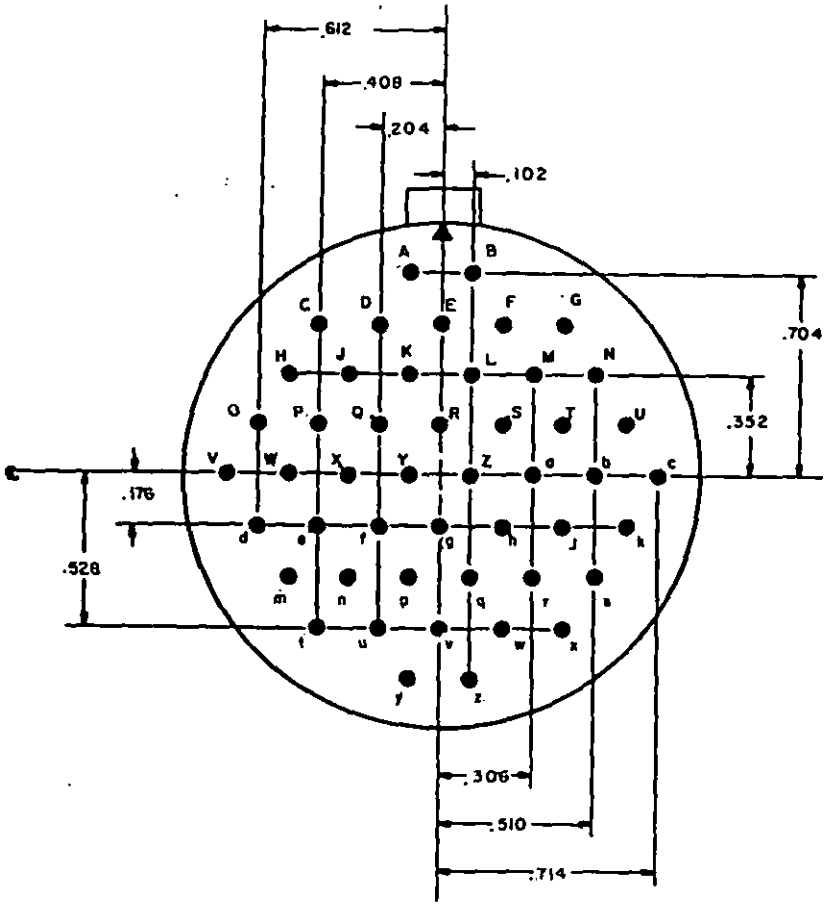
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	125	235	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-9	A	14	16	●	All	MS33690-9
			14	12	⊕		
			2	8	◕		
			1	4	◔		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



FOR OTHER INSERT POSITIONS SEE 36-11 AND 36-12.

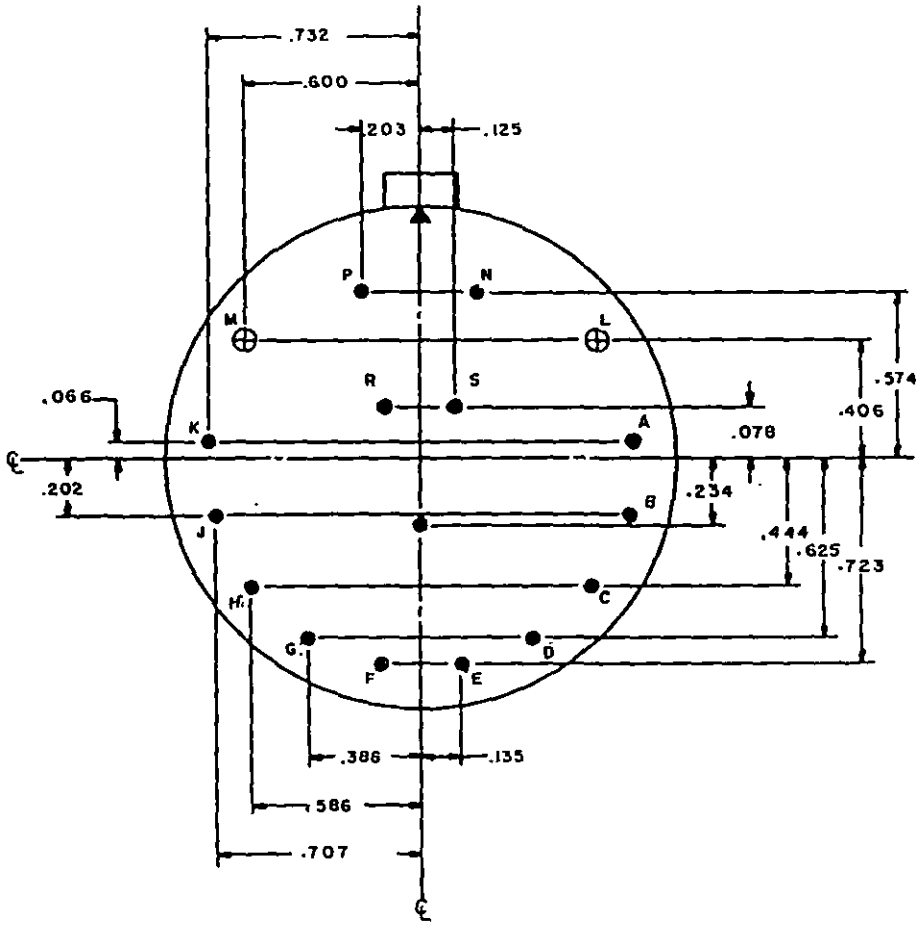
ALTERNATE INSERT POSITIONS

A DEGREES			
W	X	Y	Z
80	125	235	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-10	A	48	16	●	All	MS33690-10

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR NEW DESIGN
AFTER 17 DEC 1958



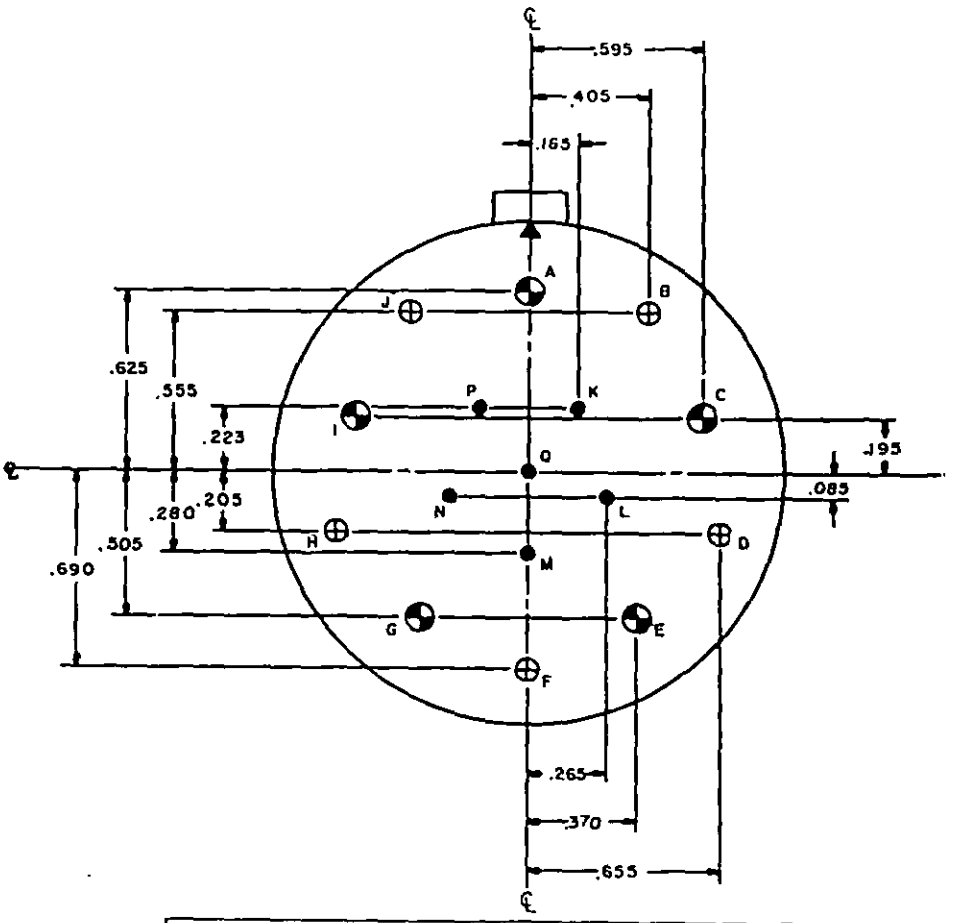
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-13	E	3	16	●	N,P,Q	MS33690-13
		A	12			All others	
			2	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

INACTIVE FOR NEW DESIGN
AFTER 17 DEC 1958

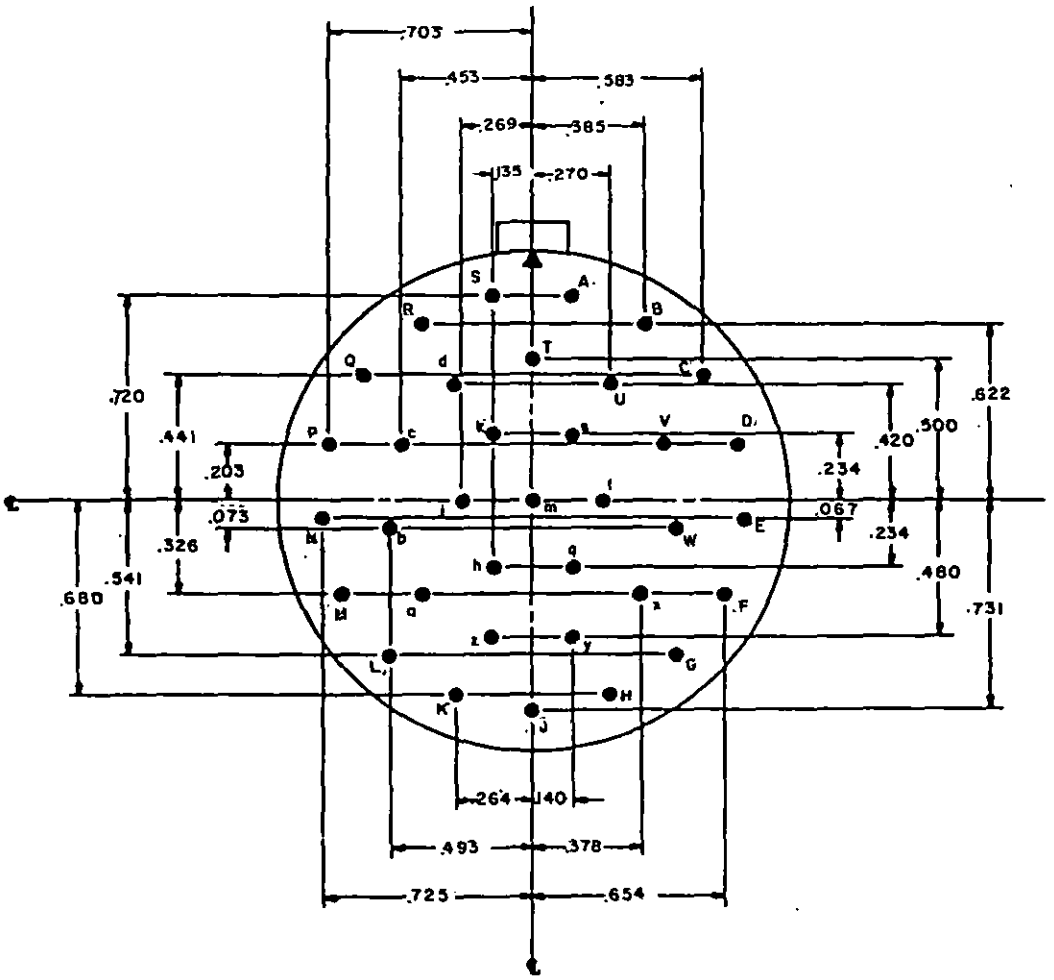


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
90	180	270	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-14	D	6	16	●	All	MS33690-14
			5	12	⊕		
			5	8	◕		

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Check the source to verify that this is the current version before use.

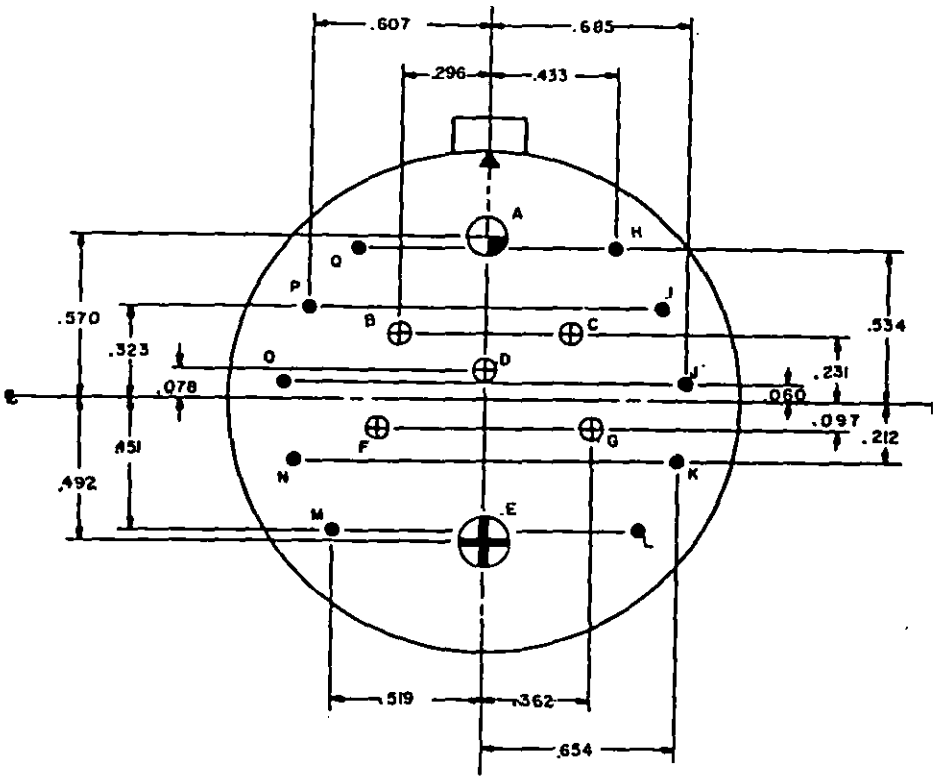


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
60	125	245	305

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-15	D	35	16	●	M	MS33690-15
		A				All others	

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INACTIVE FOR NEW DESIGN
AFTER 17 DEC 1958



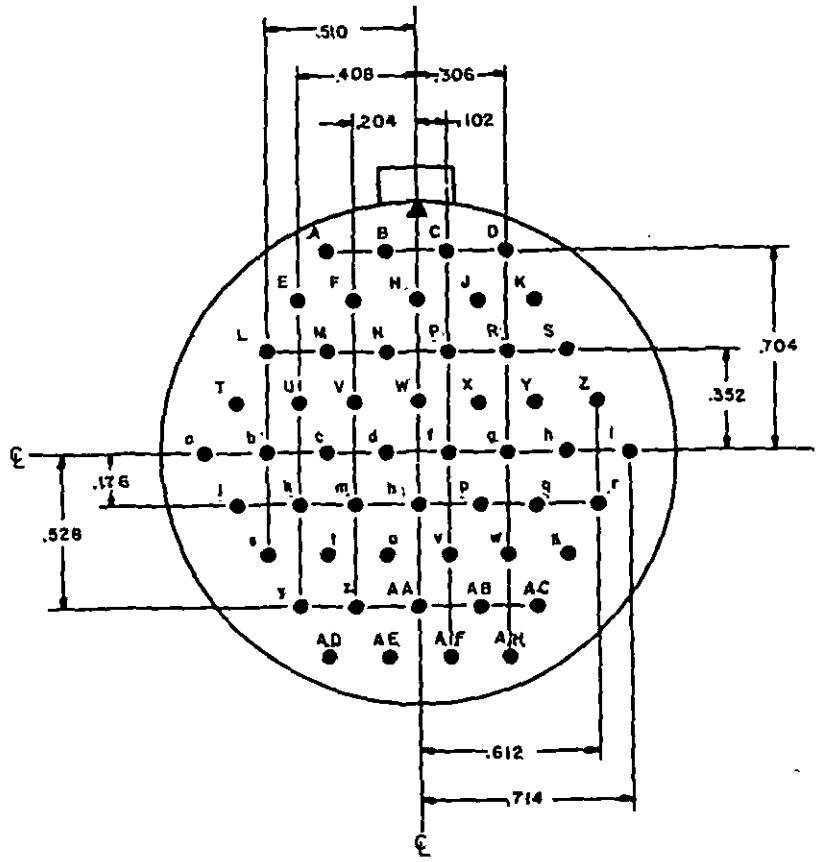
(Not applicable to MIL-C-22992)

ALTERNATE INSERT POSITIONS

A DEGREES			
W	X	Y	Z
80	110	250	280

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-19	D	10	16	○	All	MS33690-19
			5	12	⊕		
			1	4	◒		
			1	0	⊕		

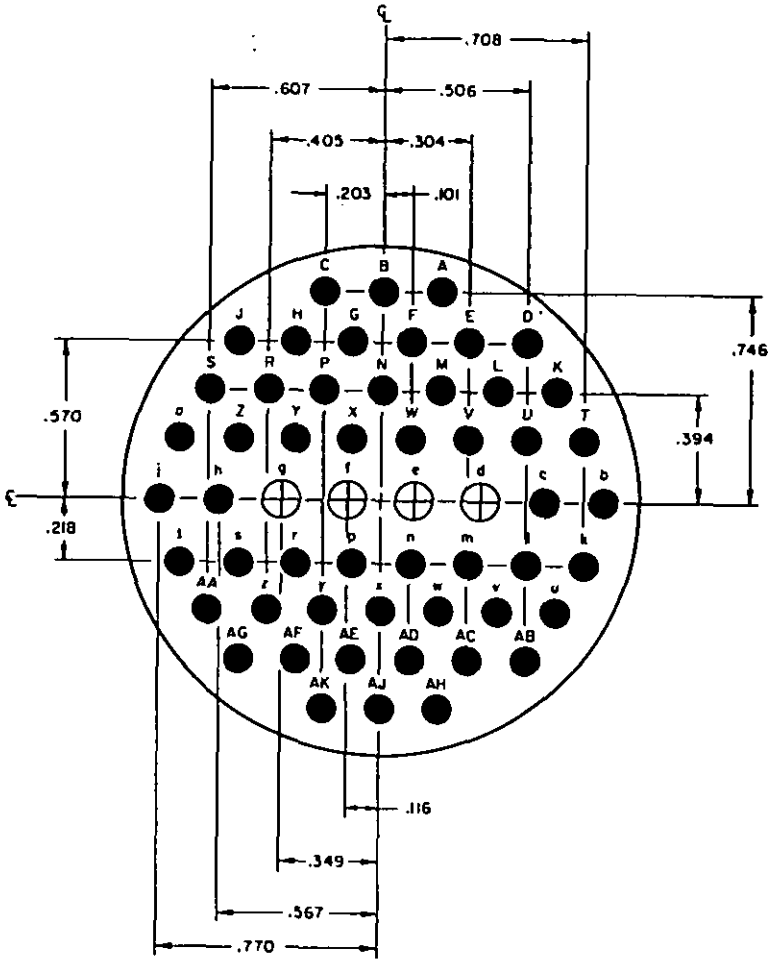
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
72	144	216	288

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-52	A	52	16	●	All	MS33690-52

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ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
110	250	260	280

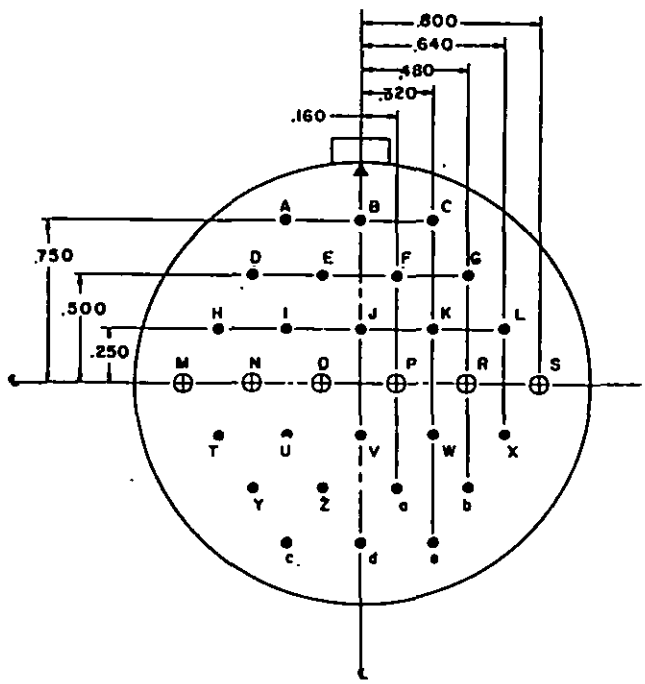
Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-66	A	52	16	●	All	MS33690-66
			4	12	⊕		

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SECTION 1300

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

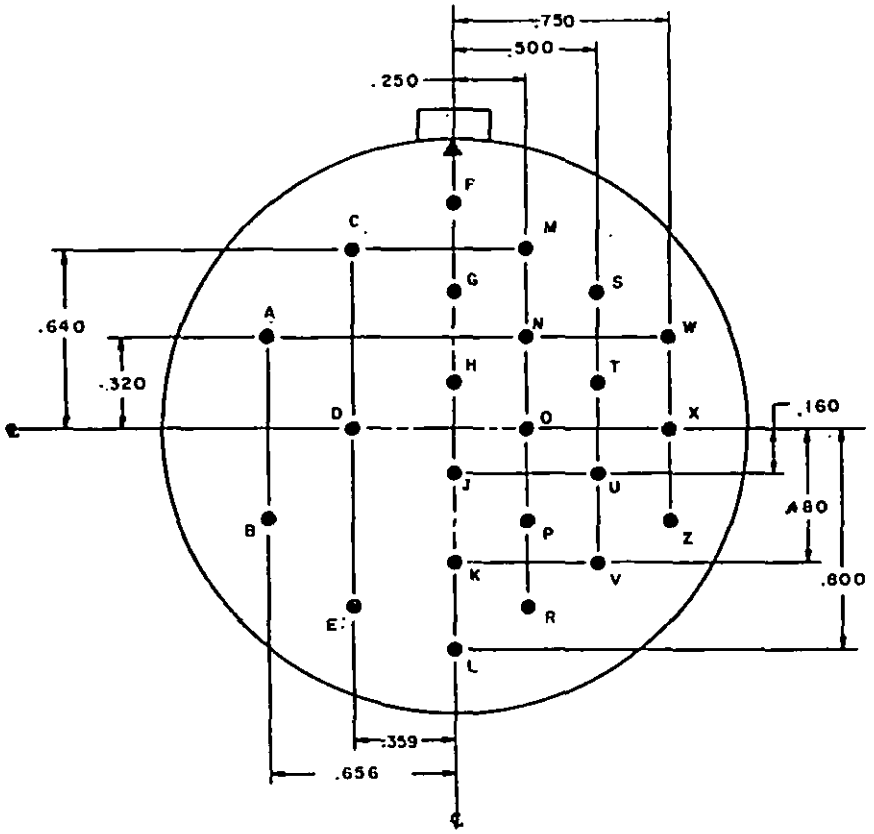
SHELL SIZE 40



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
65	130	235	300

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-1	D	24	16	●	All	MS33691-1
			6	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



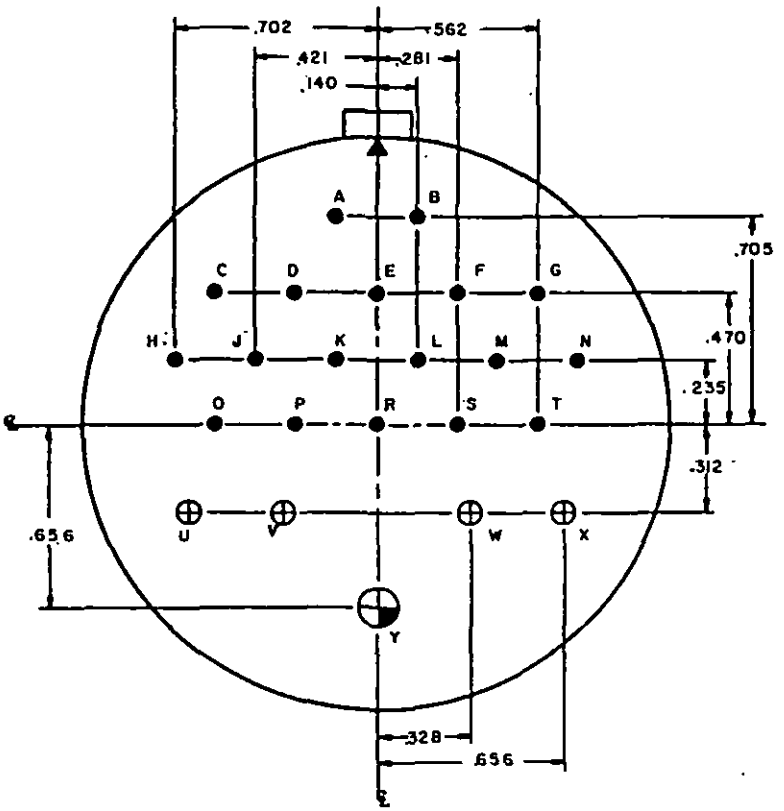
FOR OTHER INSERT POSITION SEE 40-13.

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-2	B	5	16	●	A,B,C,D,E	MS33691-2
		D	18			All others	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

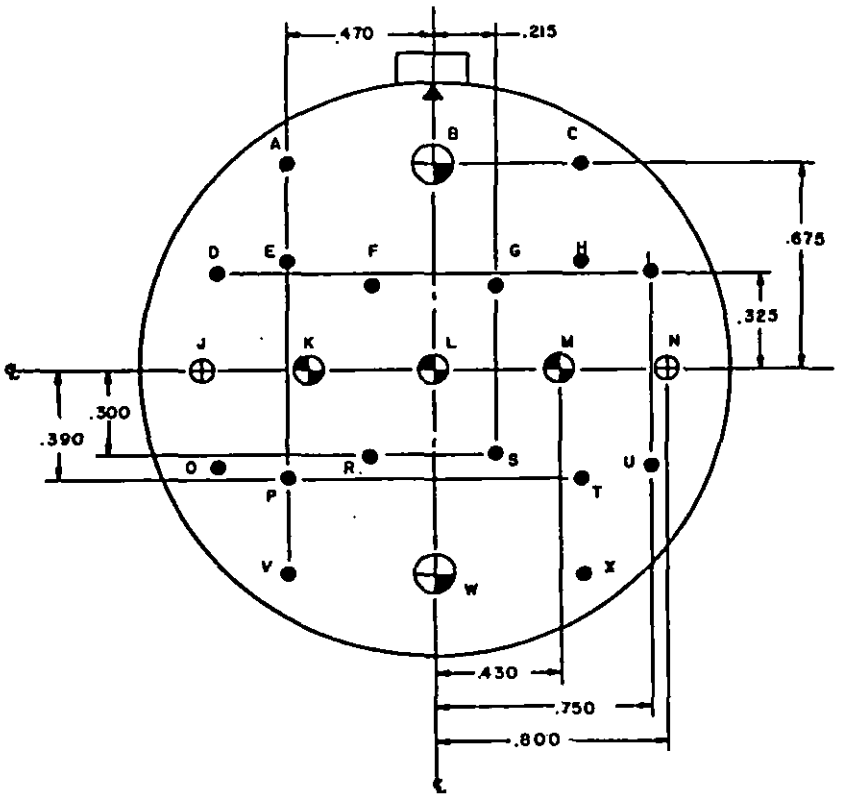


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-3	D	18	16	●	All	MS33691-3
			4	12	⊕		
			1	4	◒		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

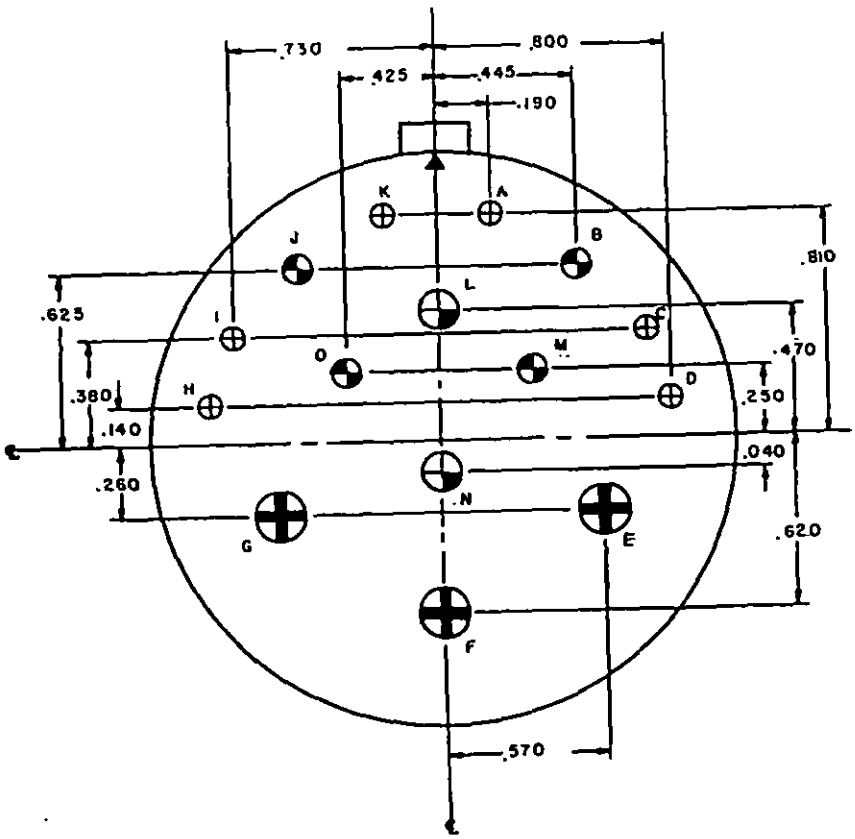


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-4	D	16	16	●	All	MS33691-4
			2	12	⊕		
			3	8	◕		
			2	4	⊕		

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Check the source to verify that this is the current version before use.

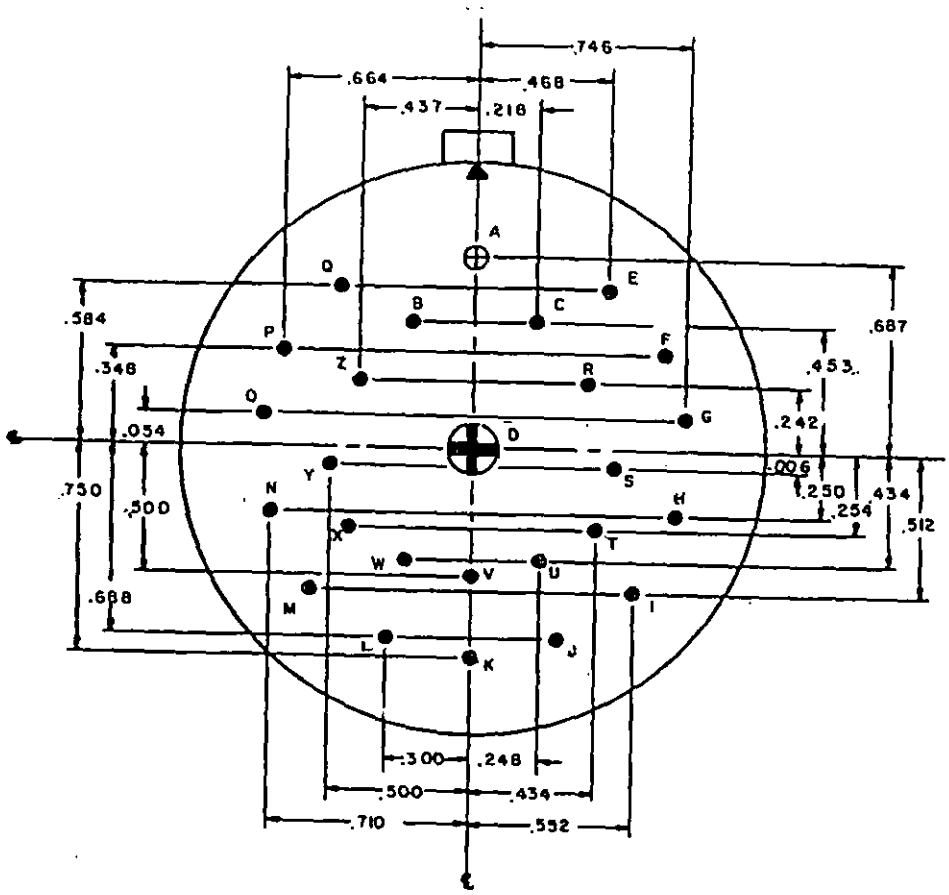


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-5	A	6	12	⊕	All	MS33691-5
			4	8	◕		
			2	4	◑		
			3	0	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

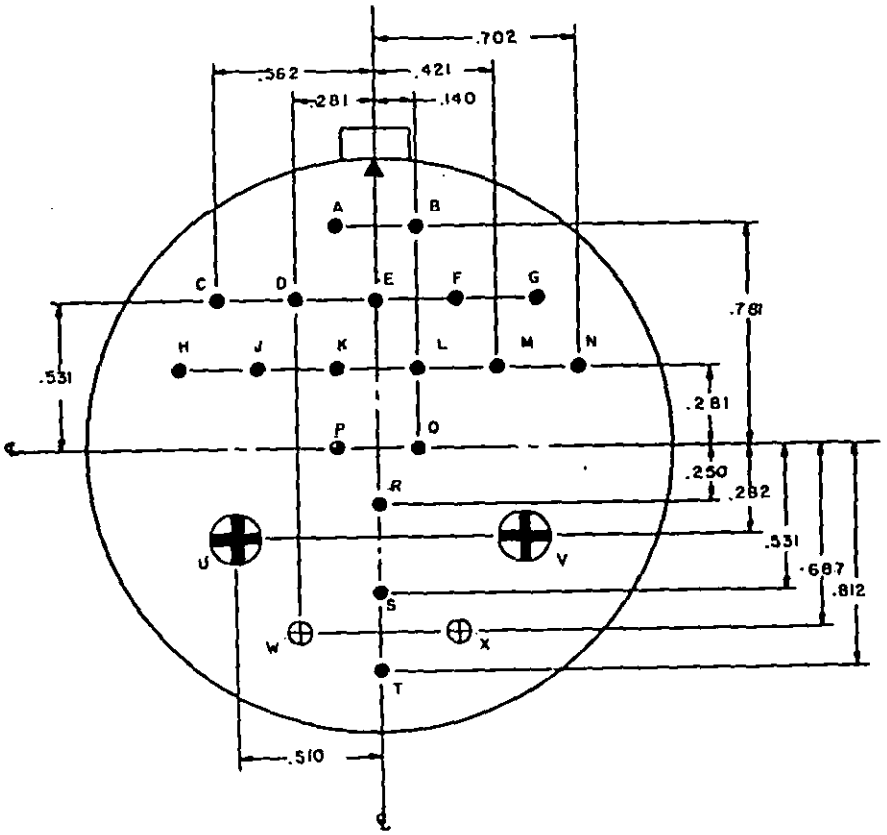


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-6	D	24	16	●	All	MS33691-6
			1	12	⊕		
			1	0	⊕		

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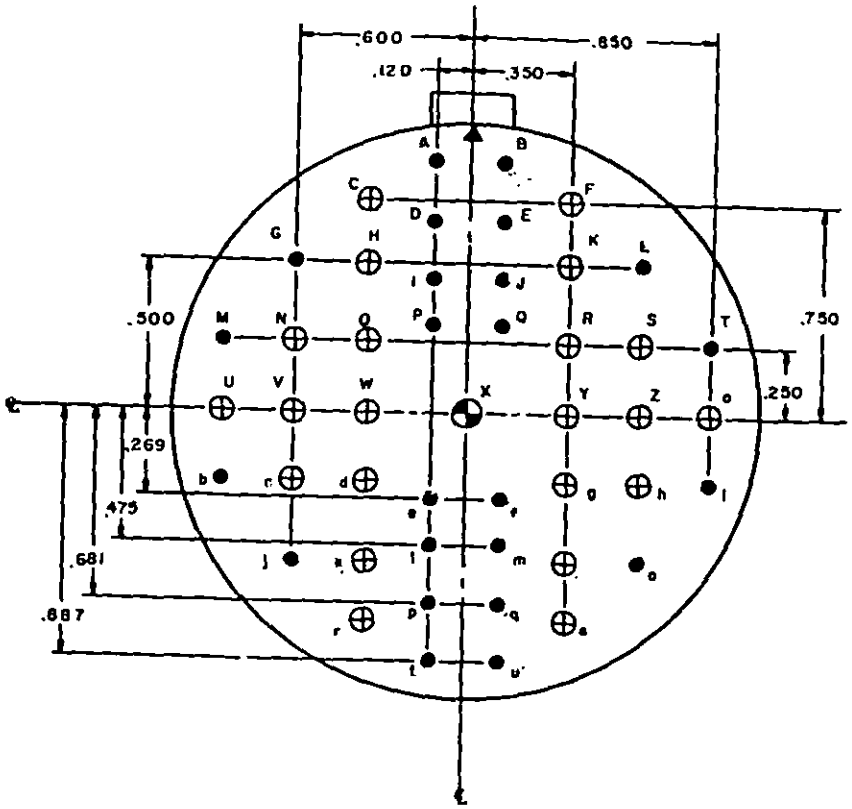


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-7	A	2	12	⊕	W,X	MS33691-7
			2	0	⊕	U,V	
			2	16	●	P,Q	
		D	16			All others	

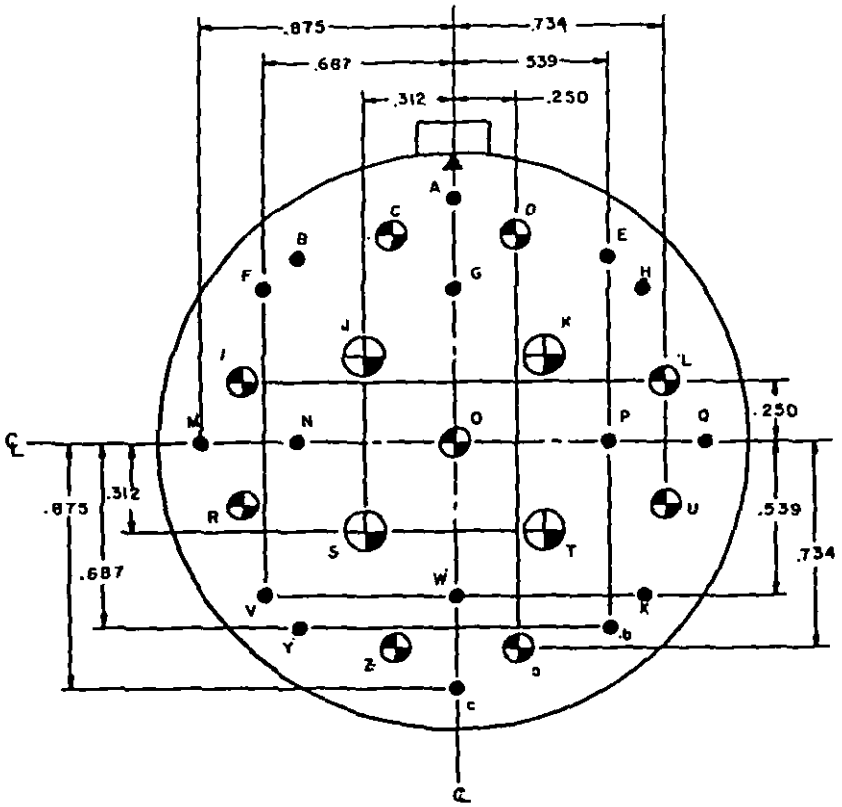
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
65	125	225	310

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-9	A	24	16	●	All	MS33691-9
			22	12	⊕		
			1	8	◒		

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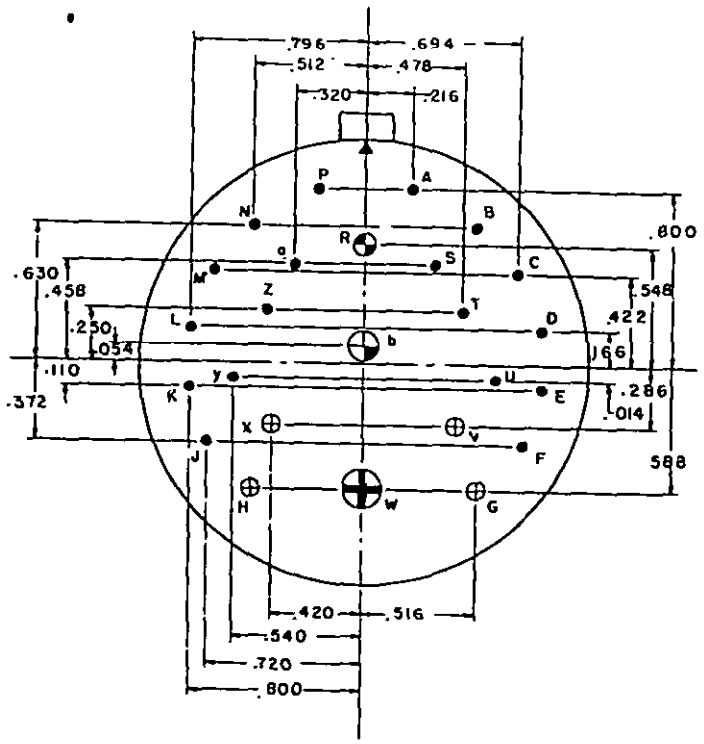


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
65	125	225	310

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-10	A	16	16	●	All	MS33691-10
			9	8	◕		
			4	4	⊕		

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Check the source to verify that this is the current version before use.

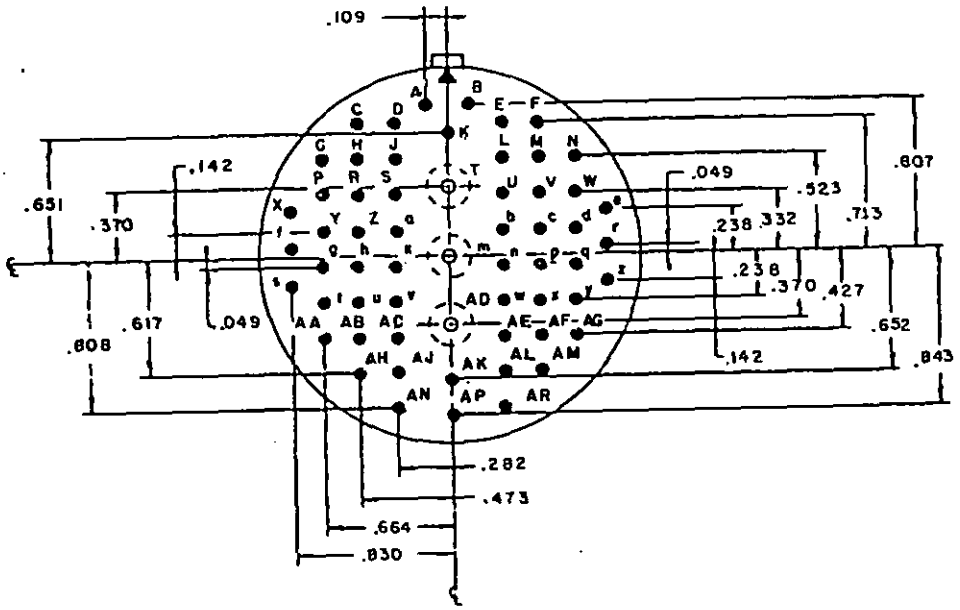


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
80	110	250	280

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-11	0	18	16	●	All	MS33691-11
			4	12	⊕		
			1	8	◒		
			1	4	◒		
			1	0	⊕		

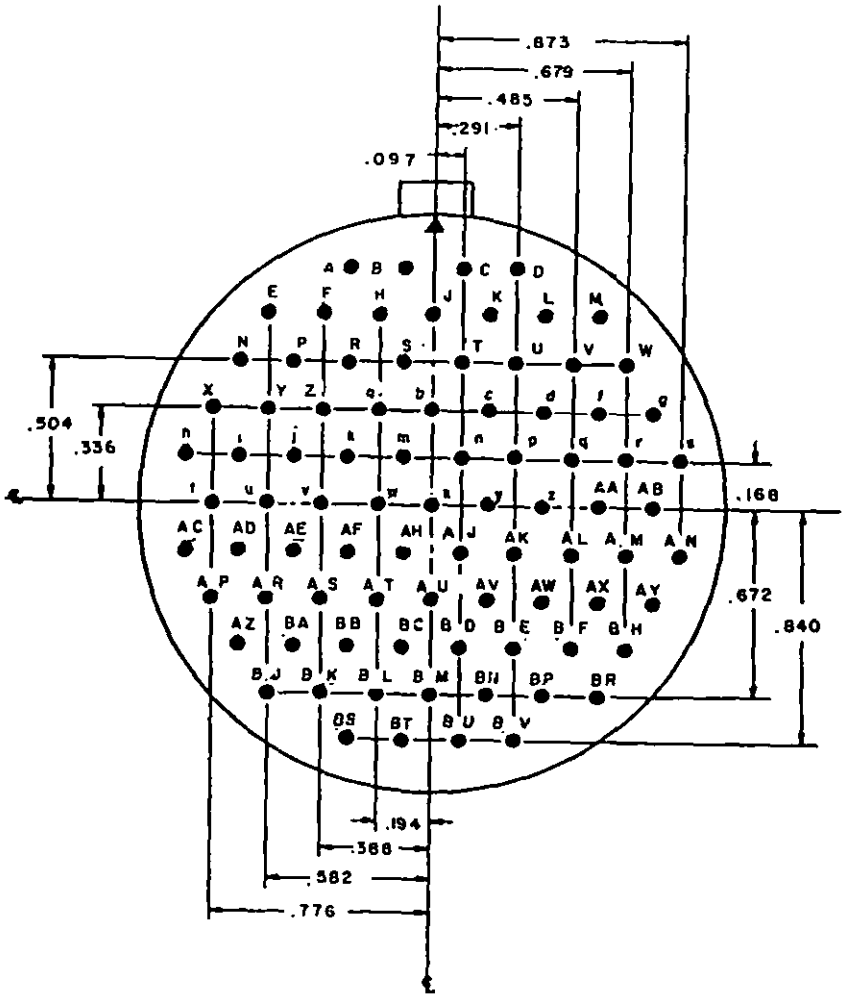
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-15		3	8 (shielded) MS14056-2	◯	T,m,AD	MS33691-15
		INST	57	16	●	All others	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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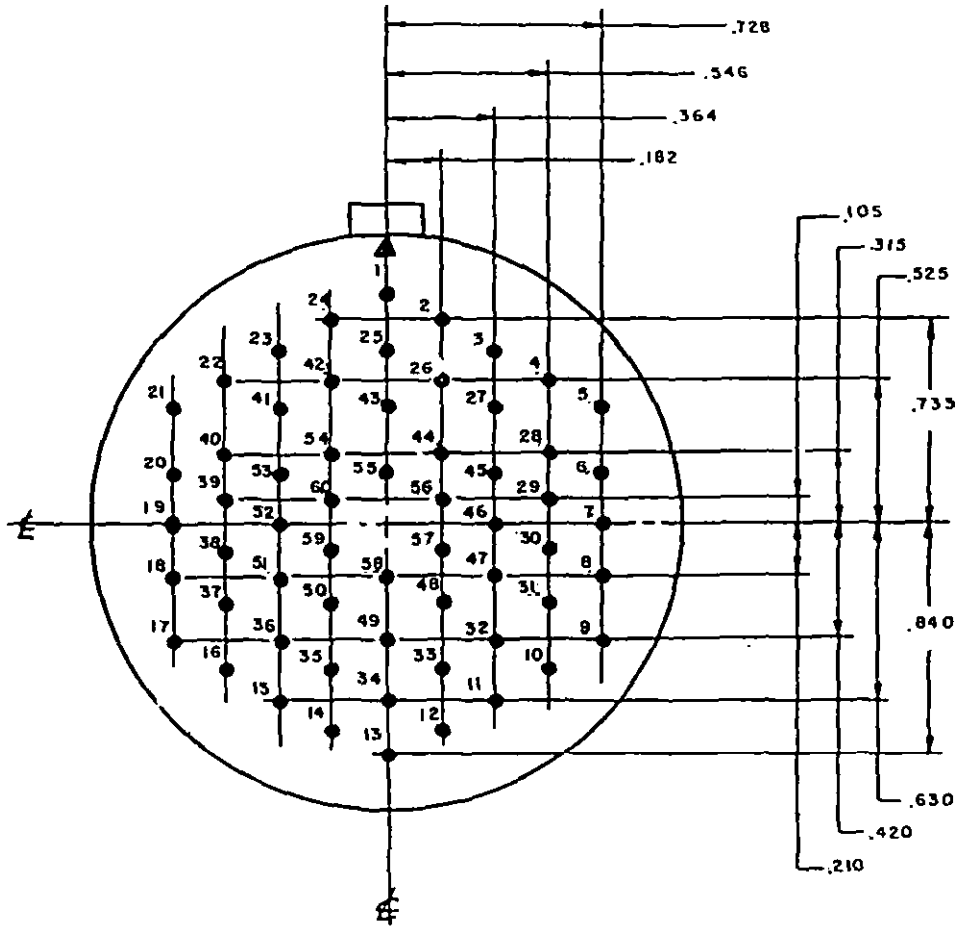


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
72	144	216	288

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-56	A	85	16	●	All	MS33691-56

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NOT APPLICABLE FOR DESIGN OF
AIR FORCE AIRBORNE EQUIPMENT.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
30	130	220	290

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-62	A	60	16	●	All	MS33691-62

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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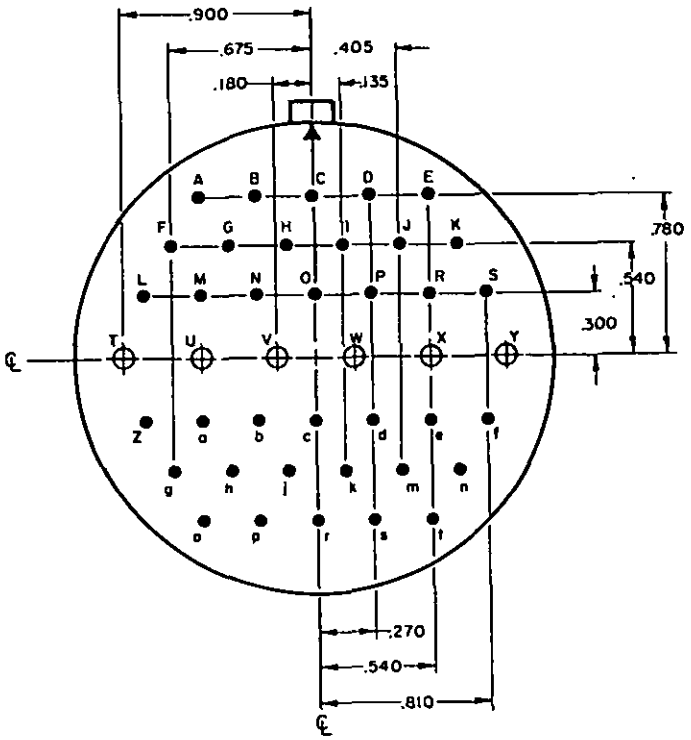
SECTION 1400

INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

SHELL SIZE 44

NOTE:

1. Shell size 44 applies to MIL-C-5015, except arrangement number 52 applies also to MIL-C-22992.



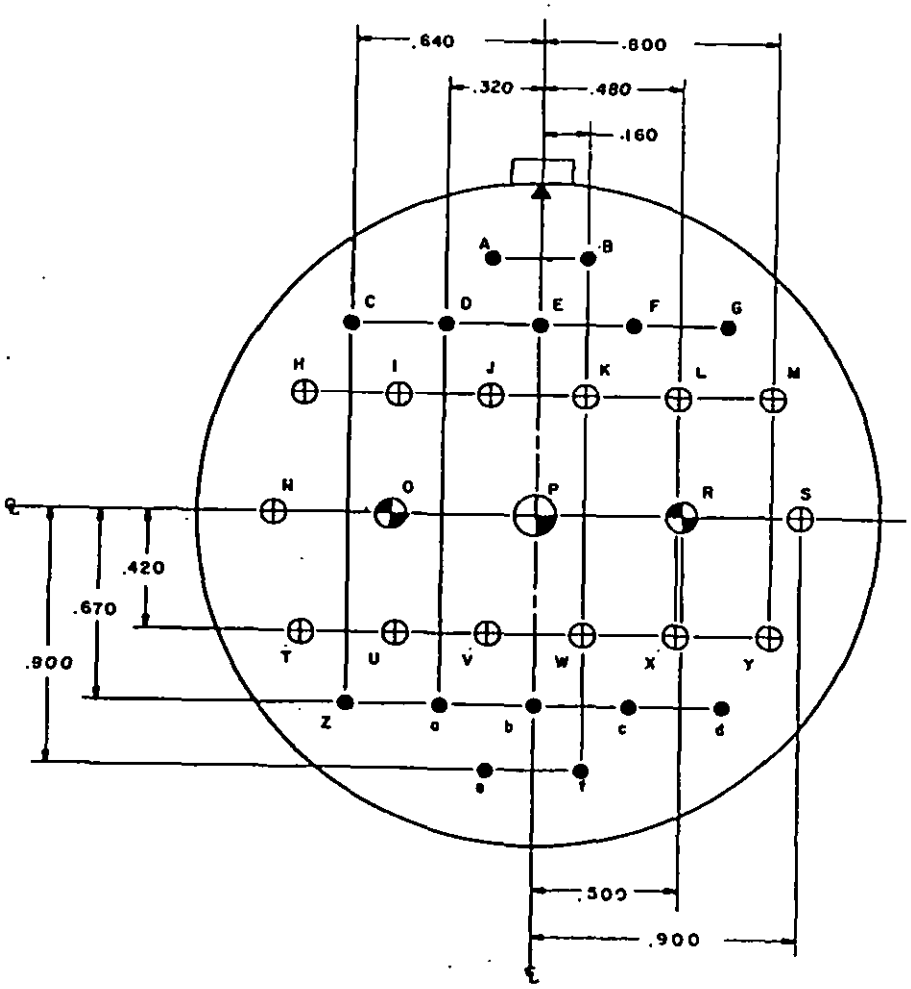
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
65	125	225	310

FOR OTHER INSERT POSITIONS SEE 44-5 AND 44-6.

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
44	-1	D	36	16	●	All	MS33692-1
			6	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

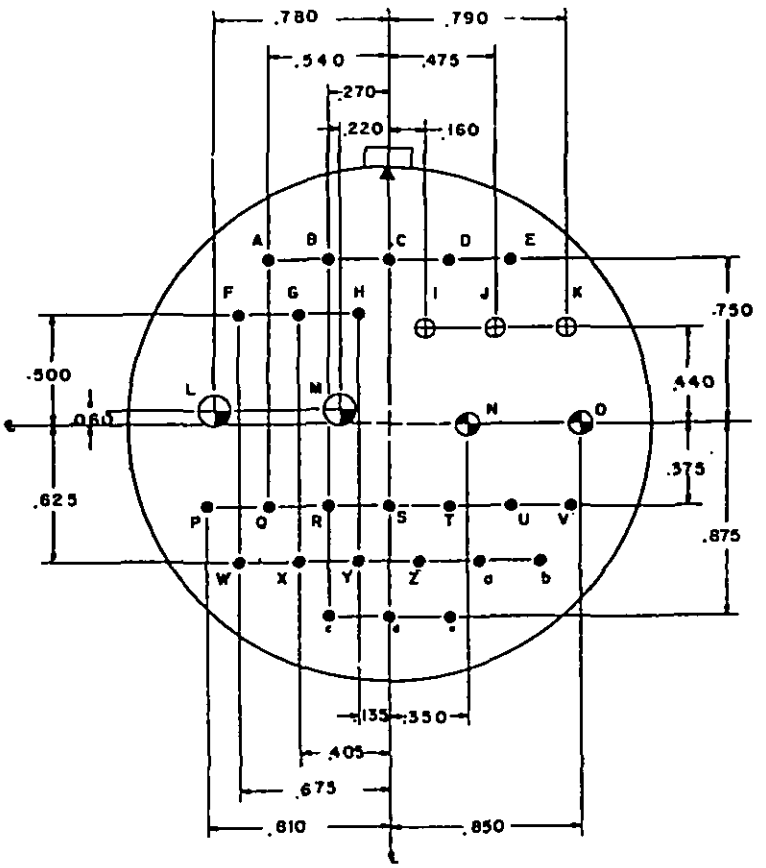


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
65	125	225	310

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
44	-2	D	14	16	●	A11	MS33692-2
			14	12	⊕		
			2	8	◕		
			1	4	◔		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

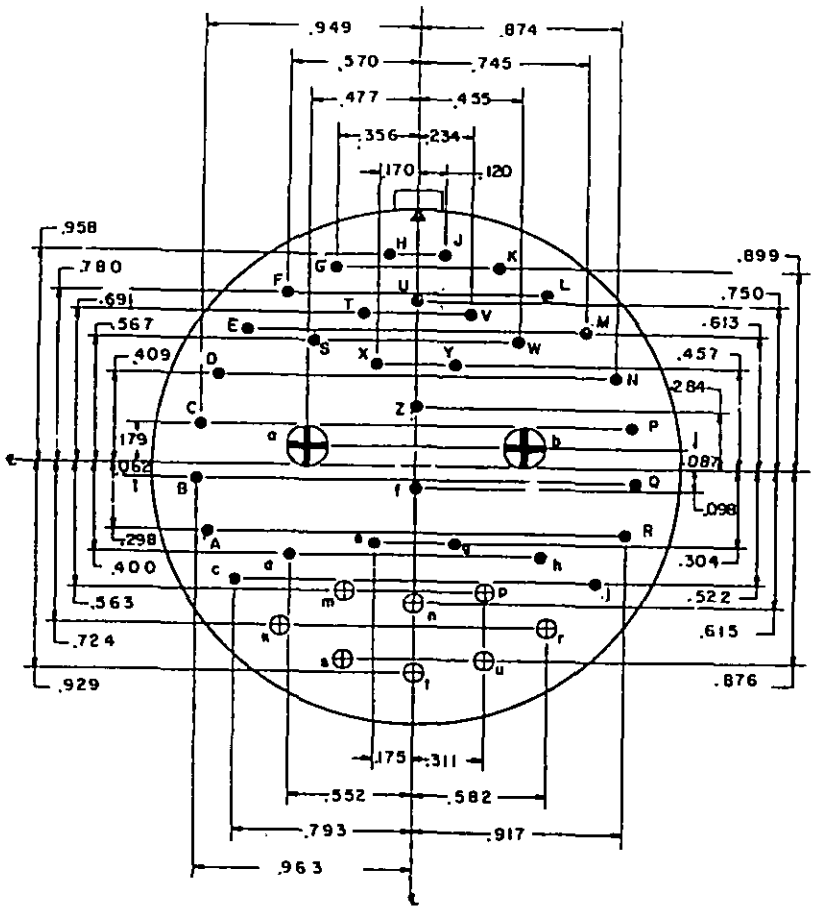


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
65	125	225	310

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
44	-3	D	24	16	●	All	MS33692-3
			3	12	⊕		
			2	8	◒		
			2	4	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

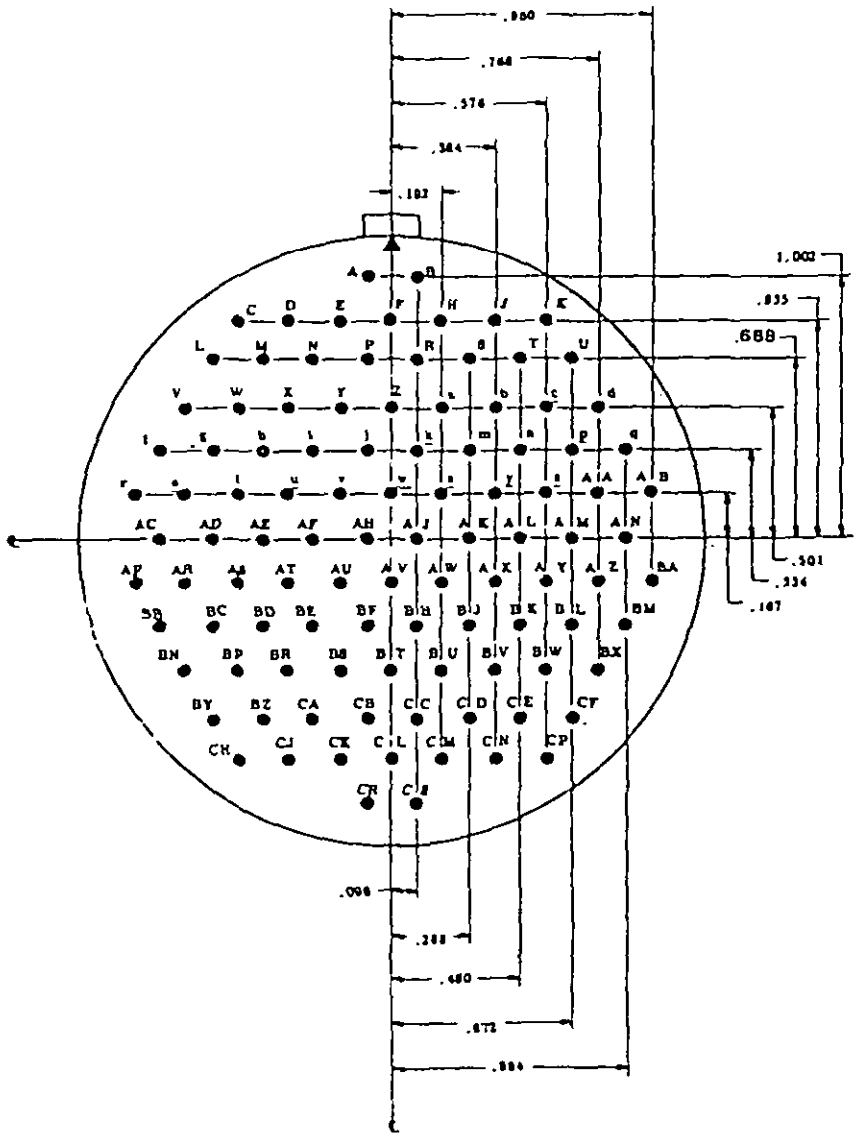


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
65	125	225	310

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
44	-4	D	31	16	●	All	MS33692-4
			8	12	⊕		
			2	0	⊕		

Source: https://assist.dla.mil - Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
72	135	225	288

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
44	-52	A	104	16	●	All	MS33692-52

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T07:50Z
Check the source to verify that this is the current version before use.

SECTION 1500

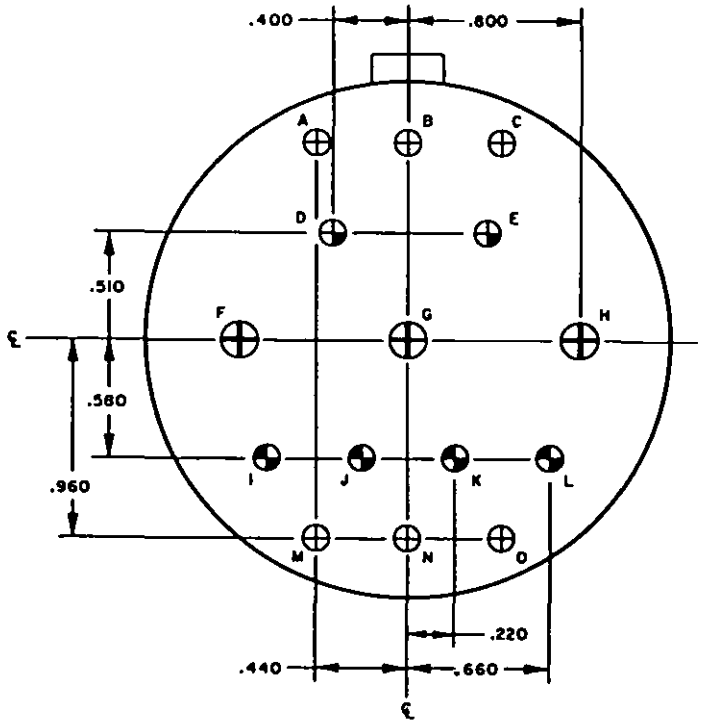
INSERT ARRANGEMENT, ELECTRICAL CONNECTORS

SHELL SIZE 48

NOTE:

1. Shell size 48 applies to MIL-C-5015 only.

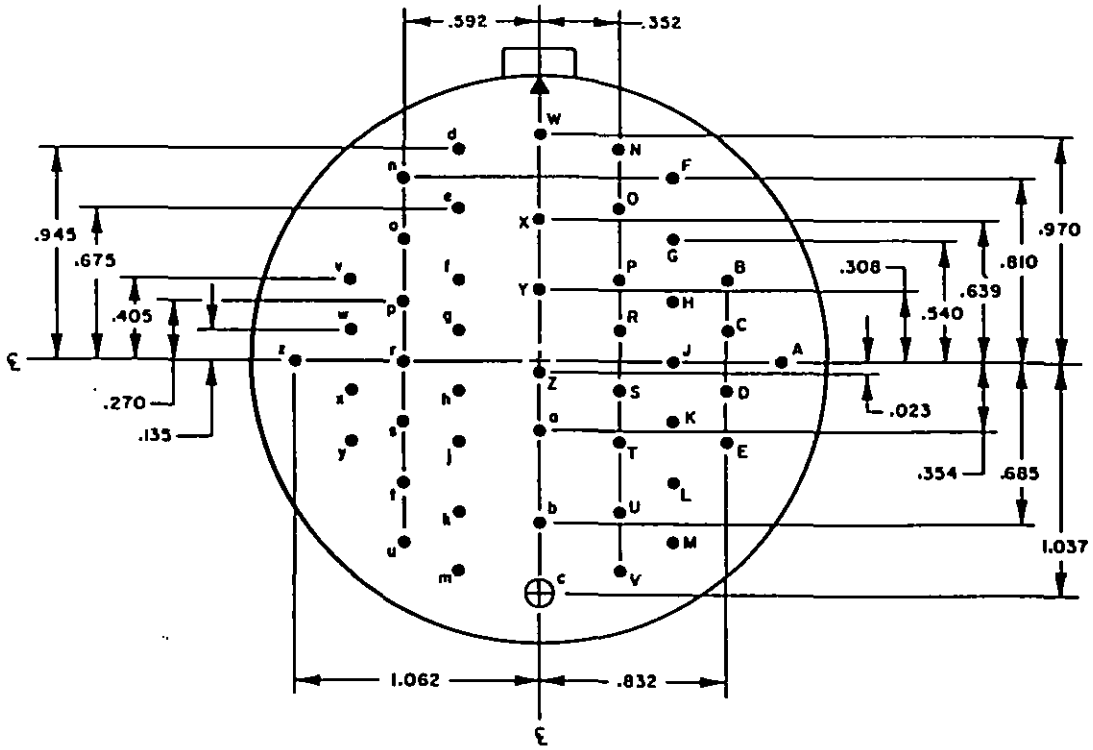
Source: https://assist.dla.mil -- Downloaded: 202610-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
65	125	225	310

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
48	-1	E	2	12	⊕	B,N	MS33693-1
		D	4			A,C,M,O	
			4	8	◒	I,J,K,L	
			2	4	⊕	D,E	
			3	0	⊕	F,G,H	

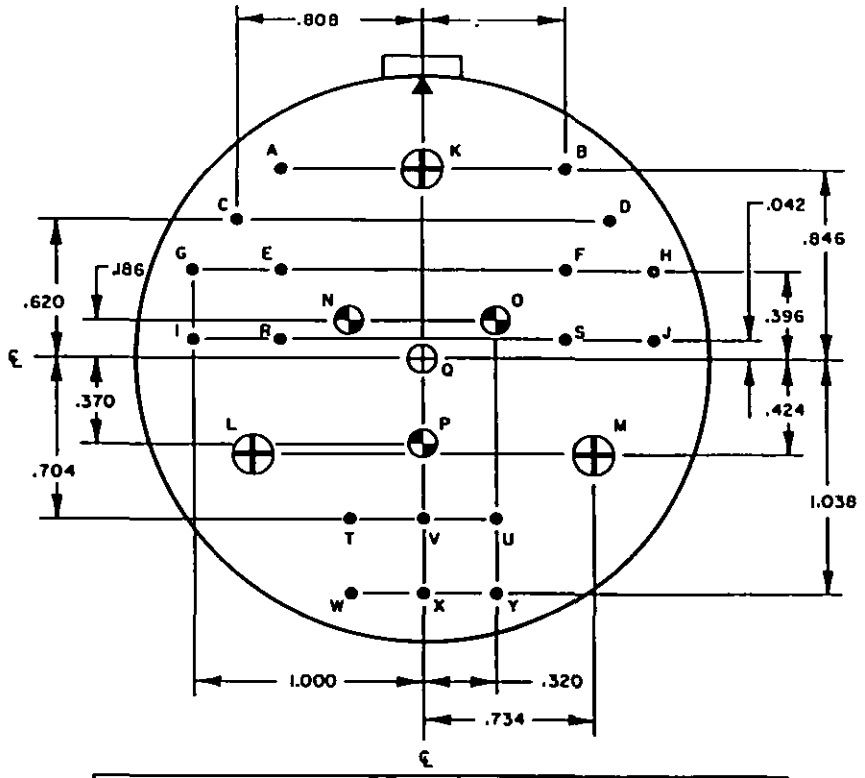
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
65	125	225	310

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
48	-2	D	1	12	⊕	C	MS33693-2
		E	6	16	●	W,X,Y,Z a, b	
		D	40			All others	

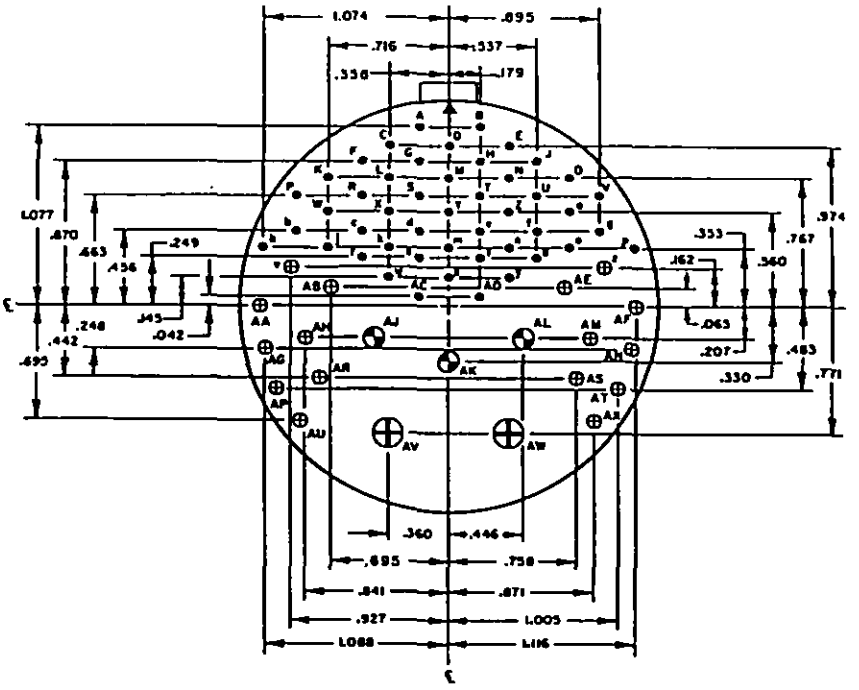
Source: https://assist.dla.mil -- Downloaded: 202610-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
65	125	225	310

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
48	-3	D	18	16	●	All	MS33693-3
			1	12	⊕		
			3	8	◒		
			3	0	⊕		

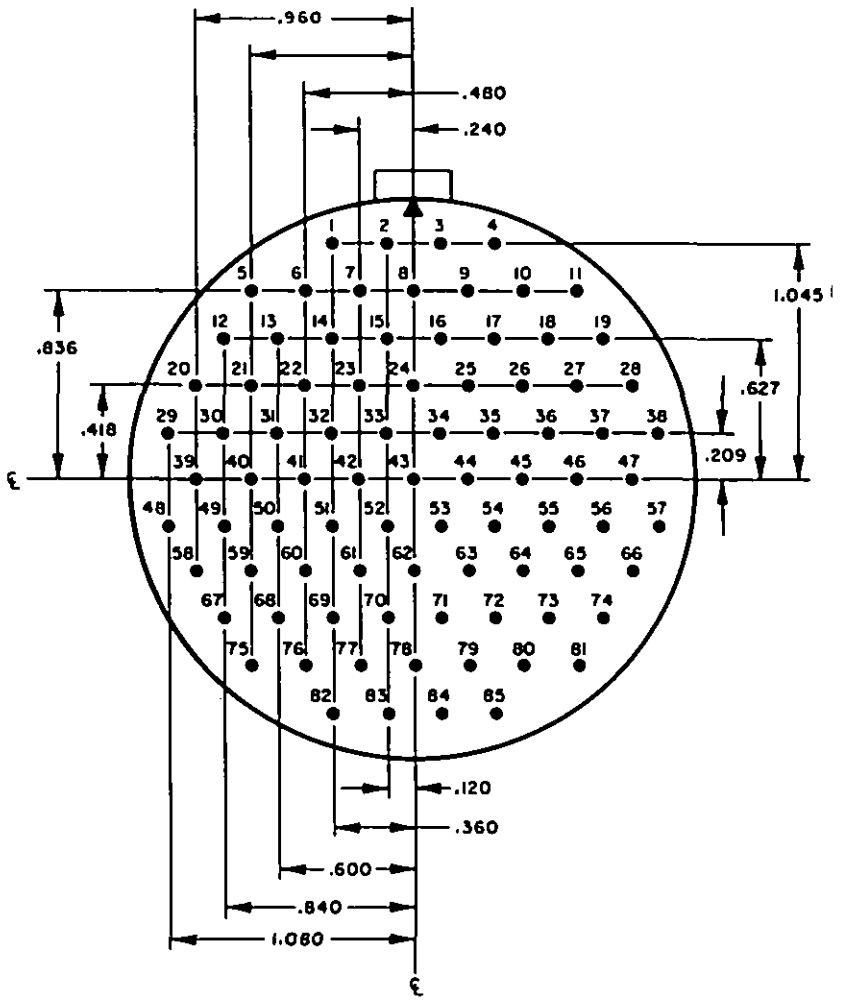
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
65	125	225	310

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
48	-4	D	3	8	◔	AJ,AK,AL	MS33693-4
			2	0	⊕	AV,AW	
		-A	47	16	●	All others.	
			16	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:50Z
Check the source to verify that this is the current version before use.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
72	144	216	288

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
48	-62	D	85	16	●	A11	MS33693-62

Source: https://assist.dla.mil - Downloaded: 2026-10-26T07:50Z
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APPENDIX B

INSERT ARRANGEMENTS CANCELLED BY REVISION A TO MIL-STD-1651

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APPENDIX B

INSERT ARRANGEMENTS CANCELLED BY REVISION A TO MIL-STD-1651

10. SCOPE

10.1 Scope. This appendix gives the data necessary for qualification of all insert arrangements cancelled by Revision A of MIL-STD-1651 Insert Arrangements for all MIL-C-5015, MIL-C-22992 (Classes C, J, R) and MIL-C-83723 (Series II) Electrical Connectors. Tables II and III list the insert arrangements cancelled by Revision A to MIL-STD-1651 along with their replacements when applicable. This appendix is a mandatory part of this standard. The information contained herein is intended for compliance.

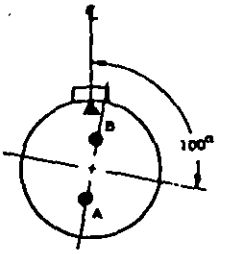
20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. DEFINITIONS

30.1 The terms used in this standard are generally accepted by the electrical and electronics industries and commonly used in electrical connector engineering practice.

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 12S, 13-1 IS CANCELLED, WITHOUT REPLACEMENT, AFTER 30 APRIL 1992. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



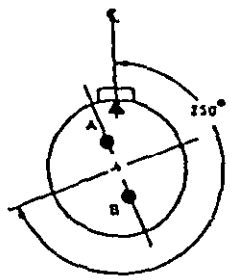
CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 12S-3 EXCEPT FOR POSITION OF INSERT IN SHELL.

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
12S, 13	-1	A	2	16 16S	●	All	MS33681-1

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* INSERT ARRANGEMENT 12S, 13-2 IS CANCELLED, WITHOUT REPLACEMENT, AFTER 30 APRIL 1992. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



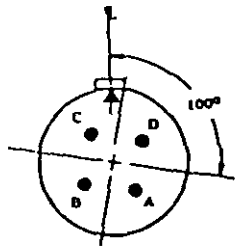
CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 12S-3 EXCEPT FOR POSITION OF INSERT IN SHELL.

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
12S, 13	-2	A	2	16 16S	●	All	MS33681-2

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* INSERT ARRANGEMENT 14S, 15-10 IS CANCELLED, WITHOUT REPLACEMENT, AFTER 30 APRIL 1992. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.

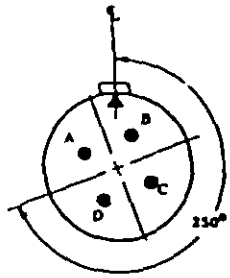


CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 14S-2 EXCEPT FOR POSITION OF INSERT IN SHELL.

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
14S, 15	-10	INST	4	16 16S	●	All	MS33682-10

* INSERT ARRANGEMENT 14S, 15-11 IS CANCELLED, WITHOUT REPLACEMENT, AFTER 30 APRIL 1992. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



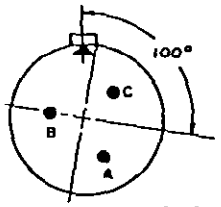
CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 14S-2 EXCEPT FOR POSITION OF INSERT IN SHELL.

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
14S, 15	11	INST	4	16 16S	●	All	MS33682-11

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 14S, 15-12 IS CANCELLED, WITHOUT REPLACEMENT, AFTER 30 APRIL 1992. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.

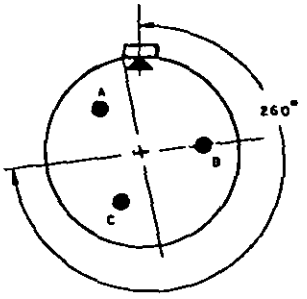


CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 14S-1 EXCEPT FOR POSITION OF INSERT IN SHELL

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
14S, 15	-12	A	3	16 16S	●	A11	MS33682-12

* INSERT ARRANGEMENT 14S, 15-13 IS CANCELLED, WITHOUT REPLACEMENT, AFTER 30 APRIL 1992. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



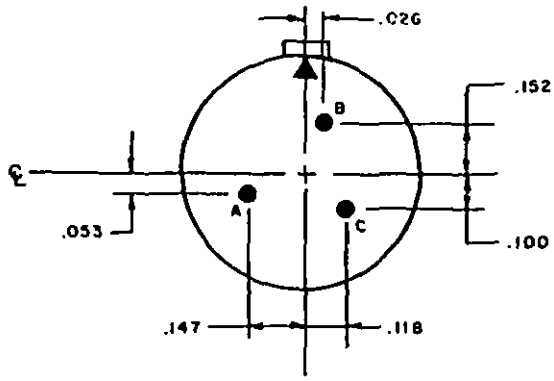
CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 14S-1 EXCEPT FOR POSITION OF INSERT IN SHELL

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
14S, 15	13	A	3	16 16S	●	A11	MS33682-13

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Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 16S, 17-14 IS CANCELLED, WITHOUT REPLACEMENT, AFTER 30 APRIL 1992. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.

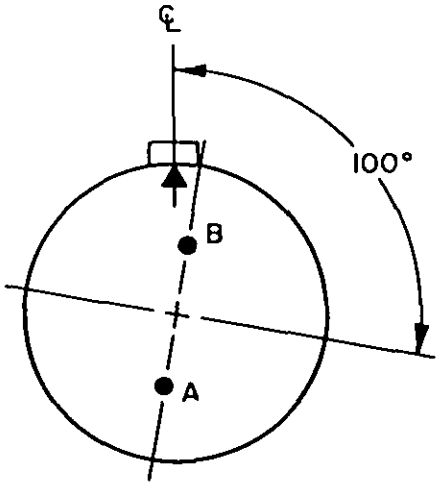


ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16S, 17	-14	D	3	16 16S	●	All	MS33683-14

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 16S, 17-15 CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 16S,17-4 FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



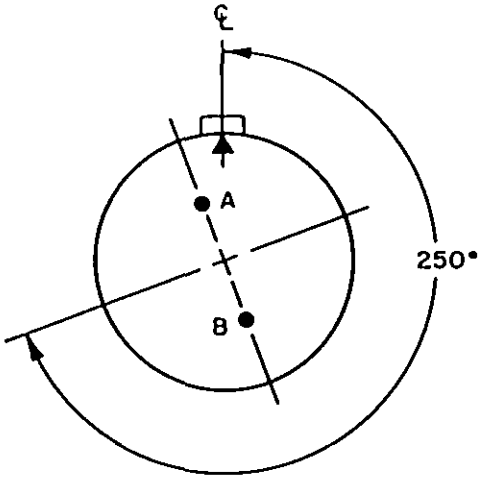
CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 16S-4 EXCEPT FOR POSITION OF INSERT IN SHELL

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16S, 17	-15	D	2	16 16S	●	All	MS33683-15

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 16S, 17-16 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 16S, 17-4. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



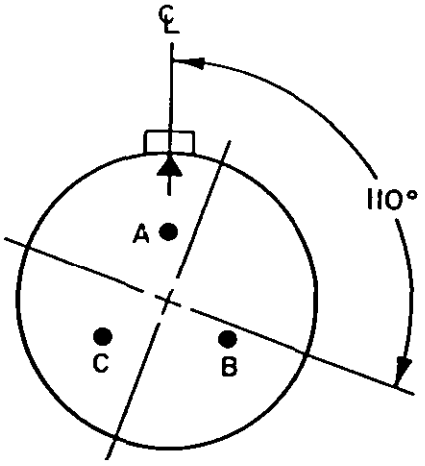
CONTACT ARRANGEMENT AND DIMENSIONS
IDENTICAL TO 16S-4 EXCEPT FOR POSITION
OF INSERT IN SHELL

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16S, 17	-16	D	2	16 16S	●	All	MS33683-16

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 16S, 17-17 IS CANCELLED, WITHOUT REPLACEMENT, AFTER 30 APRIL 1992. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



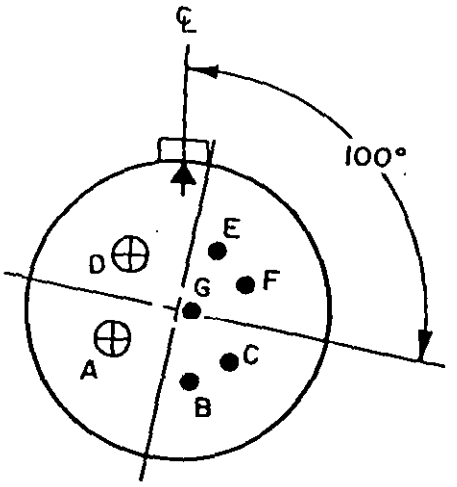
CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 16S-14 EXCEPT FOR POSITION OF INSERT IN SHELL

ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
16S, 17	-17	D	3	16 16S	●	All	MS33683-17

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 18-17 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 18-9. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 18-9 EXCEPT FOR POSITION OF INSERT IN SHELL.

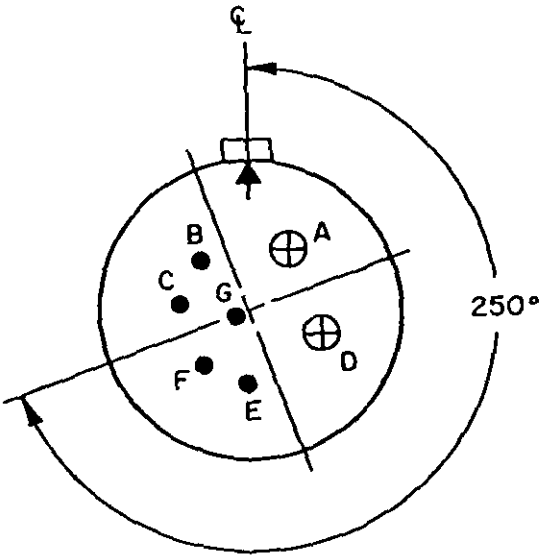
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-17	INST	5	16	●	All	MS33684-17
			2	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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* INSERT ARRANGEMENT 18-18 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 18-9. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 18-9 EXCEPT FOR POSITION OF INSERT IN SHELL.

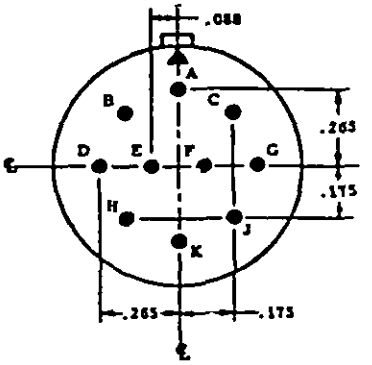
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-18	INST	5	16	●	All	MS33684-18
			2	12	⊕		

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* INSERT ARRANGEMENT 18-19 IS CANCELLED, WITHOUT REPLACEMENT, AFTER 30 APRIL 1992. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



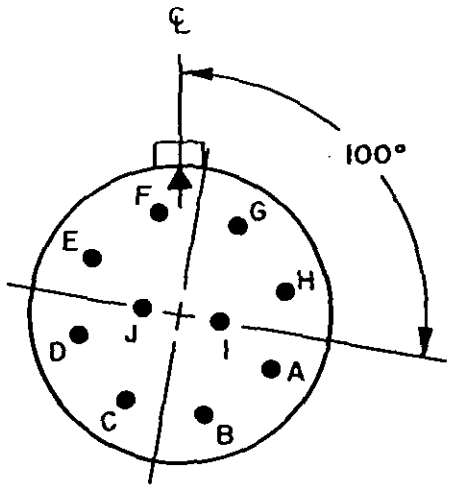
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	120	240	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-19	A	10	16	●	All	MS33684-19

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 18-23 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 18-1. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 18-1 EXCEPT FOR POSITION OF INSERT IN SHELL.

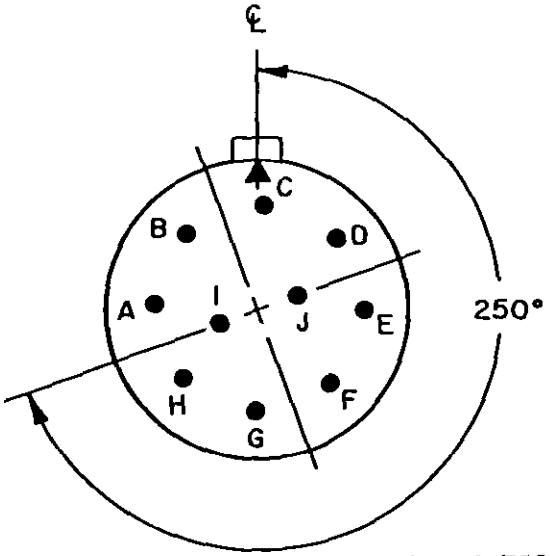
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-23	A	4	16	●	B,C,F,G	MS33684-23
		INST	6			All others	

Source: https://assist.dla.mil/-- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 18-24 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 18-1. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 18-1 EXCEPT FOR POSITION OF INSERT IN SHELL.

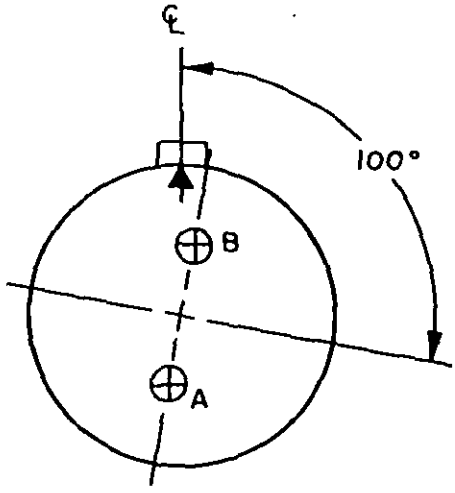
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-24	A	4	16	●	B,C,F,G	MS33684-24
		INST	6			All others	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 18-25 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 18-3. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 18-3 EXCEPT FOR POSITION OF INSERT IN SHELL.

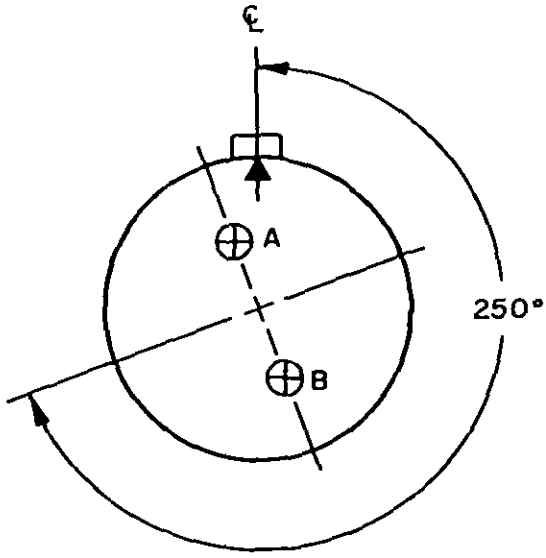
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-25	D	2	12	⊕	All	MS33684-25

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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* INSERT ARRANGEMENT 18-26 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 18-3. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 18-3 EXCEPT FOR POSITION OF INSERT IN SHELL.

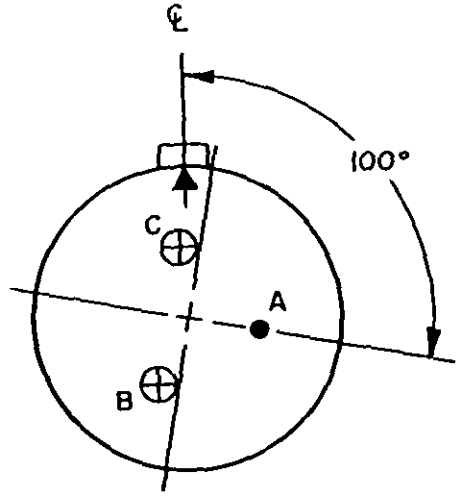
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-26	D	2	12	⊕	All	MS33684-26

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 18-27 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 18-5. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 18-5 EXCEPT FOR POSITION OF INSERT IN SHELL.

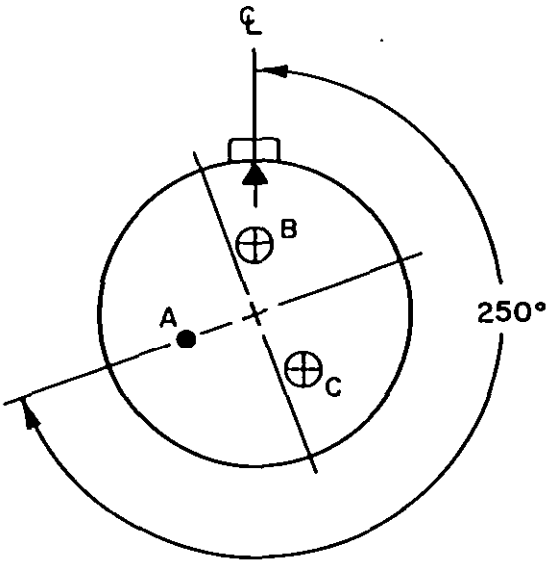
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-27	D	1	16	●	A11	MS33684-27
			2	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 18-28 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 18-5. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 18-5 EXCEPT FOR POSITION OF INSERT IN SHELL.

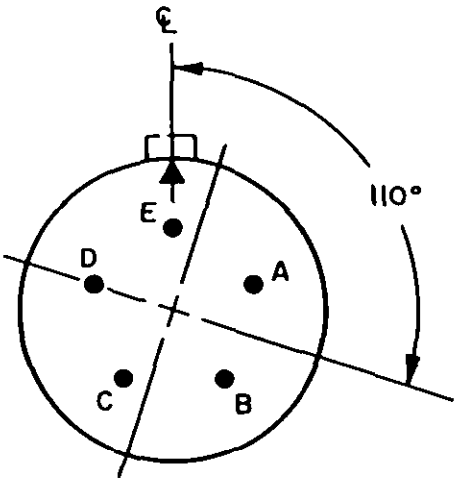
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-28	D	1	16	●	All	MS33684-28
			2	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 18-30 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 18-20. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 18-20 EXCEPT FOR POSITION OF INSERT IN SHELL.

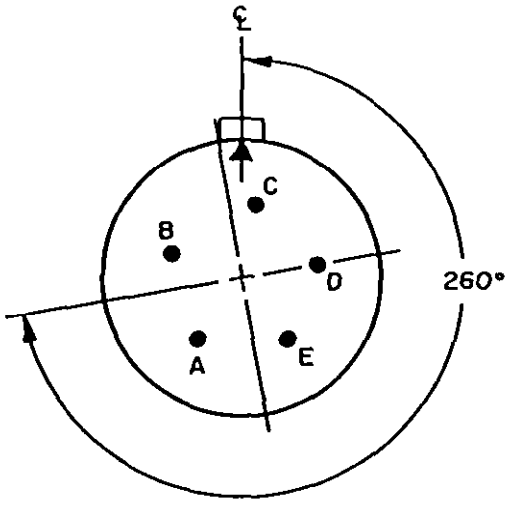
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-30	A	5	16	●	All	MS33684-30

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 18-31 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 18-20. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 18-20 EXCEPT FOR POSITION OF INSERT IN SHELL.

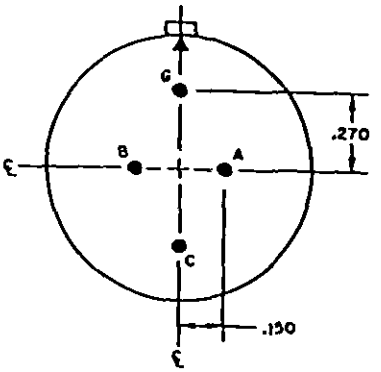
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
18	-31	A	5	16	●	All	MS33684-31

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 20-10 IS CANCELLED, WITHOUT REPLACEMENT, AFTER 30 APRIL 1992. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



SOCKET CONTACTS A AND B SHALL GROUND TO CONTACT G WHEN PIN CONTACTS ARE REMOVED

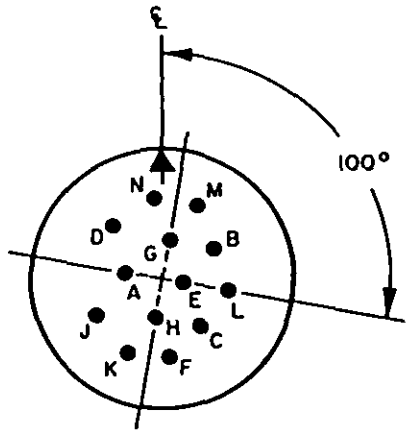
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-10	A	4	16	●	All	MS33685-10

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 20-25 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 20-11. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 20-11 EXCEPT FOR POSITION OF INSERT IN SHELL.

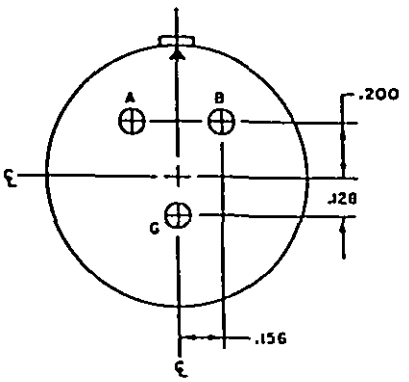
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-25	INST	13	16	●	All	MS33685-25

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 20-26 IS CANCELLED, WITHOUT REPLACEMENT, AFTER 30 APRIL 1992. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



SOCKET CONTACTS A AND B GROUND TO CONTACT G WHEN PINS ARE REMOVED

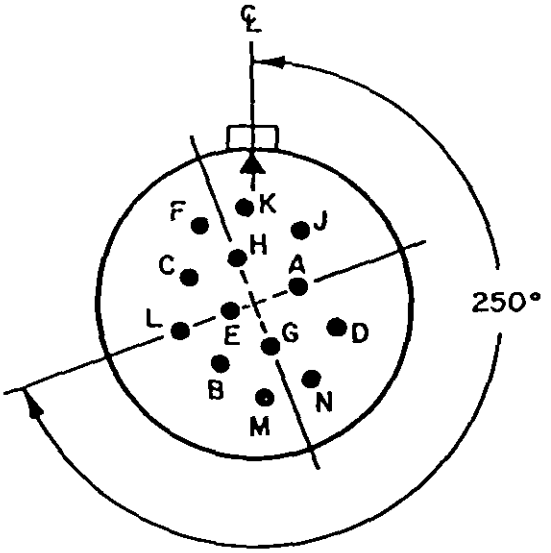
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-26	A	3	12	⊕	All	MS33685-26

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 20-30 CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 20-11. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 20-11 EXCEPT FOR POSITION OF INSERT IN SHELL.

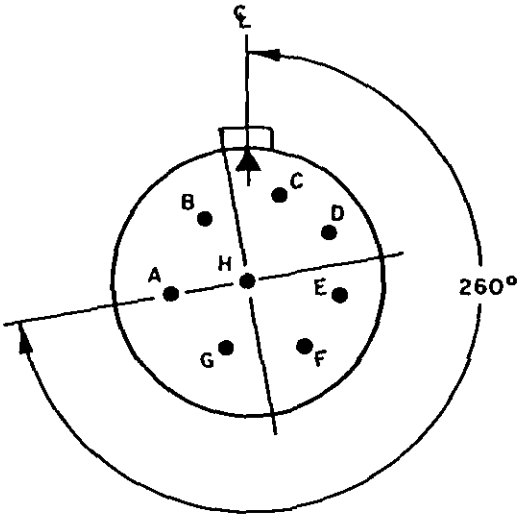
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-30	INST	13	16	●	All	MS33685-30

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 20-32 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 22-5. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 20-7 EXCEPT FOR POSITION OF INSERT IN SHELL.

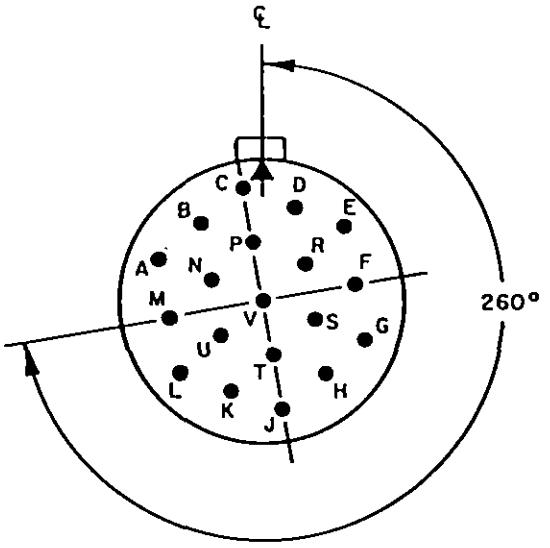
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
20	-32	D	4	16	●	A,B,G,H	MS33685-32
		A	4			All others	

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 22-30 CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 20-14. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 22-14 EXCEPT FOR POSITION OF INSERT IN SHELL.

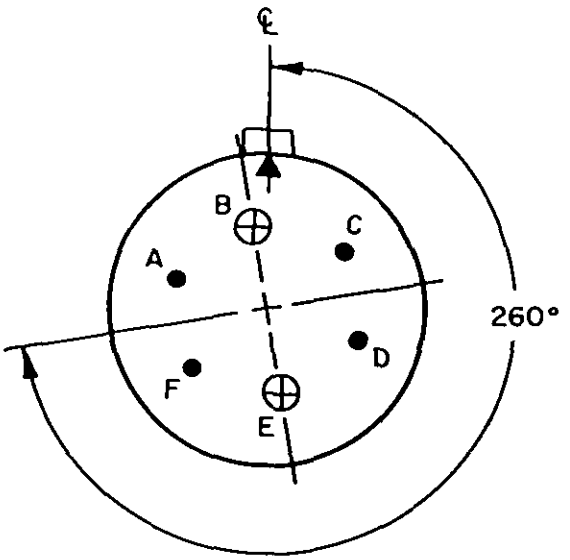
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-30	A	19	16	●	All	MS33686-30

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 22-32 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 20-7. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS ARE IDENTICAL TO 22-5 EXCEPT FOR POSITION OF INSERT IN SHELL.

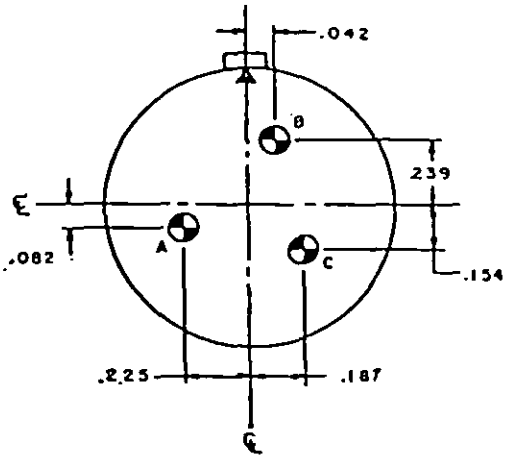
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-32	D	4	16	●	All	MS33686-32
			2	12	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 22-35 IS CANCELLED, WITHOUT REPLACEMENT, AFTER 30 APRIL 1992. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



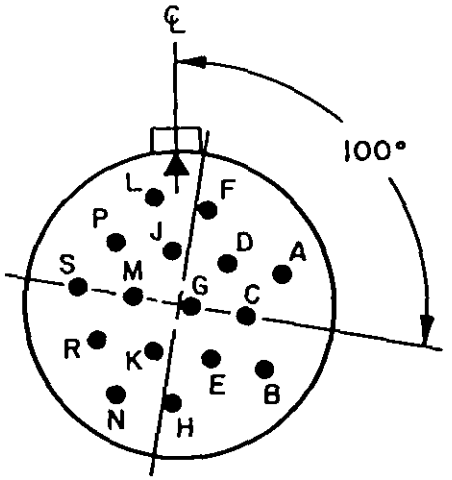
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
22	-35	D	3	8		All	MS33686-35

Source: https://assist.dla.mil - Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 24-15 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 24-5. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 24-5 EXCEPT FOR POSITION OF INSERT IN SHELL

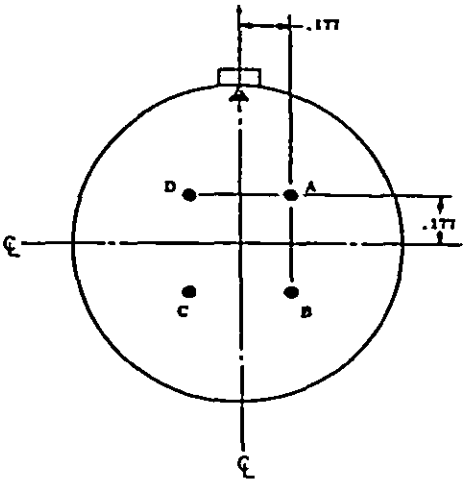
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-15	A	16	16	●	All	MS33687-15

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 24-18 IS CANCELLED, WITHOUT REPLACEMENT, AFTER 30 APRIL 1992. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

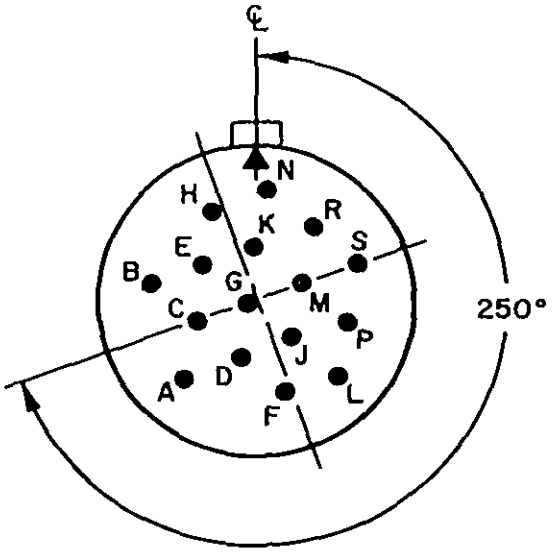
SOCKET CONTACTS A & B SHORT TOGETHER AND C & D SHORT TOGETHER WHEN PIN CONTACTS ARE REMOVED

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-18	---	4	16	●	All	MS33687-18

Source: https://assist.dla.mil - Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 24-24 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 24-5. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 24-5 EXCEPT FOR POSITION OF INSERT IN SHELL.

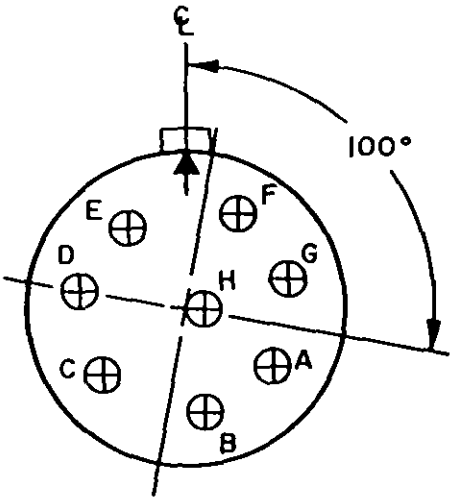
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-24	A	16	16	●	All	MS33687-24

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 24-25 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 24-6. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 24-6 EXCEPT FOR POSITION OF INSERT IN SHELL.

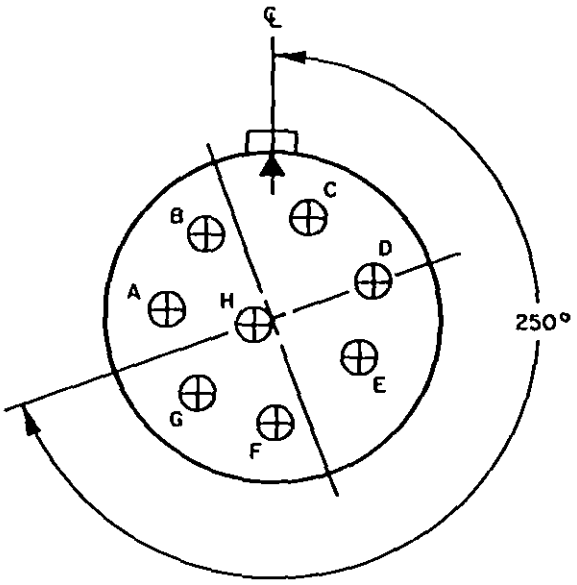
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-25	D	3	12	⊕	A,G,H	MS33687-25
		A	5			All others	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 24-26 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 24-6. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 24-6 EXCEPT FOR POSITION OF INSERT IN SHELL.

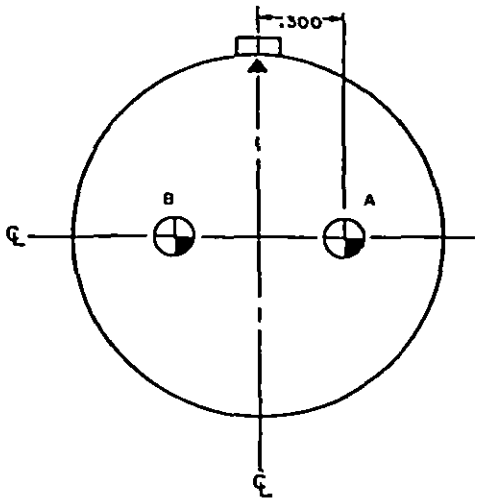
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
24	-26	D	3	12	⊕	A,G,H	MS33687-26
		A	5			All others	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 28-7 IS CANCELLED, WITHOUT REPLACEMENT, AFTER 30 APRIL 1992. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



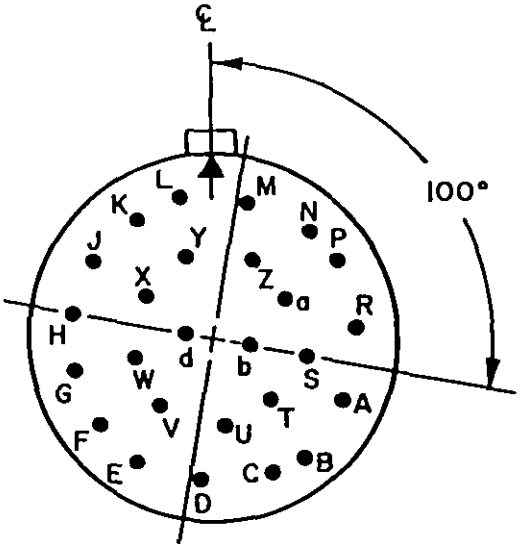
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
35	110	250	325

NOT APPLICABLE TO MIL-C-22992.

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-7	D	2	4	⊕	A11	MS33688-7

Source: https://assist.dla.mil Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 28-13 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 28-12. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 28-12 EXCEPT FOR POSITION OF INSERT IN SHELL.

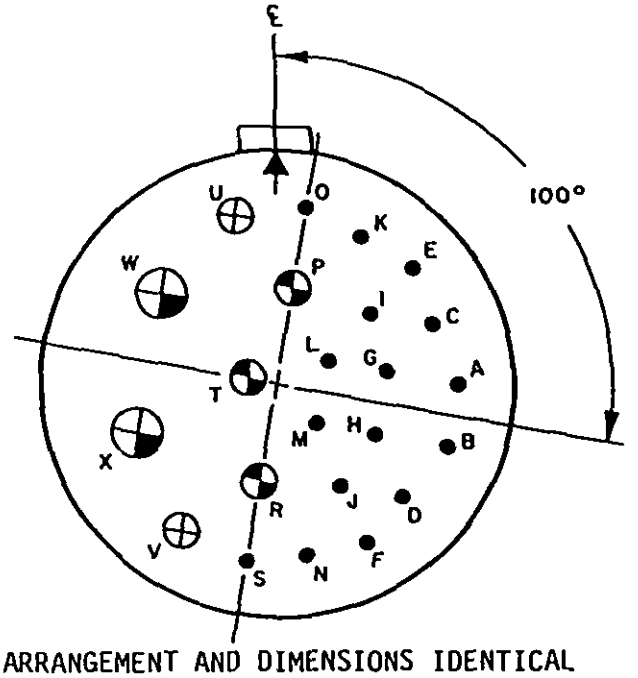
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
28	-13	A	26	16	●	All	MS33688-13

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 32-16 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 32-6. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 32-6 EXCEPT FOR POSITION OF INSERT IN SHELL.

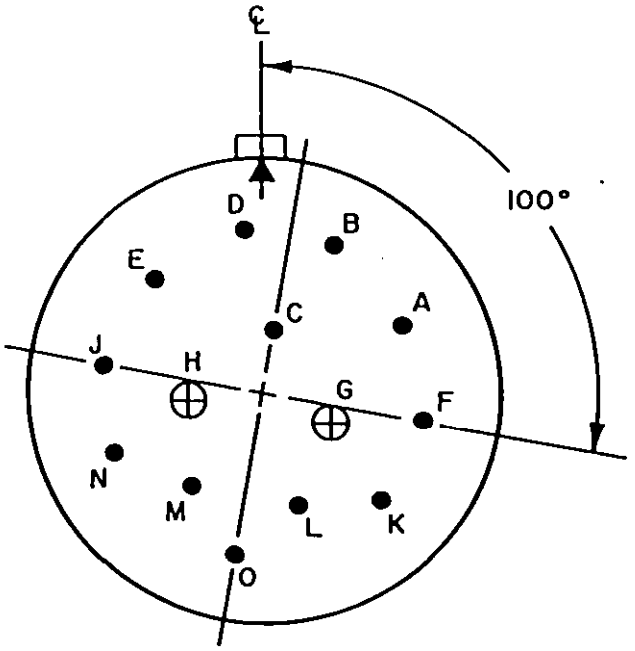
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-16	A	16	16	●	All	MS33689-16
			2	12	⊕		
			3	8	◔		
			2	4	⊕		

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 32-18 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 32-4. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 32-4 EXCEPT FOR POSITION OF INSERT IN SHELL

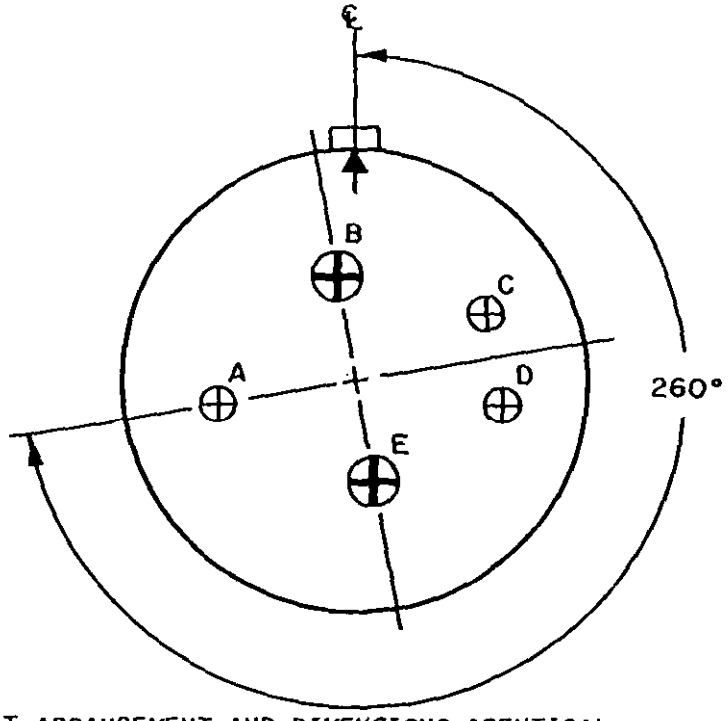
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-18	D	2	12	●	G,H	MS33689-18
		A	4	16	⊕	F,J,K,N	
		D	8			All others	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 32-19 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 32-1. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 32-1 EXCEPT FOR POSITION OF INSERT IN SHELL

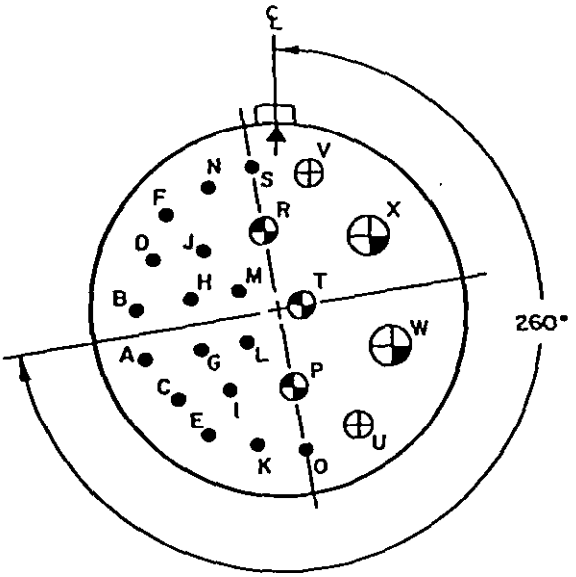
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-19	E	1	12	⊕	A	MS33689-19
			2			D,C	
		0	2	0	⊕	B,E	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 32-20 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 32-6. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 32-6 EXCEPT FOR POSITION OF INSERT IN SHELL

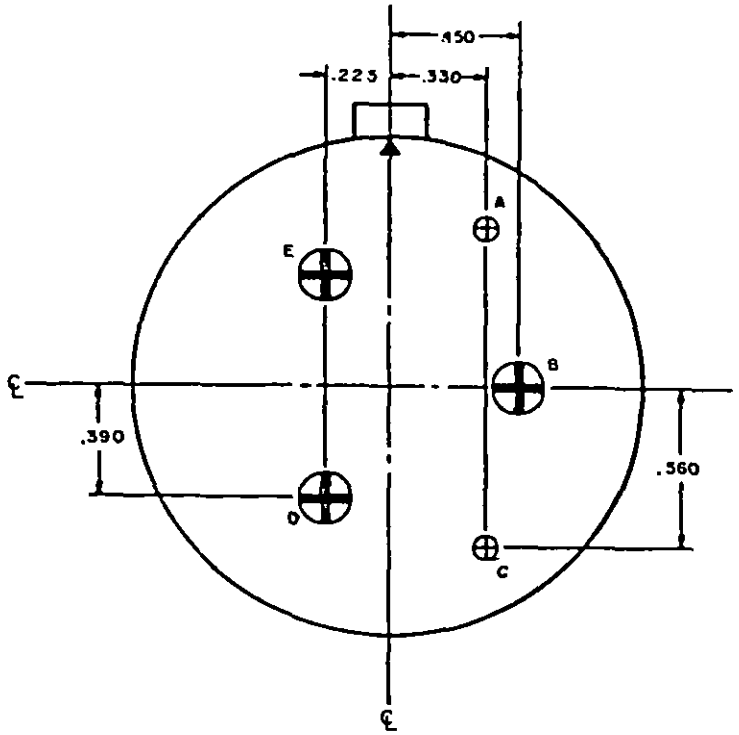
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
32	-20	A	16	16	●	All	MS33689-20
			2	12	⊕		
			3	8	◕		
			2	4	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 36-2 IS CANCELLED, WITHOUT REPLACEMENT, AFTER 30 APRIL 1992. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



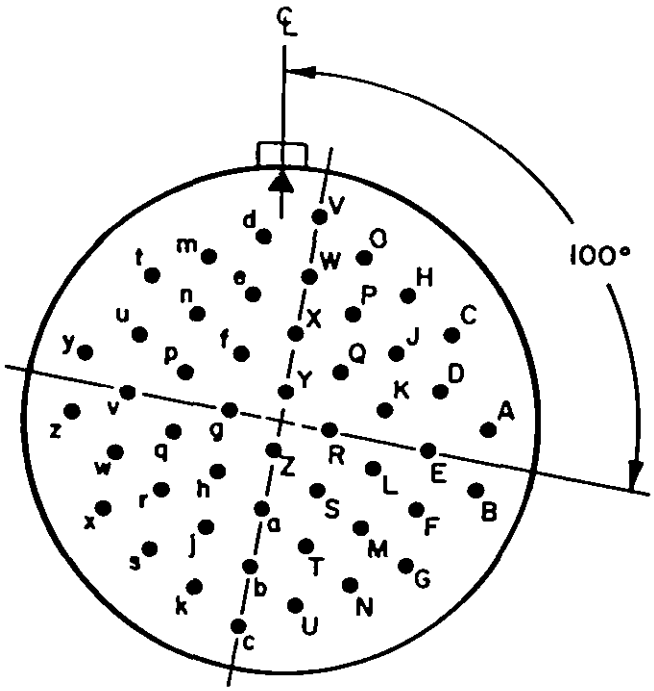
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-2	D	2	12	⊕	All	MS33690-2
			3	0	⊕		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 36-11 CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 36-10. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 36-10 EXCEPT FOR POSITION OF INSERT IN SHELL

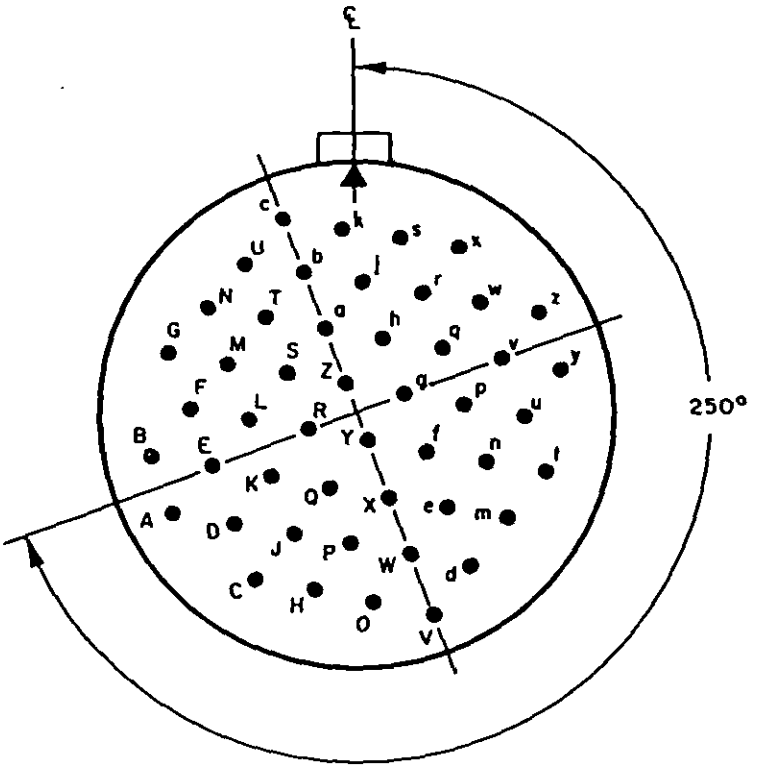
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-11	A	48	16	●	All	MS33690-11

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

* INSERT ARRANGEMENT 36-12 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 36-10 FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 36-10 EXCEPT FOR POSITION OF INSERT IN SHELL

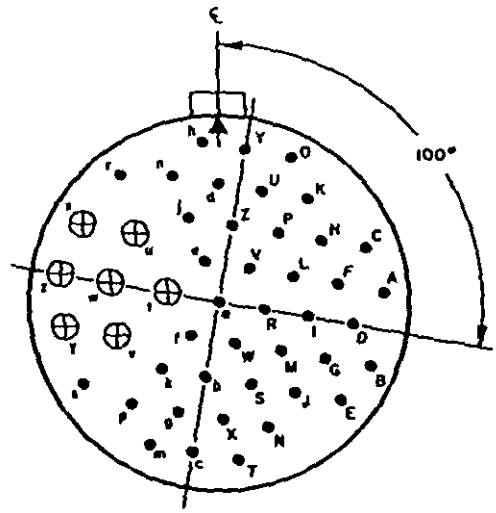
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-12	A	48	16	●	All	MS33690-12

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INSERT ARRANGEMENT 36-16 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 36-7. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 36-7 EXCEPT FOR POSITION OF INSERT IN SHELL

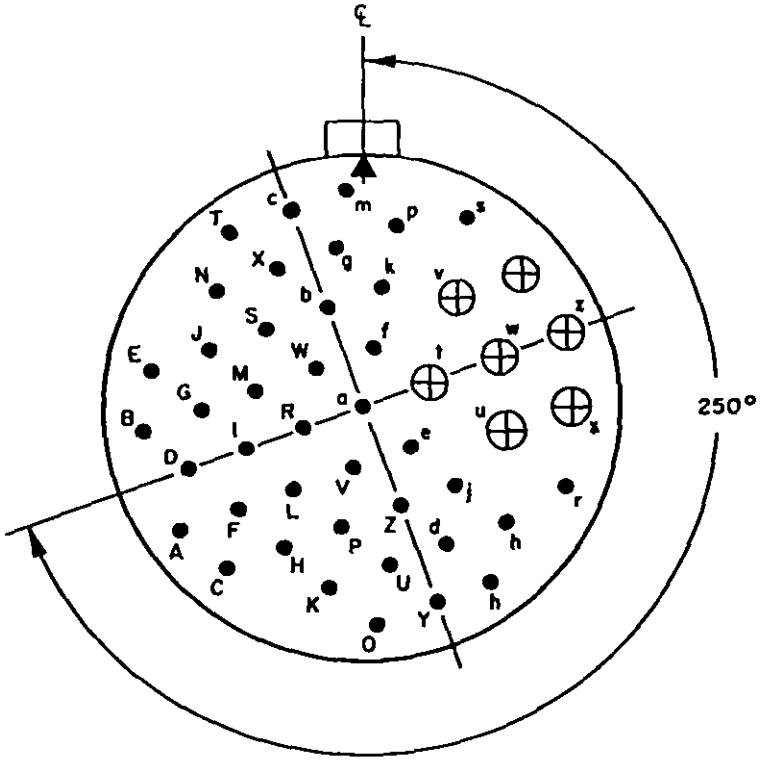
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-16	A	40	16	●	All	MS33690-16
			7	12	⊕		

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A INSERT ARRANGEMENT 36-17 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 36-7. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 36-7 EXCEPT FOR POSITION OF INSERT IN SHELL

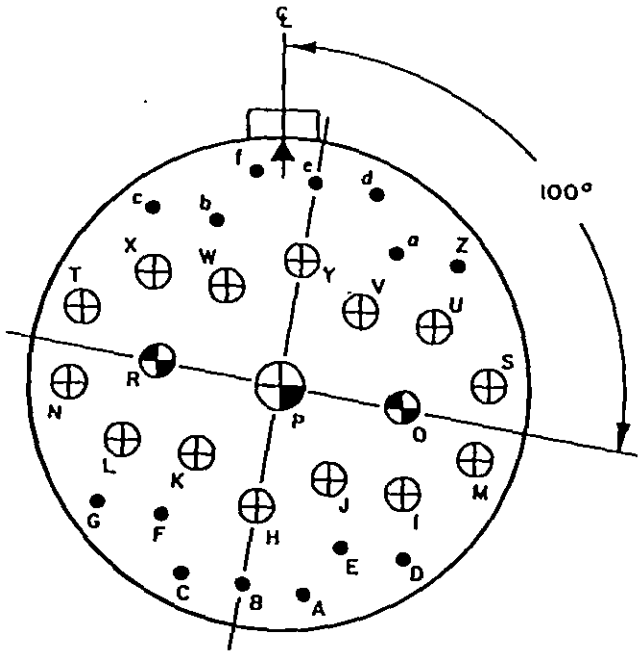
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-17	A	40	16	●	A11	MS33690-17
			7	12	⊕		

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* INSERT ARRANGEMENT 36-18 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 36-9. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 36-9 EXCEPT FOR POSITION OF INSERT IN SHELL.

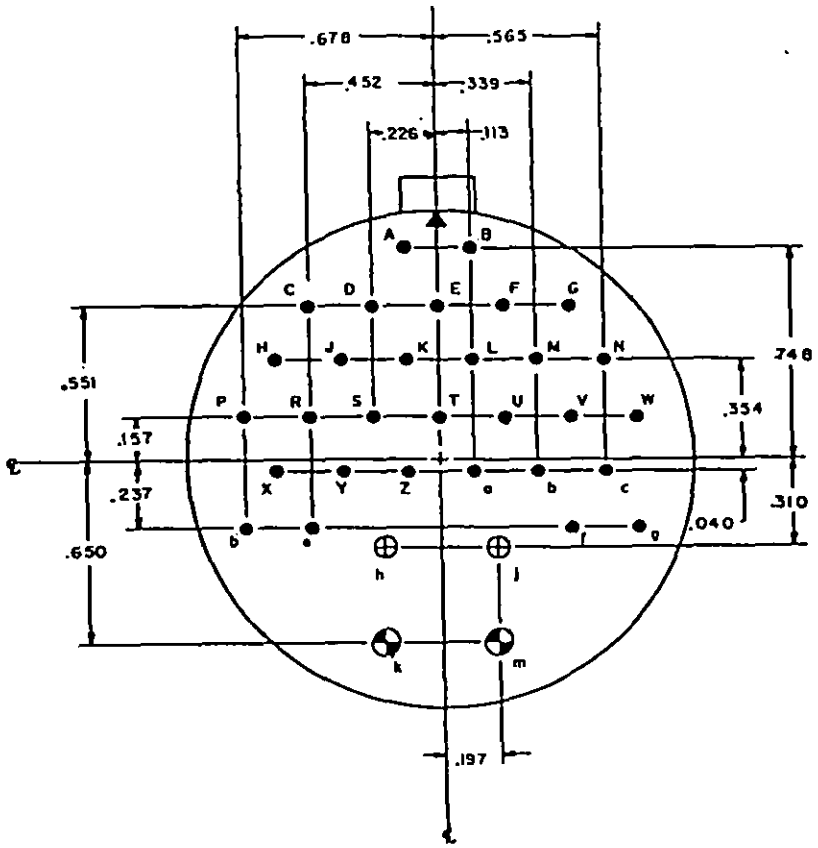
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-18	A	14	16	●	All	MS33690-18
			14	12	⊕		
			2	8	◕		
			1	4	◔		

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INSERT ARRANGEMENT 36-20 IS CANCELLED, WITHOUT REPLACEMENT, AFTER 30 APRIL 1992. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



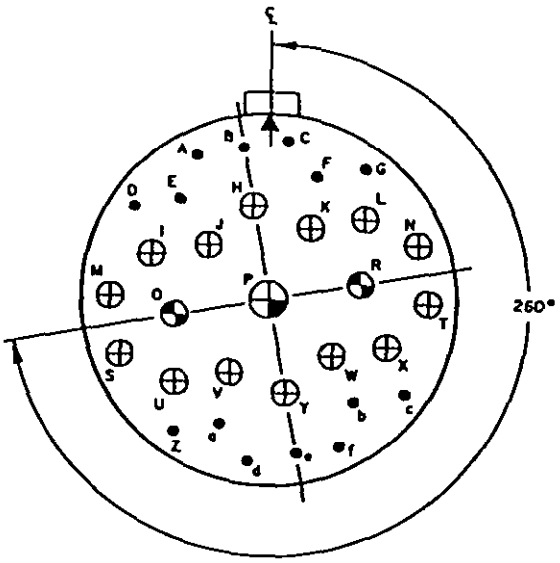
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-20	A	30	16	●	All	MS33690-20
			2	12	⊕		
			2	8	◒		

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* INSERT ARRANGEMENT 36-21 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 36-9. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 36-9 EXCEPT FOR POSITION OF INSERT IN SHELL.

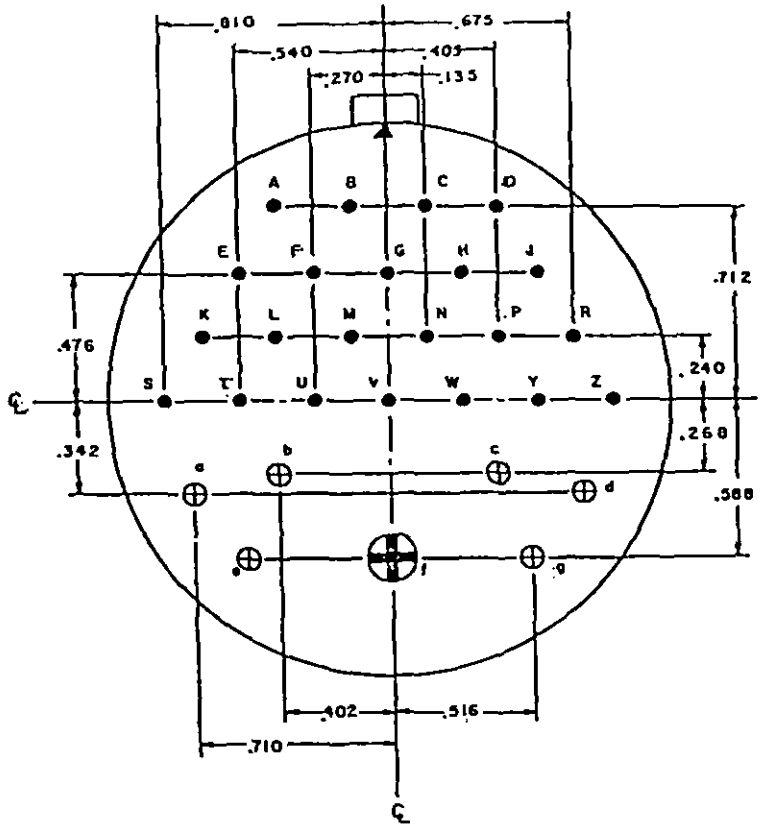
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
36	-21	A	14	16	●	All	MS33690-21
			14	12	⊕		
			2	8	◒		
			1	4	◒		

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* INSERT ARRANGEMENT 40-12 IS CANCELLED, WITHOUT REPLACEMENT, AFTER 30 APRIL 1992. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



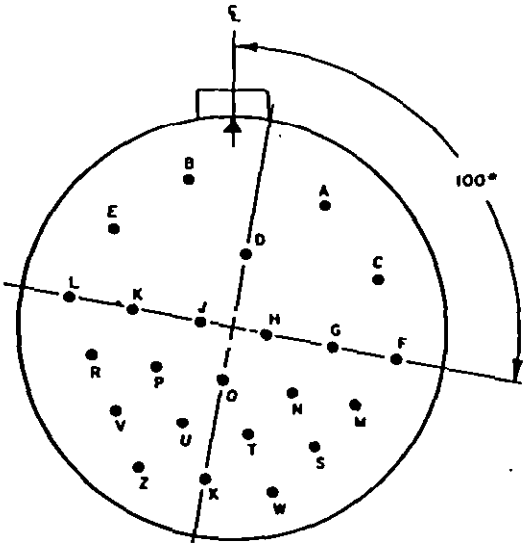
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-12	D	22	16	●	All	MS33691-12
			6	12	⊕		
			1	0	⊕		

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* INSERT ARRANGEMENT 40-13 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 40-2. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 40-2 EXCEPT FOR POSITION OF INSERT IN SHELL

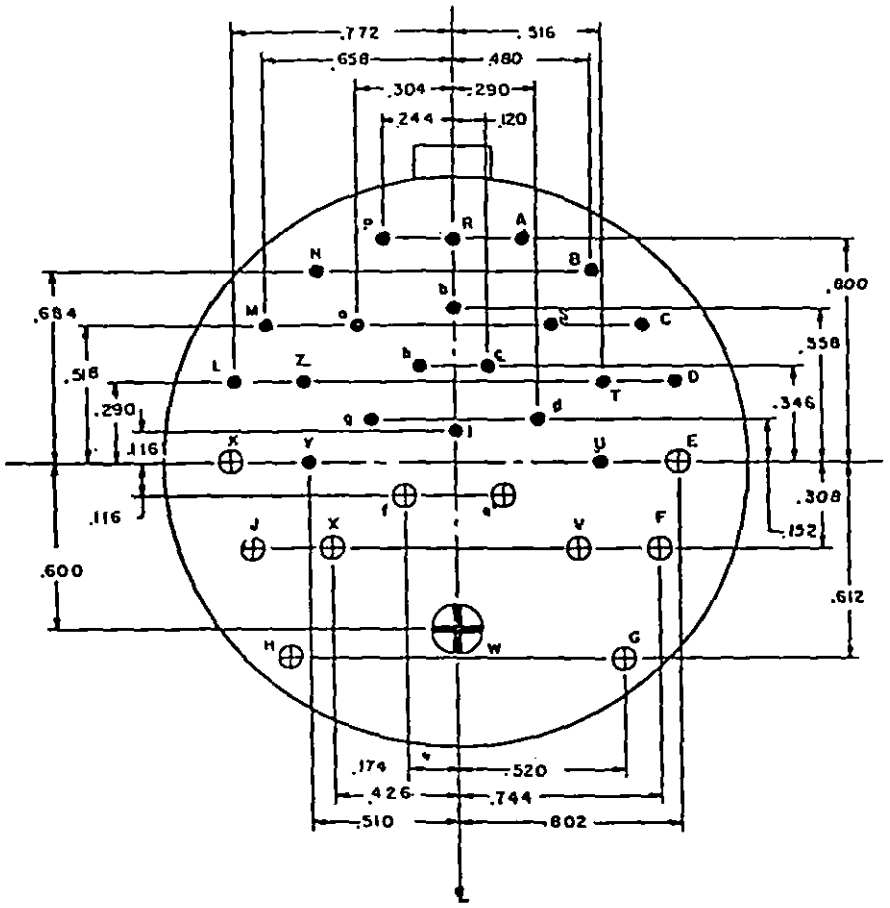
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-13	B	5	16	●	A,B,C,D,E	MS33691-13
		D	18			All others	

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INSERT ARRANGEMENT 40-14 IS CANCELLED, WITHOUT REPLACEMENT, AFTER 30 APRIL 1992. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



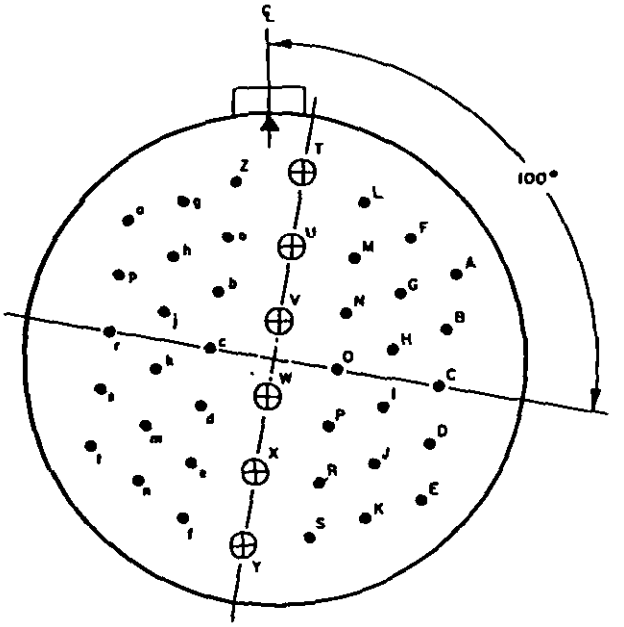
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
40	-14	0	21	16	●	All	MS33691-14
			10	12	⊕		
			1	0	⊕		

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* INSERT ARRANGEMENT 44-5 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 44-1. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 44-1 EXCEPT FOR POSITION OF INSERT IN SHELL

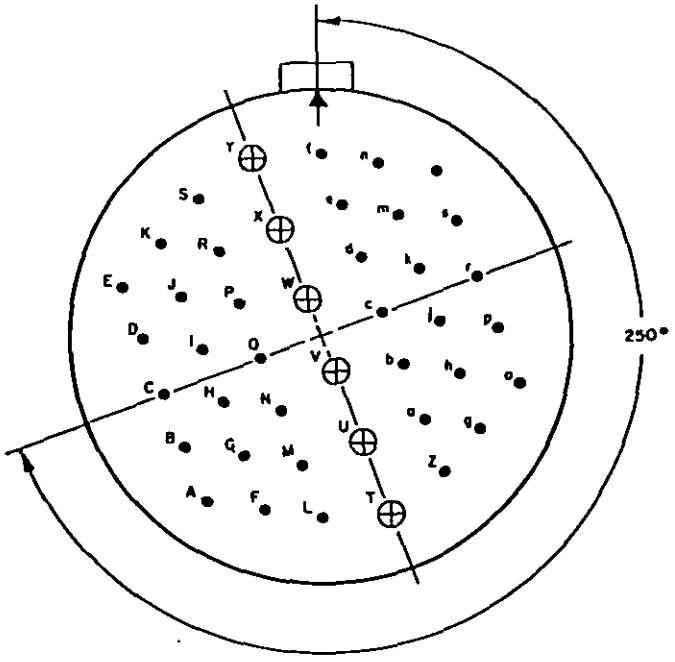
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
44	-5	D	36	16	●	All	MS33692-5
			6	12	⊕		

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* INSERT ARRANGEMENT 44-6 IS CANCELLED AFTER 30 APRIL 1992. FOR REPLACEMENT PURPOSES USE 44-1. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



CONTACT ARRANGEMENT AND DIMENSIONS IDENTICAL TO 44-1 EXCEPT FOR POSITION OF INSERT IN SHELL

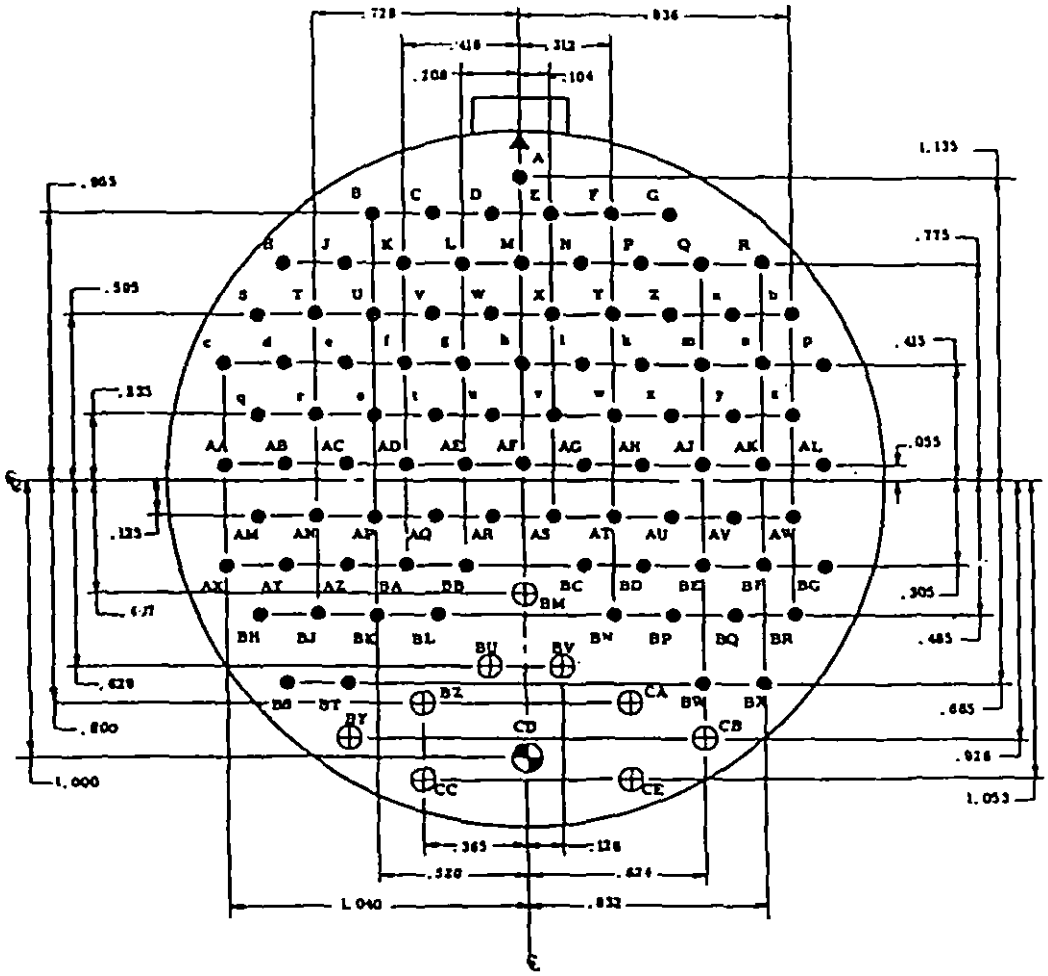
ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

(Not applicable to MIL-C-22992)

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
44	-6	D	36	16	●	All	MS33692-6
			6	12	⊕		

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* INSERT ARRANGEMENT 48-5 IS CANCELLED, WITHOUT REPLACEMENT, AFTER 30 APRIL 1992. FOR INVENTORY DEPLETION INFORMATION SEE 6.4.



ALTERNATE INSERT POSITIONS			
A DEGREES			
W	X	Y	Z
---	---	---	---

Shell size	Arrangement number	Service rating	Number of contacts	Size contacts	Symbol	Contact location	Supersedes document
48	-5	A	90	16	●	All	MS33693-5
			9	12	⊕		
			1	8	◕		

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-STD-1651A	2. DOCUMENT DATE (YYMMDD) 920403

3. DOCUMENT TITLE: Insert Arrangements for MIL-C-5015, MIL-C-22992 (Series C, J, and R) and MIL-C-83723 (Series II) Electrical Connectors

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDING OFFICER, NAVAL AIR WARFARE CENTER AIRCRAFT DIVISION LAKEHURST SYSTEMS REQUIREMENTS DEPARTMENT	b. TELEPHONE (Include Area Code) (1) Commercial: (908) 323-7488 (2) AUTOVON: 624-7488
c. ADDRESS (Include Zip Code) CODE SR3 LAKEHURST, NJ 08733-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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